U.S. DEPARTMENT OF AGRICULTURE
AGRICULTURAL RESEARCH SERVICE

EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN

A/E CONTRACT NO: 12805B23D0001
A/E ORDER NO: 1232SA25F0384

EDINBURG - KERRVILLE BLDG 4
22675 N MOOREFIELD RD
MAB6419

REGISTERED ARCHITECT
JOHN PAUL FRANCIS
STATE OF TEXAS

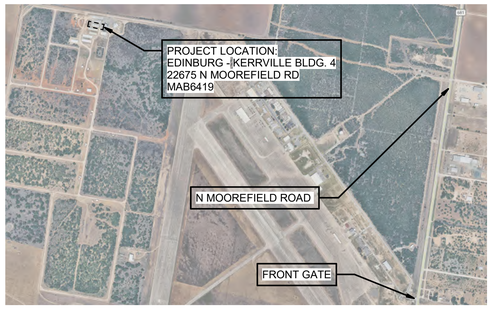
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VICINITY MAP



LOCATION MAP



BID DESCRIPTION

BASE BID: DEMOLISH EXISTING ROOFING BACK TO RIGID FRAME STRUCTURE, DEMOLISH EXISTING CANOPY AT WEST END OF STRUCTURE COMPLETE INCLUDING FOUNDATION (FILL VOID LEFT FROM FOUNDATION REMOVAL WITH SELECT FILL AND COMPACT) , DEMOLISH RAIL FENCING AND GATES COMPLETE INCLUDING CONCRETE FOUNDATIONS (FILL HOLES WITH SELECT FILL AND COMPACT), DEMOLISH CMU BLOCK STRUCTURE COMPLETE INCLUDING FOUNDATION(FILL VOID LEFT FROM FOUNDATION REMOVAL WITH SELECT FILL AND COMPACT), DEMOLISH EXISTING SUMP PIT TO EXTENTS INDICATED (BACKFILL REMAINING STRUCTURE WITH SELECT FILL AND COMPACT), DEMOLISH EXISTING WATER PIPING TO WATERING TROUGHS COMPLETE TO EXTENTS INDICATED, DEMOLISH EXISTING ELECTRICAL POWER AND LIGHTING SYSTEMS COMPLETE TO EXTENTS INDICATED. PROVIDE EXTENSION TO RIGID FRAME AS INDICATED, INSTALL 8" PURLINS AS INDICATED, PROVIDE NEW EXPOSED FASTENER METAL ROOF PANELS OVER REFLECTIVE THERMAL INSULATION, PROVIDE METAL FENCE RAILING AND GATES FOR CORRALS AS INDICATED, PROVIDE CONCRETE WALKING SURFACE AS INDICATED, PROVIDE CONCRETE ALTERATION TO THE EXISTING DIP VAT AS INDICATED, PROVIDE REINFORCED CONCRETE WALKING SURFACE WITH ROUGH BROOMED FINISH AS INDICATED, PROVIDE PLUMBING TO WATERING TROUGHS AS INDICATED, PROVIDE POWER AND LIGHTING AS INDICATED.

ADDITIVE ALTERNATE #1: PROVIDE SYSTEMS AS INDICATED ABOVE TO CREATE ADDITIONAL CATTLE CORRAL PENS 9, 10, AND 11 AS INDICATED.

ADDITIVE ALTERNATE #2: ADDITIVE ALTERNATE #2 PROVIDES GALVANIZATION FOR ALL FENCING, GATE, AND BOLLARD STEEL.

USDA

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REAL PROPERTY	xxxxx

NO.	DATE	BY	DESCRIPTION

REVISIONS

USDA Agricultural Research Service

USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
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MOORE FIELD EDINBURG, TEXAS

COVER SHEET

PROJECT NO. xxxxx	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD 20705	SHEET G100
DATE 08/31/2026		1 OF 32

STRUCTURAL GENERAL NOTES

DESIGN CRITERIA
CODES AND STANDARDS

A. BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE, AMERICAN CONCRETE INSTITUTE (ACI), ACI 318-19.
B. MINIMUM DESIGN LOADS FOR BUILDING AND OTHER STRUCTURES, ASCE 7-22.
C. INTERNATIONAL BUILDING CODE (IBC), 2024
D. AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC) STEEL CONSTRUCTION MANUAL, 15th EDITION
E. STRUCTURAL COMPONENTS IN THIS PROJECT DESIGNED USING LRFD METHODOLOGY
F. AISI AMERICAN IRON AND STEEL INSTITUTE.

1.0 MINIMUM DESIGN LOADS AND REQUIREMENTS IN ACCORDANCE WITH IBC, 2024

LIVE LOADS:	
ROOF	12 PSF
WIND DESIGN DATA:	
ULTIMATE WIND SPEED	134 MPH
RISK CATEGORY	I
WIND IMPORTANCE FACTOR	1.0
WIND EXPOSURE	
COMPONENTS AND CLADDING DESIGN PRESSURE	SEE TABLE
EARTHQUAKE DESIGN DATA:	
SEISMIC IMPORTANCE FACTOR	1.0
OCCUPANCY CATEGORY	N.A.
SS	0.11
S1	0.65
SITE CLASS	E
SDS	0.092
SD1	0.08
SEISMIC DESIGN CATEGORY	A
SEISMIC RESISTING SYSTEM	STEEL FRAMES NOT SPECIFICALLY DETAILED FOR SEISMIC RESISTANCE
DESIGN BASE SHEAR	0.057W
CS	0.057
R	3
DESIGN METHOD	EQUIVALENT LATERAL FORCE
SNOW DESIGN DATA	
GROUND SNOW LOAD	0 PSF
DESIGN UNIFORM SNOW LOAD	4.2 PSF
FROST PENETRATION	7 INCHES

2.0 MATERIALS

CONCRETE (28 DAY STRENGTH)	3,000 PSI (COMP. STRENGTH)
CONCRETE WORK EXPOSED TO FREEZING AND THAWING FOR EXTERIOR WALKS, PLATFORMS STEPS, ELEVATED SLAB, AND SLAB ON GRADE	
ALL OTHERS	3,000 PSI (COMP STRENGTH)
REINFORCING:	ASTM 615/A615M
BARS	ASTM A185
WELDED WIRE FABRIC	

STRUCTURAL STEEL SHAPES, PLATES, AND MISCELLANEOUS SHAPES:

APPLICABLE SHAPE SERIES	ASTM DESIGNATION	MIN. YIELD STRESS (FY) (KSI)
W	A992	50
M	A36	36
S	A36	36
C	A36	36
MC	A36	36
L	A36	36
HSS (RECTANGLE)	A500	46
HSS (ROUND)	A500	42
PIPE	A53	35
PLATES	A36	36
PLATES	A572	50
BARS	A572	42

BOLTS:	ASTM A325 BEARING TYPE N
CONCRETE MASONRY UNITS:	ASTM C90
MORTAR MUST BE TYPE S CONFORMING TO ASTM C270. ALL GROUT MUST BE PUDDLED OR VIBRATED IN PLACE.	
ANCHOR RODS:	ASTM F593, GRADE 36
GROUT:	2KSI

3.0 CAST-IN-PLACE CONCRETE NOTES:

3.1 CONCRETE COVER FOR REINFORCEMENT:
- A. CONCRETE DEPOSITED AGAINST AND PERMANENTLY EXPOSED TO EARTH
- B. CONCRETE EXPOSED TO EARTH OR WEATHER:
  - #5 BARS OR SMALLER
  - #5 BARS OR LARGER
- C. CONCRETE NOT EXPOSED TO WEATHER OR NOT IN CONTACT WITH GROUISLAB, WALL, AND JOISTS BEAMS

3.2 UNLESS SPECIFICALLY NOTED, SCHEDULED OR DETAILED OTHERWISE PROVIDE DEVELOPMENT LENGTH FOR REINFORCING IN CONCRETE COMPONENTS IN ACCORDANCE WITH THE SCHEDULE IN NOTE 3.3 BELOW. THIS SCHEDULE SHALL APPLY TO ALL DEVELOPMENT LENGTHS NOT OTHERWISE NOTED, DETAILED OR SCHEDULED IN THE DRAWINGS OR SPECIFICATIONS.

3.3 REINFORCING BAR DEVELOPMENT LENGTHS Ld

TENSION AND DEVELOPMENT SPLICE LENGTHS - f'c = 3000 psi CONCRETE

BAR SIZE	DEVELOPMENT LENGTH		SPLICE LENGTH		STANDARD 90 DEGREE HOOK		
	TOP BAR (INCHES)	OTHER BAR (INCHES)	TOP BAR (INCHES)	OTHER BAR (INCHES)	EMBED (INCHES)	LEG LENGTH (INCHES)	BEND DIA. (INCHES)
#3	22	17	28	22	6	6	2.25
#4	29	22	37	29	8	8	3
#5	36	28	47	36	10	10	3.75
#6	43	33	56	43	12	12	4.5
#7	63	48	81	63	14	14	5.25
#8	72	55	93	72	16	16	6

TENSION AND DEVELOPMENT SPLICE LENGTHS - f'c = 4000 psi CONCRETE

BAR SIZE	DEVELOPMENT LENGTH		SPLICE LENGTH		STANDARD 90 DEGREE HOOK		
	TOP BAR (INCHES)	OTHER BAR (INCHES)	TOP BAR (INCHES)	OTHER BAR (INCHES)	EMBED (INCHES)	LEG LENGTH (INCHES)	BEND DIA. (INCHES)
#3	19	15	24	22	6	6	2.25
#4	25	19	32	25	8	8	3
#5	31	24	40	31	10	10	3.75
#6	37	29	48	37	12	12	4.5
#7	54	42	70	54	14	14	5.25
#8	62	48	80	25	16	16	6

NOTES:
1. STRAIGHT DEVELOPMENT LENGTH AND SPLICE LENGTHS ARE BASED ON UNCOATED BARS ASSUMING CENTER TO CENTER BAR SPACING GREATER THAN 3 BAR DIAMETERS WITHOUT TIES OR STIRRUPS OR 2 BAR DIAMETERS WITH TIES OR STIRRUPS, AND CLEAR COVER OF 1 BAR DIAMETER.
2. TOP BARS ARE HORIZONTAL BARS WITH MORE THAN 12 INCHES OF CONCRETE BELOW THE BAR.
3. STANDARD 90 DEGREE HOOK EMBEDMENT LENGTH ARE BASED ON SIDE COVER GREATER THAN 2.5 INCHES AND END COVER GREATER THAN 2 INCHES WITHOUT TIES AROUND HOOKS.

3.4 LAP SPLICE LENGTHS FOR REINFORCING BARS MUST BE THE SAME AS SHOWN IN THE TABLE.

3.5 WHEN TWO BARS OF DIFFERENT SIZES ARE LAPPED, THE SMALLER SIZE GOVERNS THE LAP LENGTH UNLESS SPECIFICALLY NOTED OTHERWISE.

3.6 WHEN REINFORCING STEEL IS NOTED AS CONTINUOUS REINFORCING IN GRADE BEAMS, WALLS, SLABS AND/OR BEAMS, SPLICE CONTINUOUS REINFORCING STEEL ONLY WHEN UNAVOIDABLE DUE TO STOCK LENGTHS. STAGGER ALL SPLICES A MINIMUM OF 4'-0". ADJACENT BAR SPLICES ARE NOT ACCEPTABLE. LOCATE THE TOP BAR SPLICES WITHIN THE MIDDLE HALF OF THE SPAN AND LOCATE THE BOTTOM BAR SPLICES AT SUPPORTS, OR BETWEEN SUPPORTS AND 1/3 SPAN POINT, UNLESS NOTED OTHERWISE ON PLANS, DETAILS OR SCHEDULES.

3.7 HORIZONTAL JOINTS WILL NOT BE PERMITTED IN CONCRETE CONSTRUCTION EXCEPT AS SHOWN ON THE DRAWINGS. VERTICAL JOINTS MUST OCCUR AT LOCATIONS INDICATED.

3.8 AT CONSTRUCTION JOINTS, CONTACT SURFACES MUST BE CLEAN AND FREE OF LATENCIES AND INTENTIONALLY ROUGHENED TO A FULL AMPLITUDE OF APPROXIMATELY 1/4 INCH

3.9 PROVIDE FULL EMBEDMENT WITH 90-DEGREE HOOKS FOR ALL DOWELS IF NOT OTHERWISE NOTED.

3.10 CHAMFER ALL EXPOSED TO VIEW CORNERS 3/4", U.N.O.

4.0 FOUNDATION:

4.1 DESIGN FOOTINGS FOR A MAXIMUM ALLOWABLE BEARING PRESSURE OF 1000 PSF.

4.2 PROTECT PIPES AND CONDUITS RUNNING THROUGH WALLS AND SLABS WITH 1/2 INCH EXPANSION MATERIAL.

4.3 MAINTAIN SUBGRADE AND FILL MOISTURE CONTENT UNTIL FOUNDATIONS ARE PLACED.

4.4 DO NOT PLACE SLAB AGAINST SUBGRADE CONTAINING FREE WATER, FROST, OR ICE.

4.5 MAINTAIN PROPER SITE DRAINAGE DURING CONSTRUCTION TO ENSURE SURFACE RUNOFF AWAY FROM STRUCTURES AND TO PREVENT PONDING OF SURFACE RUNOFF NEAR THE STRUCTURES.

4.6 KEEP OPEN EXCAVATIONS AROUND BUILDING DRY. BACKFILL AGAINST GRADE BEAMS AS SOON AS PRACTICAL. PUMP WATER OUT OF OPEN EXCAVATIONS IF FLOODING OCCURS PRIOR TO BACKFILLING.

4.7 ARRANGE FOR INDEPENDENT TESTING AGENCY TO MONITOR CUT AND FILL OPERATIONS, AND PERFORM FIELD DENSITY AND MOISTURE CONTENT TESTS TO VERIFY COMPACTION AND APPROVE FOOTING SUBGRADE PRIOR TO PLACING CONCRETE.

5.0 GENERAL:

5.1 WHERE A SECTION OR DETAIL IS SHOWN FOR ONE CONDITION, IT MUST ONLY APPLY TO LIKE OR SIMILAR CONDITIONS.

6.0 STRUCTURAL STEEL NOTES

6.1 DIMENSIONS: TO CENTERLINES OF COLUMNS AND BEAMS AND TO TOP SURFACES, TO TOP FLANGES OF BEAMS, AND BACK OF CHANNELS AND ANGLES UNLESS SHOWN OTHERWISE.

6.2 ELEVATIONS: REFERS TO TOP SURFACE OF FLANGE OR MEMBER, UNLESS SHOWN OTHERWISE.

6.3 WELD SIZES NOT INDICATED ON DRAWINGS: PROVIDE MINIMUM WELD CONNECTIONS IN ACCORDANCE WITH AISC. WELD IN ACCORDANCE WITH AWS REQUIREMENTS.

6.4 STRUCTURAL STEEL OR MISCELLANEOUS STEEL: STRUCTURAL OR MISCELLANEOUS STEEL MUST BE DETAILED, FABRICATED AND ERECTED IN ACCORDANCE WITH THE AISC SPECIFICATIONS FOR BUILDINGS. SHOP CONNECTIONS MUST BE MADE WITH A MINIMUM 3/4" DIAMETER ASTM A325 TYPE N BOLTS, UNLESS OTHERWISE NOTED.

TABLE 1
DESIGN WIND PRESSURE COMPONENTS AND CLADDING (C & C)

NOTE:
THE DESIGN WIND PRESSURES SHOWN BELOW ARE BASED UPON THE ULTIMATE DESIGN WIND SPEED OF 134 MPH. THE WIND PRESSURES SHOWN ARE TO BE USED WITH THE LOAD AND RESISTANCE FACTOR (LRFD) METHOD. ALLOWABLE STRENGTH DESIGN (ASD) PRESSURES SHOULD BE CALCULATED BASED UPON A NOMINAL VELOCITY OF 85.2 MPH.

	CLEAR WIND FLOW						
	WIND ZONE	ZONE 3		ZONE 2		ZONE 1	
		POS	NEG	POS	NEG	POS	NEG
WIND PRESSURE	≤ 9SF	78.9 PSF	-117.8 PSF	60.0 PSF	-59.8 PSF	39.4 PSF	-39.3 PSF
	> 9, ≤ 36SF	60.0 PSF	-59.8 PSF	60.0 PSF	-59.8 PSF	39.4 PSF	-39.3 PSF
	> 36SF	39.4 PSF	-39.3 PSF	39.4 PSF	-39.3 PSF	39.4 PSF	-39.3 PSF

NOTES:
1. DESIGN WIND PRESSURES INDICATED MUST BE USED IN THE DESIGN OF COMPONENTS & CLADDING ELEMENTS COMPRISING THE BUILDING ENVELOPE.
2. POSITIVE PRESSURES ACT INWARD, TOWARD THE WIND SURFACE. NEGATIVE PRESSURES ACT OUTWARD, AWAY FROM THE WIND SURFACE.
3. LINEAR INTERPOLATION IS PERMITTED FOR INTERMEDIATE EFFECTIVE WIND AREAS.

7.0 DELEGATED DESIGN NOTES:

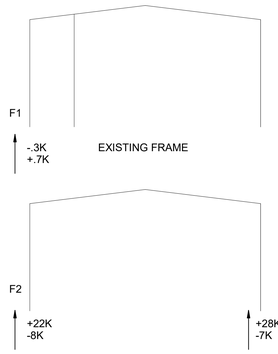
7.1 THE PRE-ENGINEERED METAL BUILDING (PEMB) SYSTEM INCLUDING PRIMARY FRAMING, SECONDARY FRAMING,BRACING, CONNECTIONS, METAL PANELS, ANCHOR BOLTS, AND PEMB ACCESSORIES MUST BE DELEGATED DESIGN TO THE PEMB MANUFACTURER.

7.2 DESIGN MUST BE PERFORMED UNDER THE RESPONSIBLE CHARGE OF A PROFESSIONAL ENGINEER LICENSED IN THE STATE OF TEXAS.

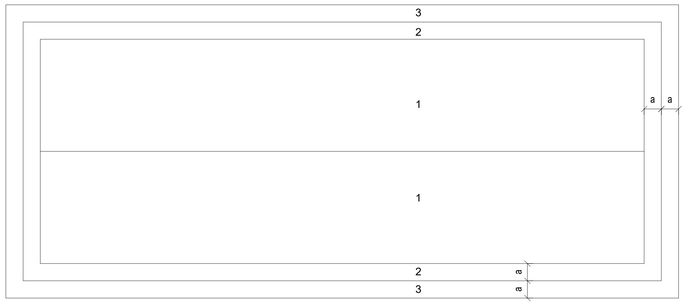
7.3 DELEGATED DESIGN MUST CONFORM TO THE GEOMETRY, LOADS, AND PERFORMANCE CRITERIA INDICATED IN THE CONTRACT DOCUMENTS.

7.4 SUBMIT SIGNED AND SEALED PEMB CALCULATIONS AND SHOP DRAWINGS OF THE FOLLOWING: PEMB DRAWINGS, ANCHOR ROD REACTIONS, AND DESIGN ASSUMPTIONS. FABRICATION MUST NOT PROCEED UNTIL SUBMITTALS ARE REVIEWED BY THE EOR.

7.5 PEMB MANUFACTURER MUST DESIGN THE BUILDING SUCH THAT COLUMN REACTIONS DO NOT EXCEED THOSE REASONABLY INFERRED FROM THE DESIGN CRITERIA AND PRELIMINARY REACTIONS SHOWN IN THE CONTRACT DOCUMENTS. INCREASE IN REACTIONS EXCEEDING 10% MUST BE REPORTED TO THE EOR PRIOR TO FABRICATION AND MAY REQUIRE REDESIGN AT NO ADDITIONAL COST TO THE GOVERNMENT IF CAUSED BY PEMB OPTIMIZATION OR SUBSTITUTION.



NOTE:
1. THE LOADS INDICATED ABOVE ARE THE FACTORED LOADS FOR THE FRAMES.
2. HORIZONTAL NEGATIVE LOADS ACT AWAY FROM THE STRUCTURE.
3. VERTICAL NEGATIVE LOADS INDICATE UPLIFT.
4. THE LOADS SHOWN ARE PROVIDED FOR BIDDING PURPOSES. IF FINAL PEMB LOADS FROM PMB MFR. DIFFER FROM THESE VALUES BY MORE THAN 10 PERCENT, THE CONTRACTOR MUST RE-SIZE THE FOOTINGS FOR THE REVISED LOADS.



a = 3'-0"

1 S001 C&C ZONE ROOF DIAGRAM
3/32" = 1'-0"

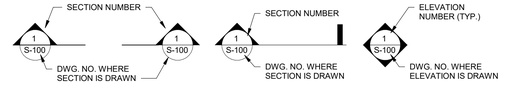
ABBREVIATIONS

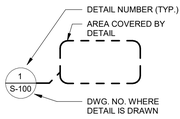
AFF	ABOVE FINISHED FLOOR
AISC	AMERICAN INSTITUTE OF STEEL CONSTRUCTION
APPROX.	APPROXIMATE
ARCH.	ARCHITECTURAL
@	AT
AHU	AIR HANDLING UNIT
AWS	AMERICAN WELDING SOCIETY
BC	BOTTOM CHORD
bf	FLANGE WIDTH
BFF	BELOW FINISHED FLOOR
BOT. OR B	BOTTOM OR BOTTOM MOST
BRG	BEARING
C&C	COMPONENT AND CLADDING
CJ OR CONST. JT.	CONSTRUCTION JOINT
CL	CENTERLINE
CFSS	COLD FORMED STEEL STUD
C.G.	CENTER OF GRAVITY
CLR.	CLEAR
CMU	CONCRETE MASONRY UNIT
COL.	COLUMN
COMP.	COMPOSITE
CONC.	CONCRETE
CONT.	CONTINUOUS
CRSI	CONCRETE REINFORCING STEEL INSTITUTE
D	DEPTH
DIA.	DIAMETER
db	BAR DIAMETER
DCJ	DOWELED CONTROL JOINT
DL.	DEAD LOAD
DO	DITTO
DWG.	DRAWING
EL.	ELEVATION
E.O.S.	EDGE OF SLAB
EQ.	EQUALLY
E.S.	EACH SIDE
E.W.	EACH WAY
EXP.	EXPANSION
FD	FLOOR DRAIN
FIN.	FINISHED, FINISH
F.F.	FINISHED FLOOR
GALV.	GALVANIZED
GR	GALVANIZED STEEL GRATING
HSS	HOLLOW STEEL SECTION
I_P	POSITIVE MOMENT OF INERTIA
I_N	NEGATIVE MOMENT OF INERTIA
JST	JOIST
JT.	JOINT
K	KIP (KILO POUND)
KSF	KIPS PER SQUARE FOOT
LL	LIVE LOAD
LLH	LONG LEG HORIZONTAL
LLV	LONG LEG VERTICAL
MAX.	MAXIMUM
MECH.	MECHANICAL
MFR.	MANUFACTURER
MIN.	MINIMUM
NO. OR #	NUMBER
O.C.	ON CENTER
OF	OUTSIDE FACE
PL	PLATE
PSF	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
RP	RADIUS POINT
REQ'D	REQUIRED
S_P	POSITIVE SECTION MODULUS
S_N	NEGATIVE SECTION MODULUS
SCH	SCHEDULE
SJ	SAW JOINT
STIFF.	STIFFENER
T	TRUSS
t	THICKNESS
TC	TOP CHORD
T.O.C.	TOP OF COLUMN
T.O.S.	TOP OF STEEL
TYP.	TYPICAL
U.N.O.	UNLESS OTHERWISE NOTED
W/	WITH
WWF	WELDED WIRE FABRIC
W.P.	WORKING POINT

GENERAL NOTE:

CONTRACTOR MUST FIELD VERIFY SITE CONDITIONS AND DIMENSIONS PRIOR TO BEGINNING WORK.

SYMBOLS:





CALLOUT HEAD

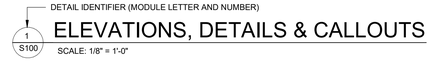
PLAN SYMBOLS:



KEYNOTE TAG

NORTH ARROWS

VIEW TITLES:



FLOOR PLANS, ELEVATIONS, SECTIONS AND DETAIL VIEWS TITLE

STEPHEN H. TOLLISON
REGISTERED PROFESSIONAL
ENGINEER
#10338
8/31/26
STATE OF MISSISSIPPI

11 0 11 21 32
SCALE: 3/32" = 1'-0"

Johnson ⁜M Adams PA
ARCHITECTS/ENGINEERS — PLANNING — SURVEYING — MANAGEMENT
P.O. BOX 513, GREENWOOD, MISSISSIPPI — (662) 455-4943

NO.	DATE	BY	DESCRIPTION

REVISIONS

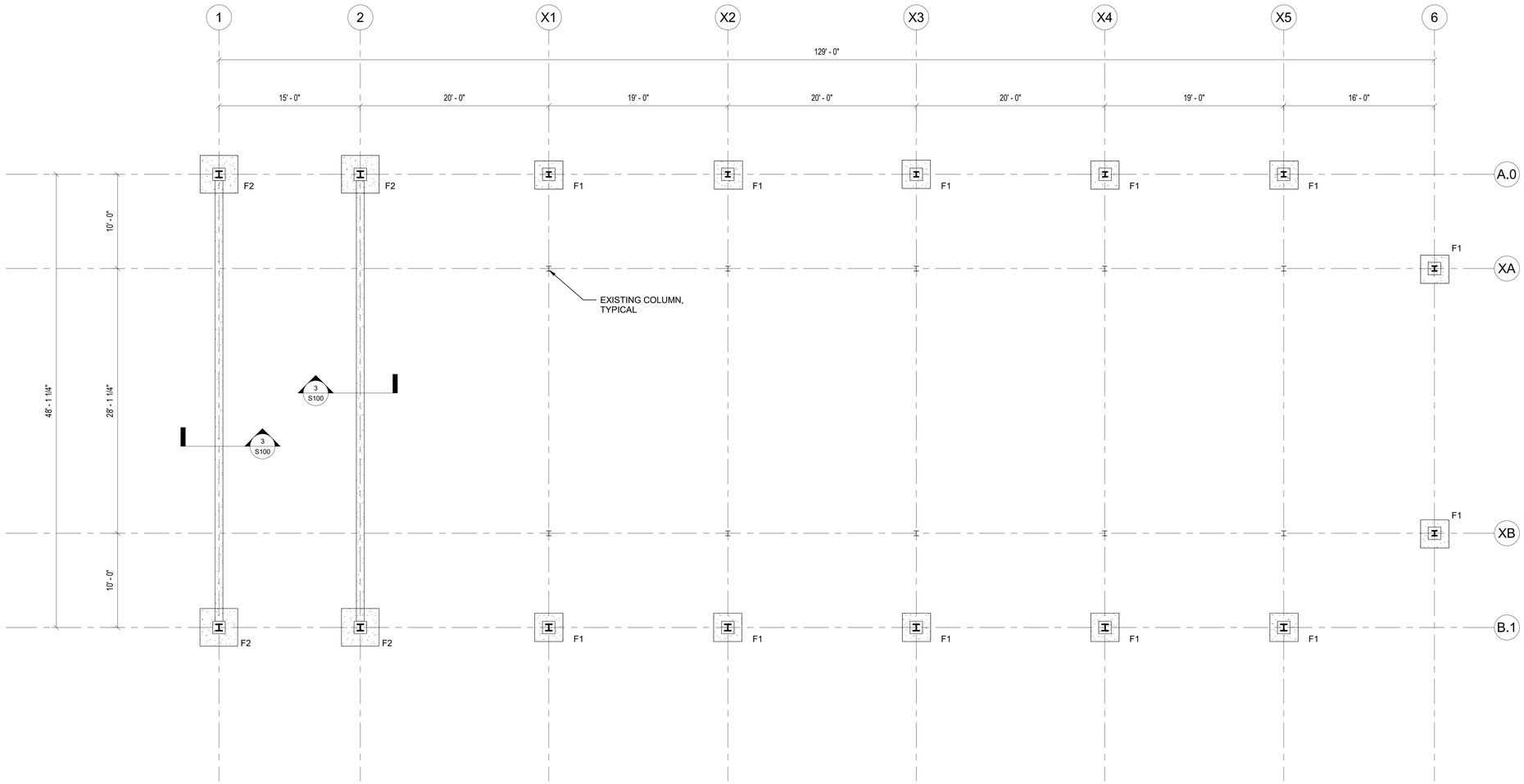
A-E FIRM	USDA
PROJECT MANAGER: REH	EPM: B. DEAN
DESIGNER: THT	PPM: xxxxx
CHECKED BY: DRL	SAFETY & HEALTH: D. CORDRAY
JOB #: 4125	REAL PROPERTY: xxxxx

USDA Agricultural Research Service

USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

STRUCTURAL ABBR. AND LEGEND

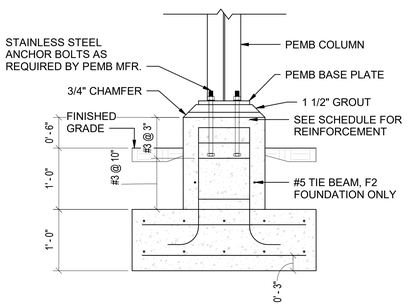
PROJECT NO. xxxxx	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD 20705	SHEET S001
DATE 08/31/2026		2 OF 32



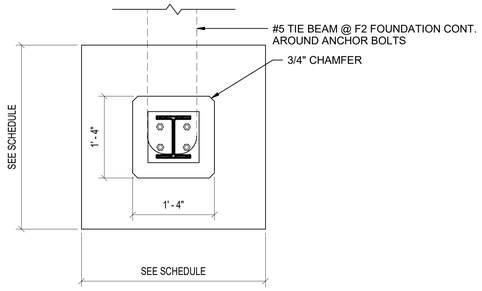
1 FOUNDATION PLAN
S100 — 3/16" = 1'-0"

PLAN NORTH

NOTE: PROVIDE COATING ON ALL EXPOSED STEEL AS SPECIFIED.

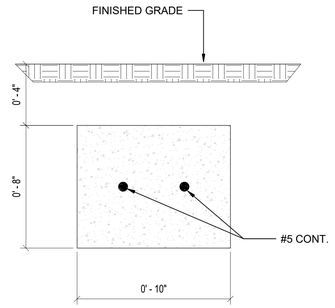


ELEVATION



PLAN

2 FOUNDATION DETAILS
S100 — 1" = 1'-0"



3 CONCRETE BEAM
S100 — 3" = 1'-0"

FOOTING SCHEDULE

	FOOTING			PEDESTAL			
MARK	DIMENSIONS L X W X T	TOP REINFORCING	BOTTOM REINFORCING	DIMENSIONS (SEE NOTE 1) L X W X T	VERT. REINFORCING	STIRRUPS	ANCHOR BOLTS (SEE NOTE 2)
F1	36" X 36" X 12"	(5) #5	(5) #5	16" X 16" X 18"	(4) #5	AS SHOWN	(4) 3/4"
F2	48" X 48" X 12"	(7) #5	(7) #5	16" X 16" X 18"	(4) #5	AS SHOWN	(4) 3/4"

NOTES:
1. DIMENSIONS SHOWN FOR PEDESTALS ARE MINIMUM. MINIMUM CLEARANCE OF 1" REQUIRED BETWEEN ANCHOR BOLTS AND REINFORCING. MINIMUM COVER OF 2" REQUIRED FOR REINFORCING. PEDESTAL DIMENSIONS MUST BE ADJUSTED ACCORDINGLY IF PEMB BASE PLATE DOES NOT ALLOW THESE CLEARANCES.
2. ANCHOR BOLTS TO BE SUPPLIED BY PEMB SUPPLIER.

THOMAS H. TOLLEY — REGISTERED PROFESSIONAL ENGINEER — #10338 — 08/31/26 — STATE OF MISSISSIPPI

SCALE: 3" = 1'-0"

SCALE: 3/16" = 1'-0"

SCALE: 1" = 1'-0"

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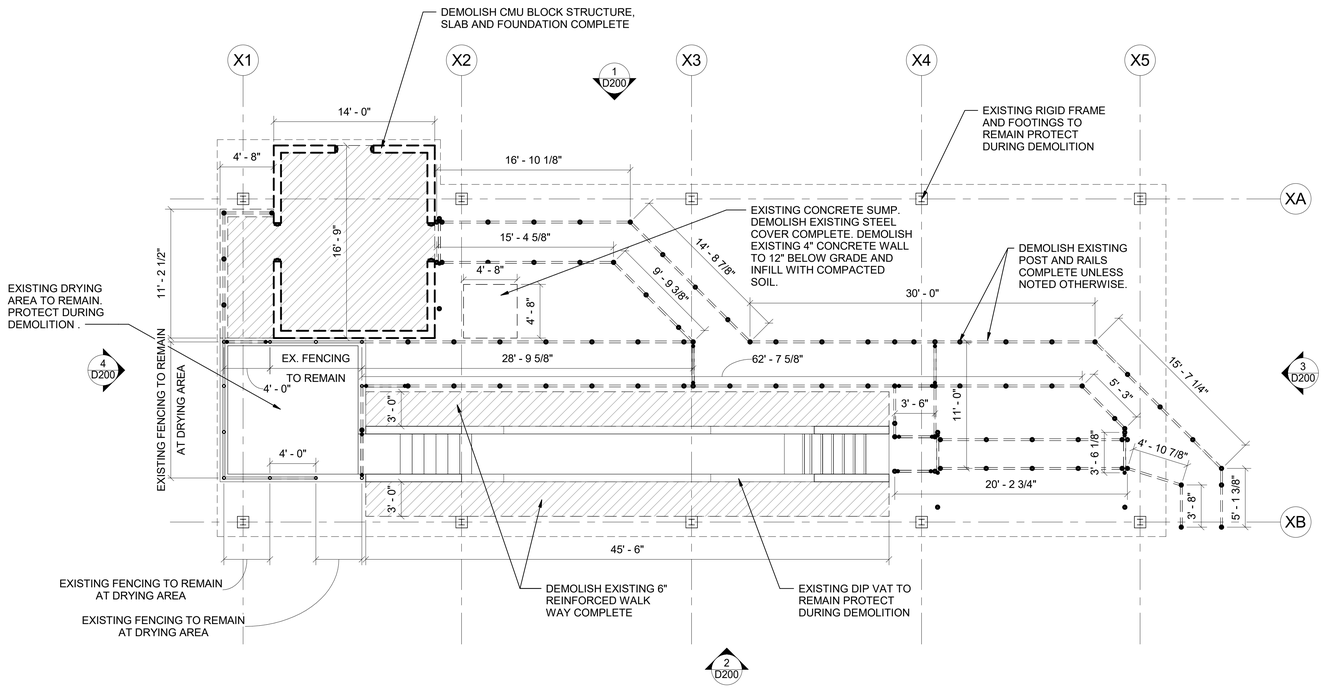
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DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

FOUNDATION PLAN

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	THT
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026

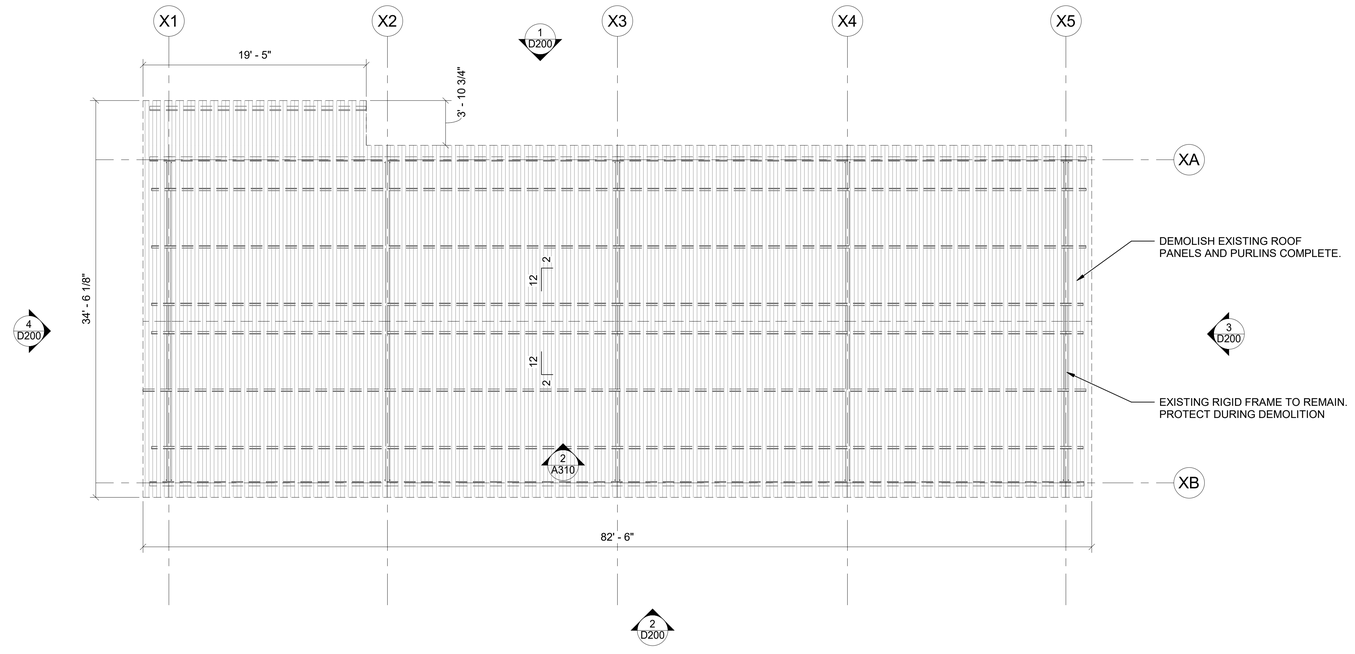
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5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD 20705

SHEET **S100**
3 OF 32



1 D100 **DEMOLITION PLAN**
3/16" = 1'-0"

PLAN NORTH



2 D100 **DEMOLITION ROOF PLAN**
3/16" = 1'-0"

PLAN NORTH

2 0 2 4 6
SCALE: 3/8" = 1'-0"

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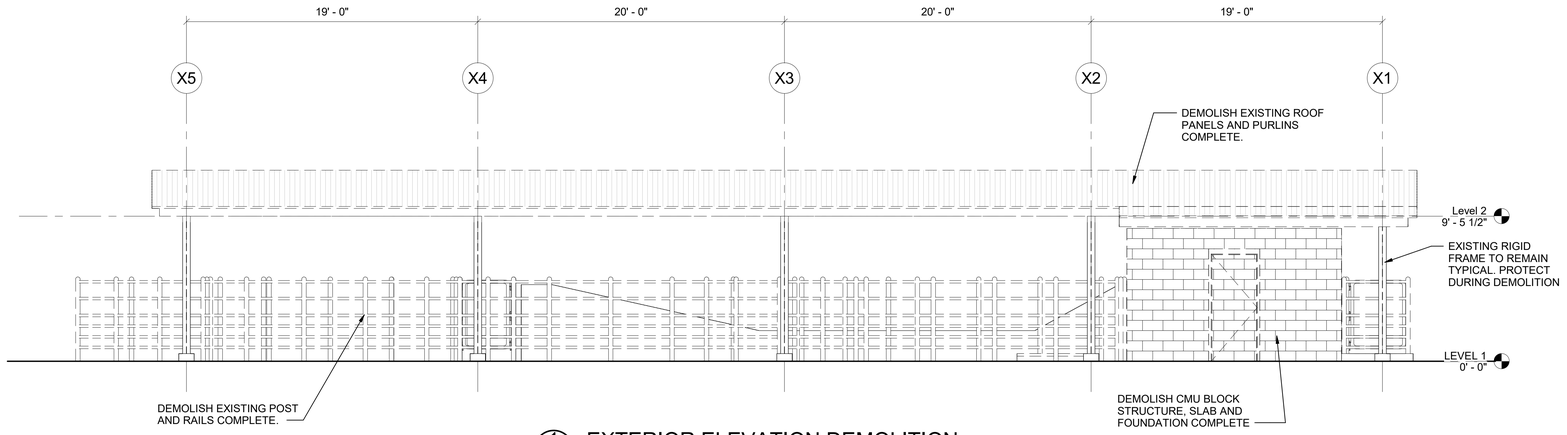
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DEMOLITION PLANS

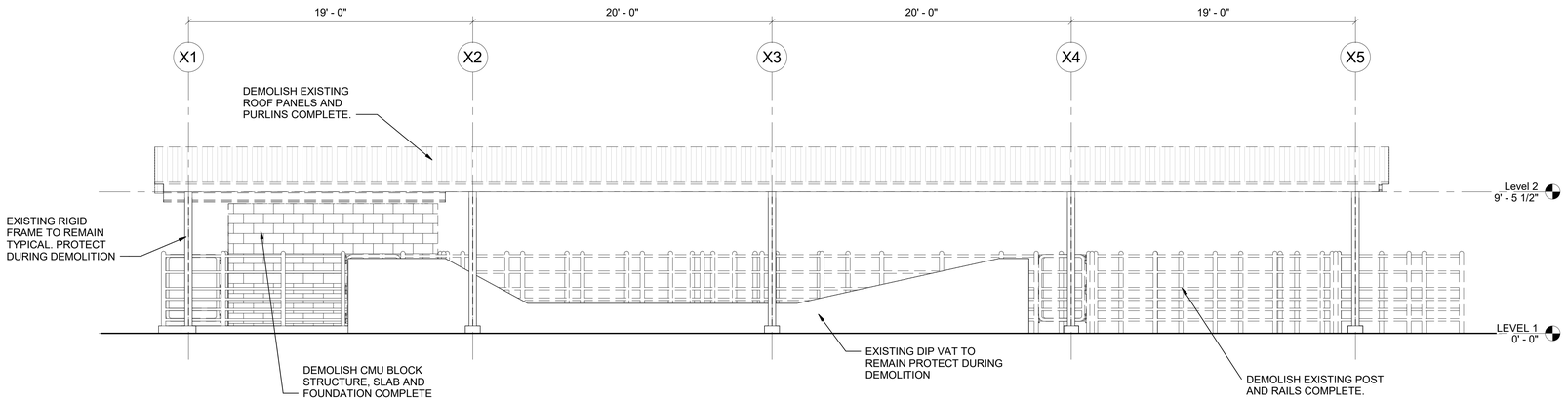
A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026

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BELTSVILLE MD 20705

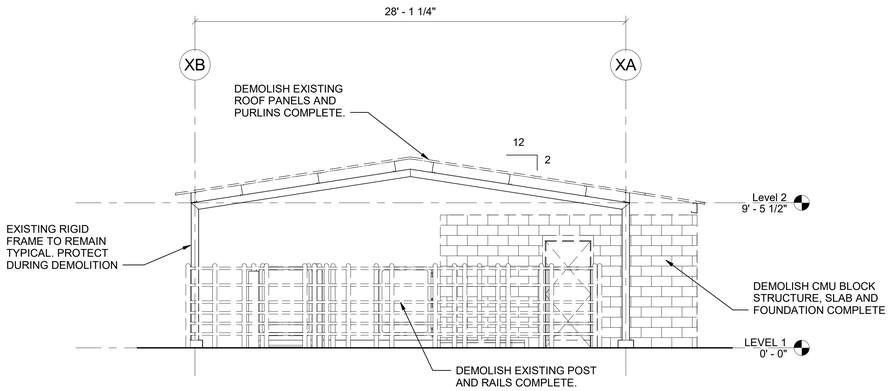
SHEET **D100**



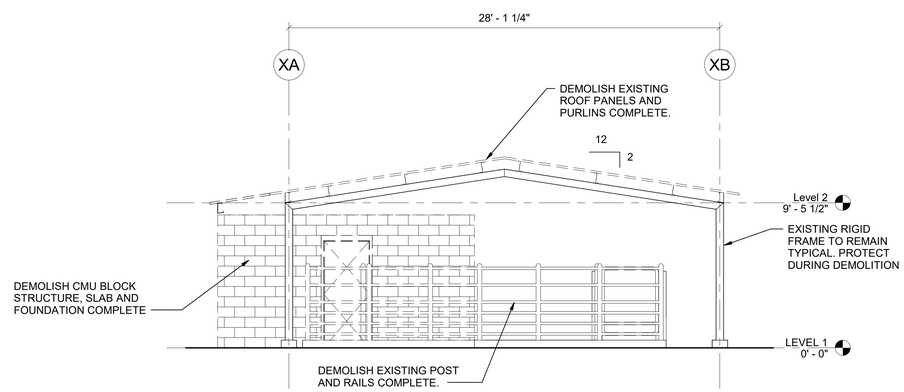
1 D200 EXTERIOR ELEVATION DEMOLITION
1/4" = 1'-0"



2 D200 EXTERIOR ELEVATION DEMOLITION
1/4" = 1'-0"



3 D200 EXTERIOR ELEVATION DEMOLITION
1/4" = 1'-0"



4 D200 EXTERIOR ELEVATION DEMOLITION
1/4" = 1'-0"

4 0 4 8
SCALE: 1/4" = 1'-0"

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			REVISIONS

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PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx

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EXTERIOR ELEVATIONS DEMOLITION

PROJECT NO.		SHEET
xxxxx	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD 20705	D200
DATE		5 OF 32
08/31/2026		

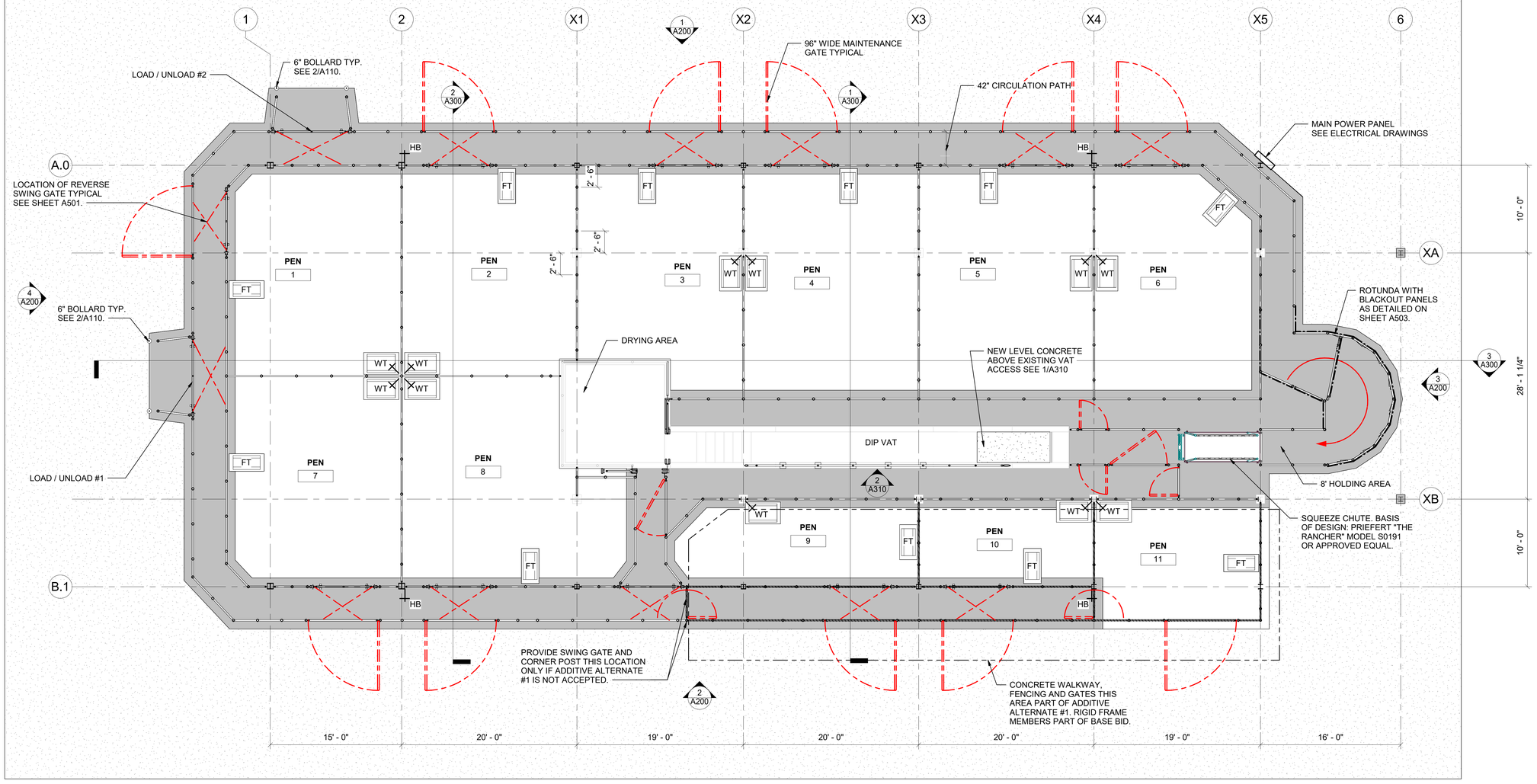
REGISTERED ARCHITECT — JOHN R. HOLT — STATE OF TEXAS

LEGEND

FT	FEED TROUGH. NOT IN CONTRACT.
WT	WATER TROUGH NOT IN CONTRACT.
	WATER TROUGH FILL VALVE DROP DOWN FROM OVERHEAD
HB	WASH DOWN HOSE BIB DROP DOWN FROM OVERHEAD TO 6' ABOVE WALK SURFACE.
X1	EXISTING COLUMN LINE
1	NEW COLUMN LINE COLUMN LINE
	BLACK OUT PANELS

PEN SCHEDULE

NUMBER	NAME	AREA
1	PEN	478 SF
2	PEN	479 SF
3	PEN	464 SF
4	PEN	533 SF
5	PEN	533 SF
6	PEN	488 SF
7	PEN	477 SF
8	PEN	528 SF
9	PEN	277 SF
10	PEN	200 SF
11	PEN	263 SF



1 A100 OVERALL PLAN
3/16" = 1'-0"

PLAN NORTH

4 0 4 8 12
SCALE: 3/16" = 1'-0"

REGISTERED ARCHITECT JOHN S. HOOD 24,172 STATE OF TEXAS

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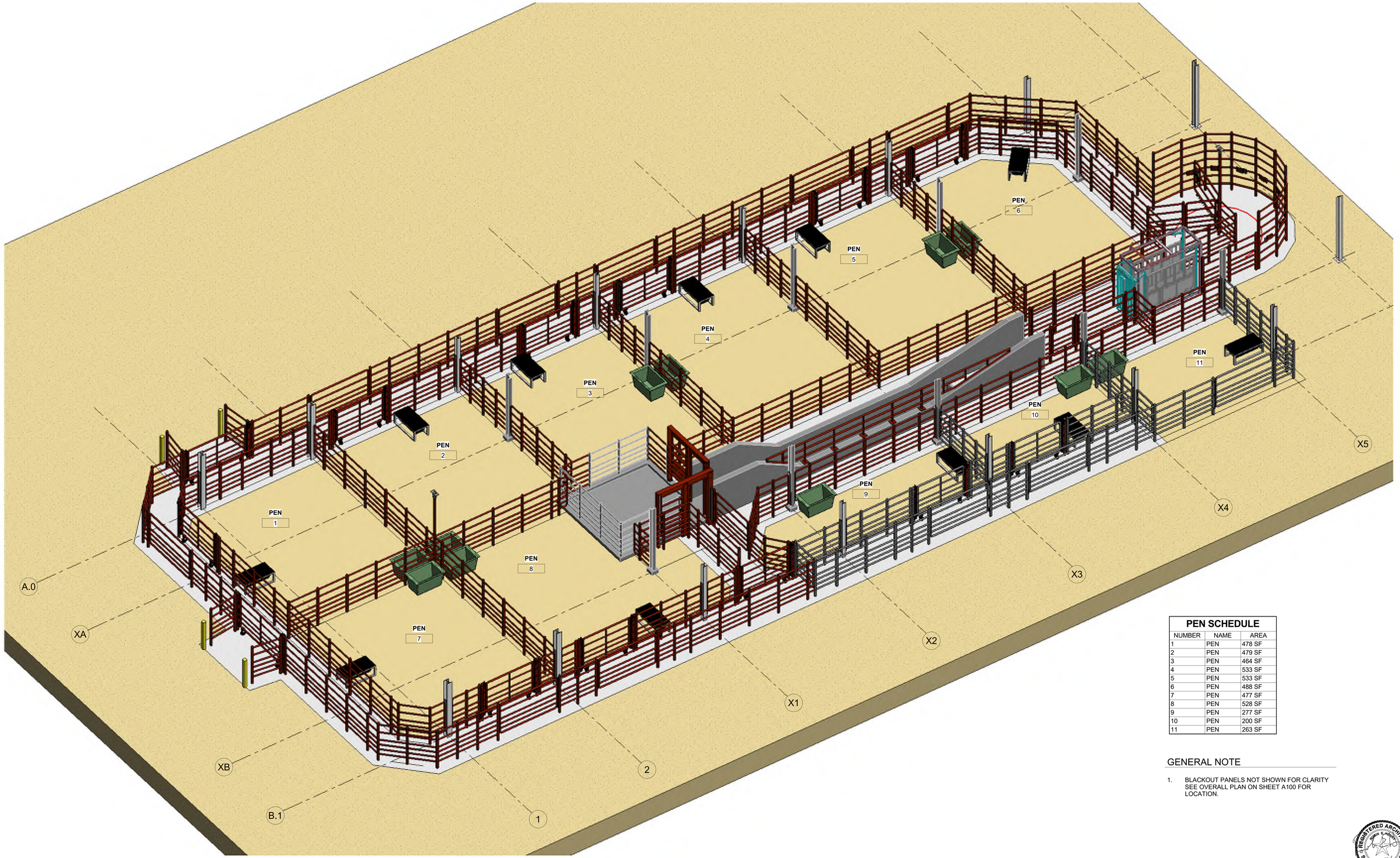
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MOORE FIELD EDINBURG, TEXAS

OVERALL PLAN

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026

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SHEET **A100**
6 OF 32



PEN SCHEDULE

NUMBER	NAME	AREA
1	PEN	478 SF
2	PEN	479 SF
3	PEN	464 SF
4	PEN	533 SF
5	PEN	533 SF
6	PEN	488 SF
7	PEN	477 SF
8	PEN	528 SF
9	PEN	277 SF
10	PEN	200 SF
11	PEN	263 SF

GENERAL NOTE

1. BLACKOUT PANELS NOT SHOWN FOR CLARITY SEE OVERALL PLAN ON SHEET A100 FOR LOCATION.

1 / A101 OVERALL 3D VIEW

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			REVISIONS

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx

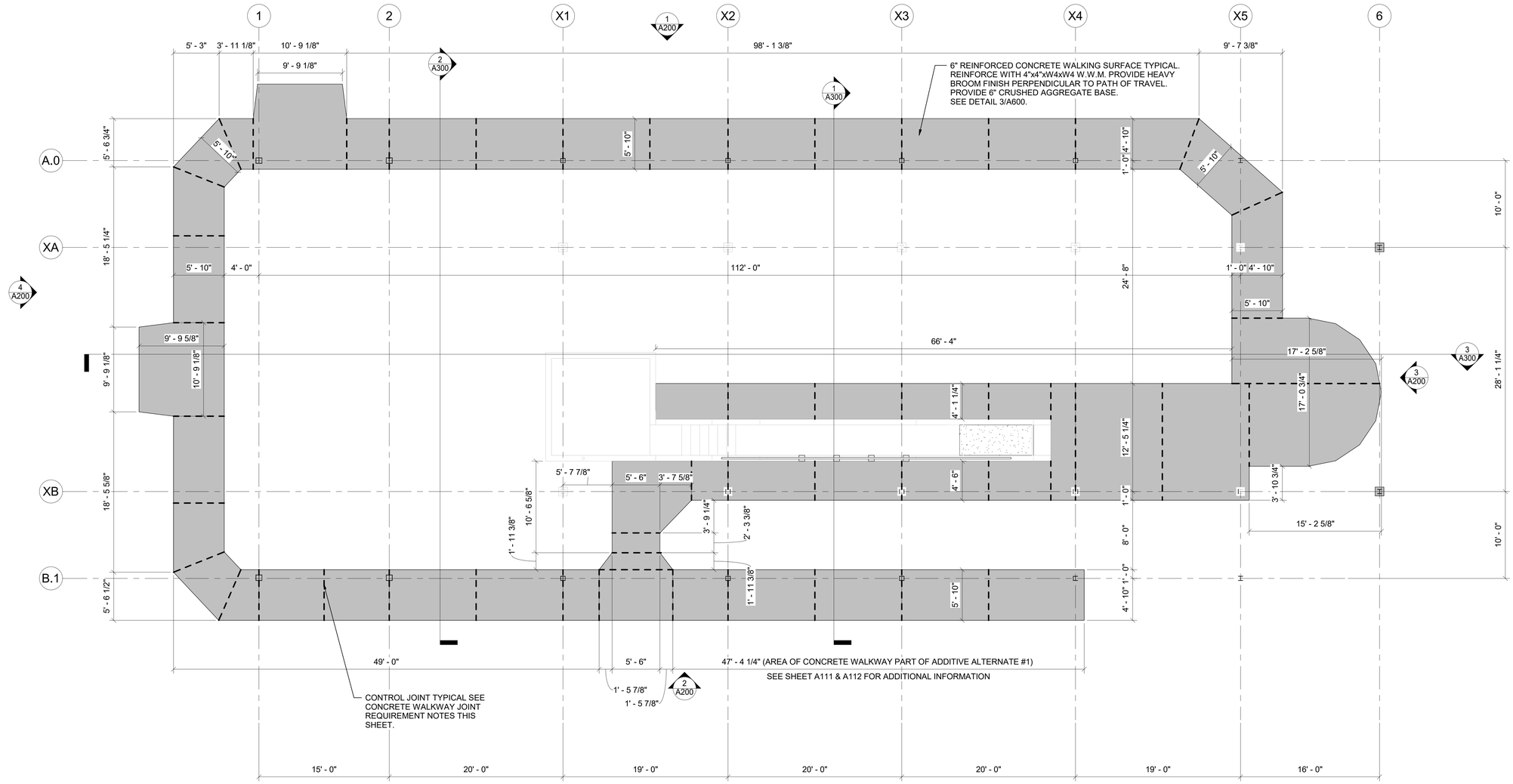
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OVERALL 3D VIEW

PROJECT NO.	DATE	SHEET
xxxxx	08/31/2026	A101
		7 OF 32

AGRICULTURAL RESEARCH SERVICE
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1 A102 WALKWAY LAYOUT PLAN

3/16" = 1'-0"

CONCRETE WALKWAY JOINT REQUIREMENTS:

PROVIDE CONTRACTION JOINTS IN CONCRETE WALKS AT 10'-0" O.C. MAXIMUM AND AT LOCATIONS SHOWN. SPACE JOINTS TO FORM APPROXIMATELY SQUARE PANELS WITH A MAXIMUM PANEL ASPECT RATIO OF 1.5:1. SAWCUT JOINTS TO A DEPTH OF NOT LESS THAN ONE-FOURTH OF THE SLAB THICKNESS. PROVIDE FULL-DEPTH ISOLATION JOINTS WHERE WALKS ABUT BUILDINGS, EXISTING CONCRETE, FOUNDATIONS, COLUMNS, EQUIPMENT PADS, AND OTHER FIXED CONSTRUCTION. CONTINUE REINFORCING THROUGH CONTRACTION JOINTS UNLESS THE STRUCTURAL DESIGN REQUIRES OTHERWISE; DISCONTINUE REINFORCEMENT AT ISOLATION JOINTS.

4 0 4 8 12

SCALE: 3/16" = 1'-0"

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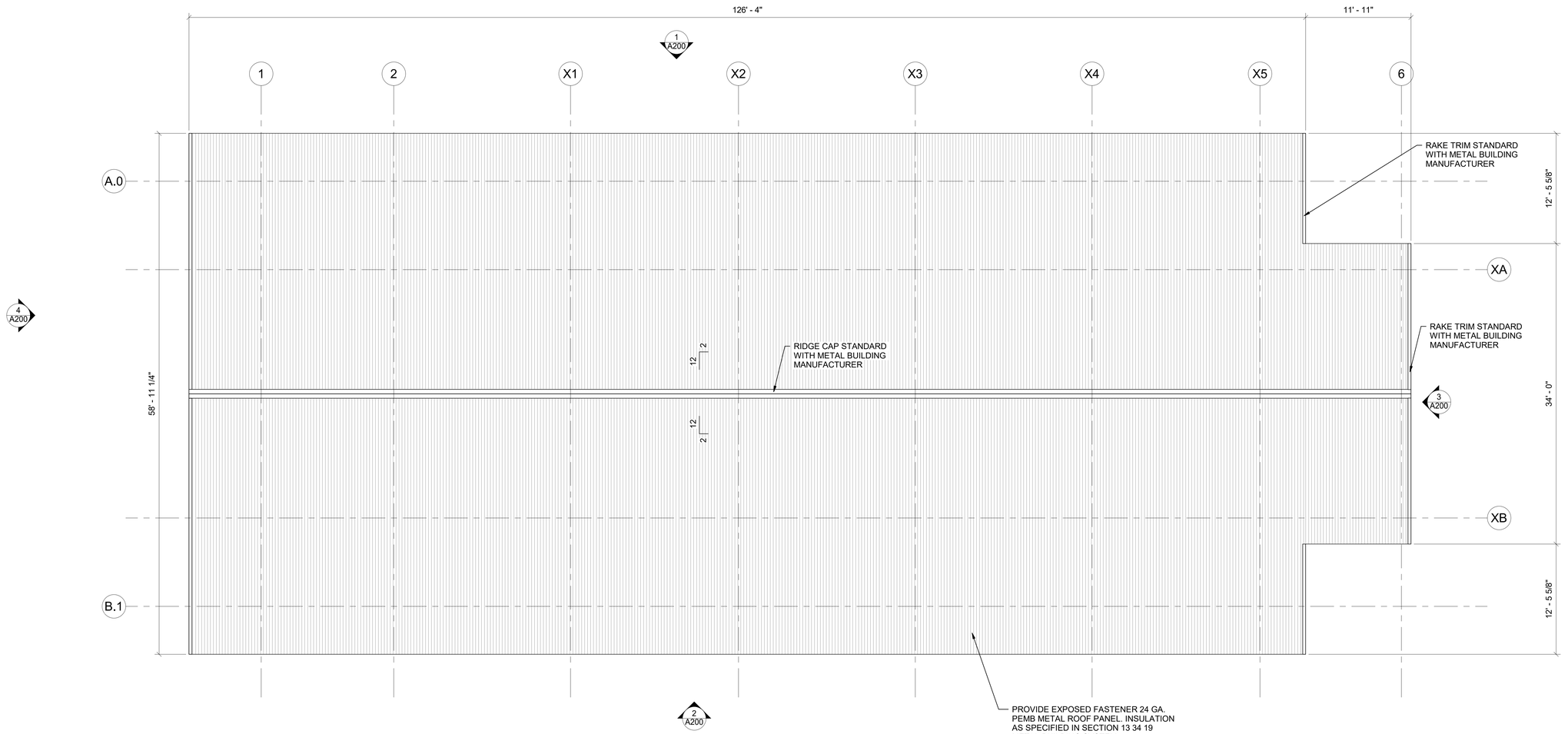
A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	xxxxx
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	xxxxx

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WALKWAY LAYOUT PLAN

PROJECT NO. xxxxx	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD 20705	SHEET A102
DATE 08/31/2026		8 OF 32



1 A103 ROOF PLAN
3/16" = 1'-0"

PLAN NORTH

4 0 4 8 12
SCALE: 3/16" = 1'-0"

REGISTERED ARCHITECT JOHN R. HARRIS 21 STATE OF TEXAS

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MOORE FIELD EDINBURG, TEXAS

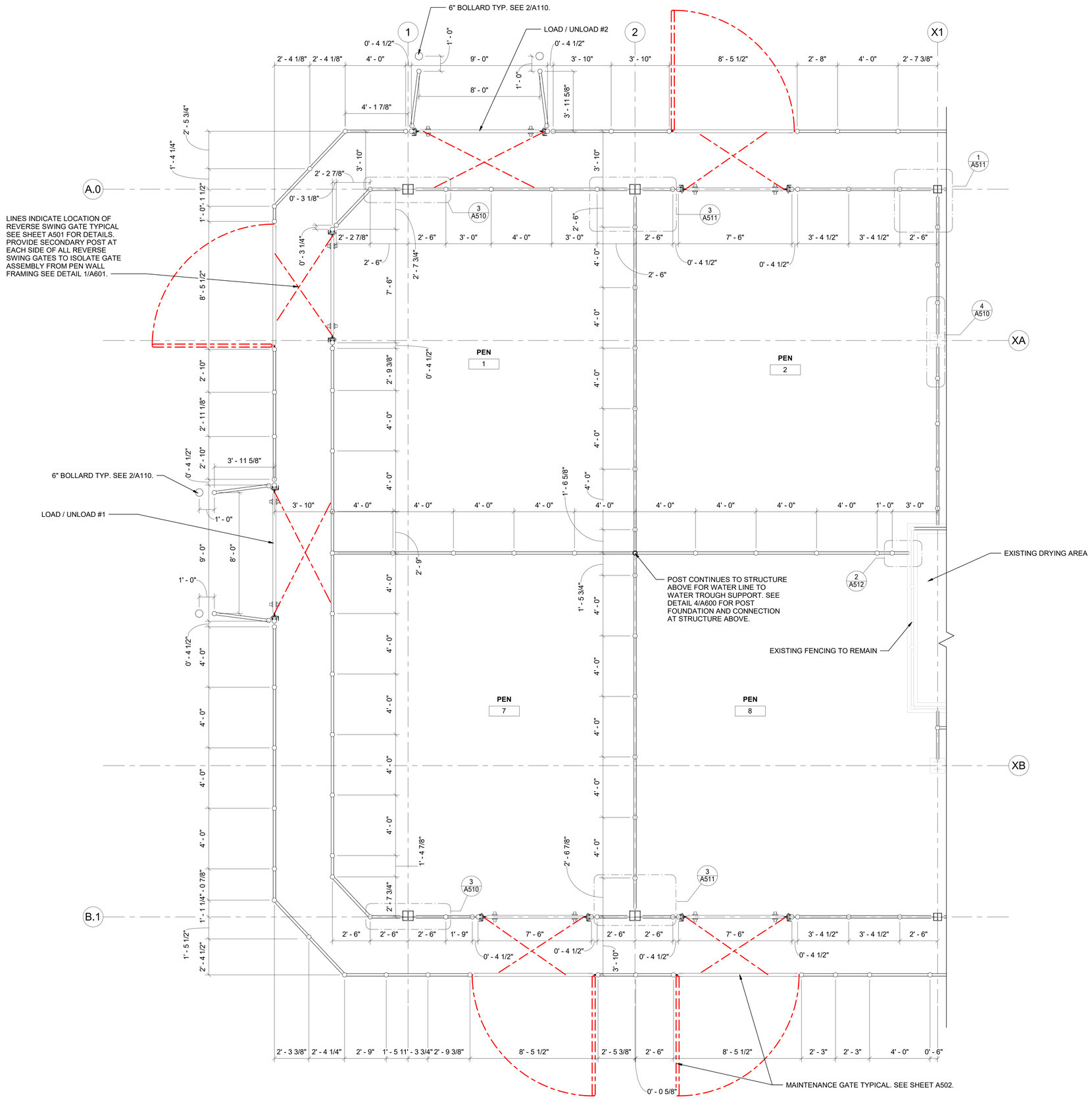
ROOF PLAN

A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	xxxxx
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	xxxxx

PROJECT NO.	xxxxx
DATE	08/31/2026

AGRICULTURAL RESEARCH SERVICE
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SHEET
A103
9 OF 32



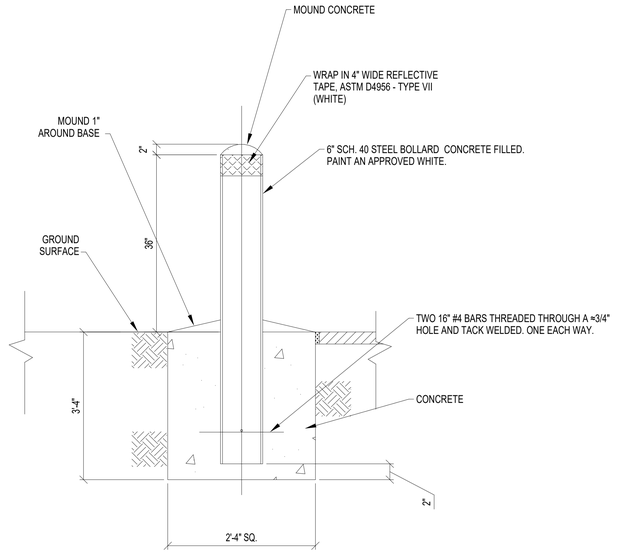
1 A110 ENLARGED PEN LAYOUT PLAN AREA A
3/8" = 1'-0"

PLAN NORTH

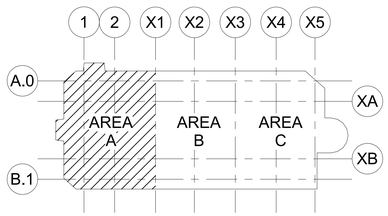
PEN SCHEDULE AREA A		
NUMBER	NAME	AREA
1	PEN	478 SF
2	PEN	479 SF
7	PEN	477 SF
8	PEN	528 SF

GENERAL NOTES

1. SEE DETAIL 1/A600 FOR POST AND RAIL DETAIL AT NATURAL GRADE.
2. SEE DETAIL 2/A600 FOR POST AND RAIL DETAIL AT WALKWAY SURFACE.



2 A110 BOLLARD DETAIL
1 1/2" = 1'-0"



KEY PLAN
N.T.S

SCALE: 1 1/2" = 1'-0"

SCALE: 3/8" = 1'-0"

NO.	DATE	BY	DESCRIPTION
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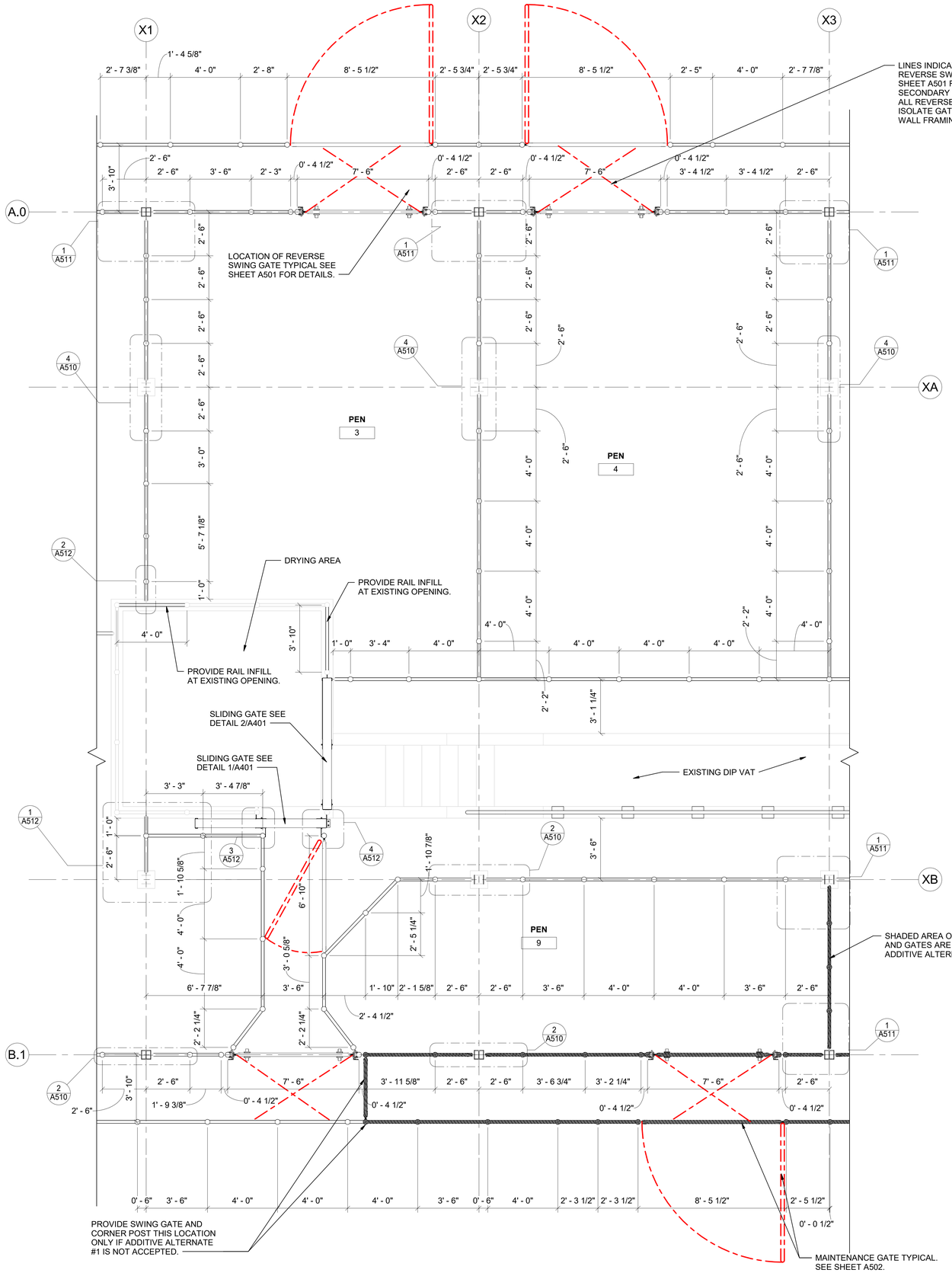
A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx

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ENLARGED PEN LAYOUT PLAN AREA A

PROJECT NO.	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD 20705	SHEET
xxxxx		A110
DATE		10 OF 32
08/31/2026		



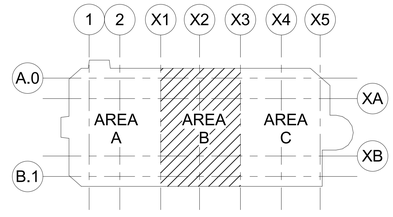
1 A111 ENLARGED PEN LAYOUT PLAN AREA B
3/8" = 1'-0"

PLAN NORTH

PEN SCHEDULE AREA B		
NUMBER	NAME	AREA
3	PEN	464 SF
4	PEN	533 SF
9	PEN	277 SF

GENERAL NOTES

1. SEE DETAIL 1/A600 FOR POST AND RAIL DETAIL AT NATURAL GRADE.
2. SEE DETAIL 2/A600 FOR POST AND RAIL DETAIL AT WALKWAY SURFACE.



KEY PLAN
N.T.S

2 0 2 4 6
SCALE: 3/8" = 1'-0"

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A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	xxxxx
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	xxxxx

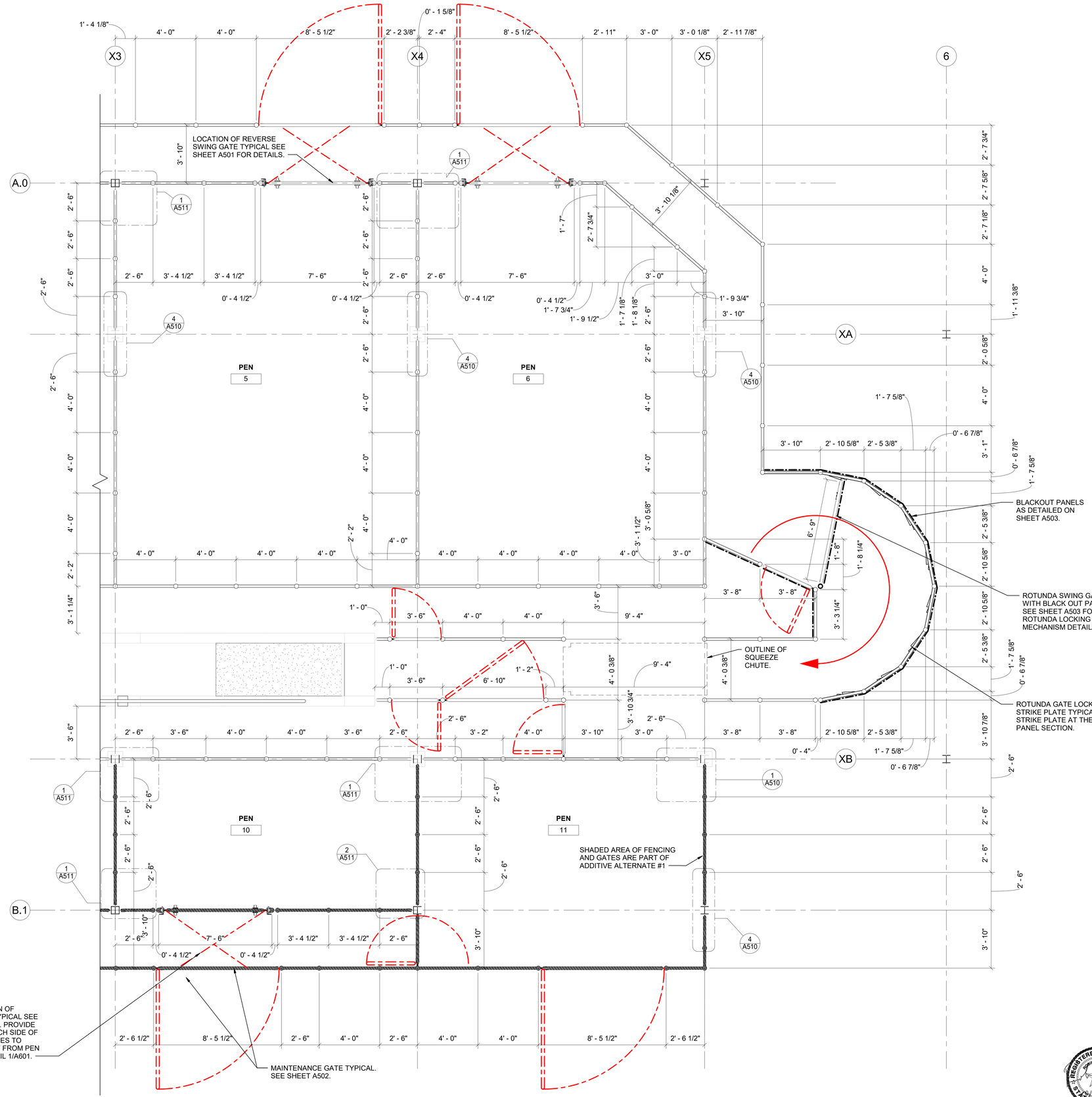
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DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

ENLARGED PEN LAYOUT PLAN AREA B

PROJECT NO.	SHEET
xxxxx	A111
DATE	11 OF 32
08/31/2026	

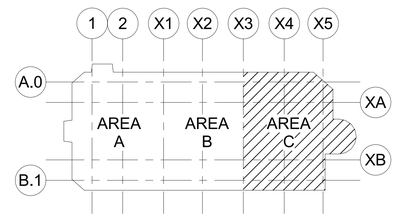
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BELTSVILLE MD 20705



PEN SCHEDULE ARE...		
NUMBER	NAME	AREA
5	PEN	533 SF
6	PEN	488 SF
10	PEN	200 SF
11	PEN	263 SF

GENERAL NOTES

1. SEE DETAIL 1/A600 FOR POST AND RAIL DETAIL AT NATURAL GRADE.
2. SEE DETAIL 2/A600 FOR POST AND RAIL DETAIL AT WALKWAY SURFACE.



KEY PLAN
N.T.S

1 / A112 ENLARGED PEN LAYOUT PLAN AREA C
3/8" = 1'-0"

PLAN NORTH

2 0 2 4 6
SCALE: 3/8" = 1'-0"

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REGISTERED ARCHITECT JOHN L. FOSTER STATE OF TEXAS

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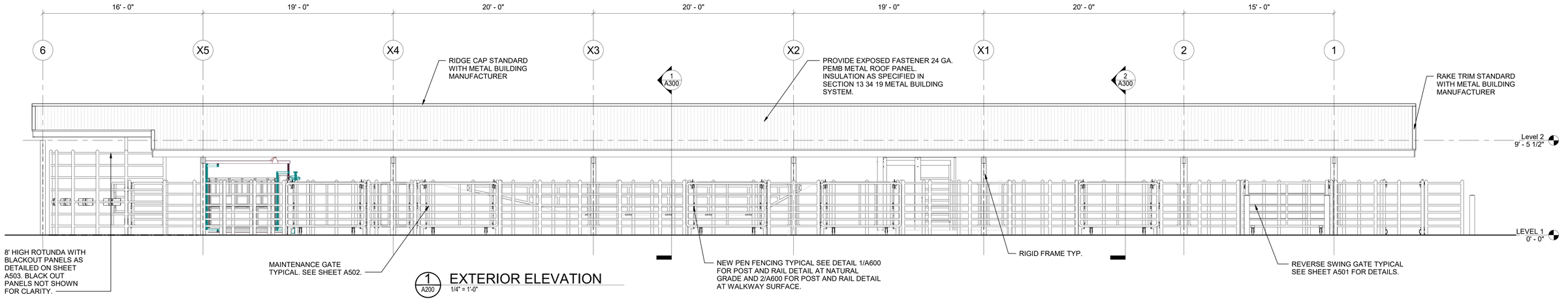
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MOORE FIELD EDINBURG, TEXAS

ENLARGED PEN LAYOUT PLAN AREA C

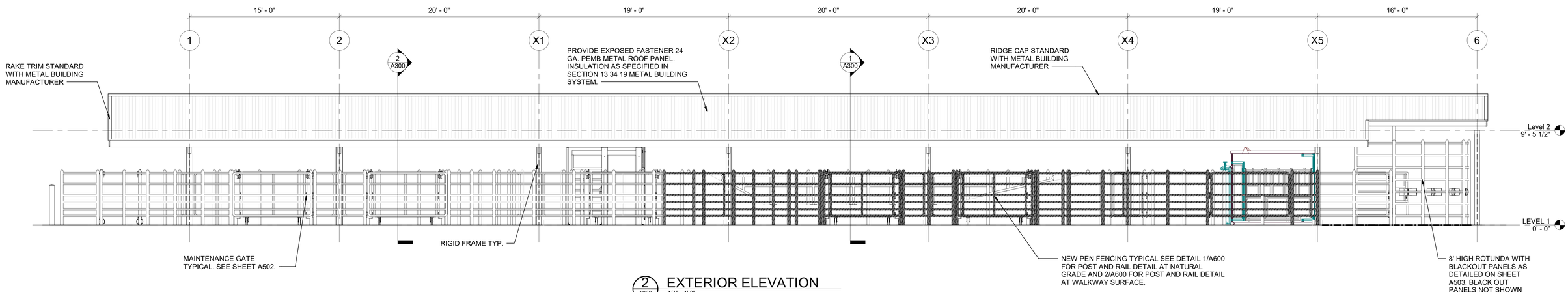
A-E FIRM	USDA		
PROJECT MANAGER	REH	EPM	B. DEAN
DESIGNER	JLF	PPM	xxxxx
CHECKED BY	DRL	SAFETY & HEALTH	D. CORDRAY
JOB #	4125	REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx		
DATE	08/31/2026		

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD 20705

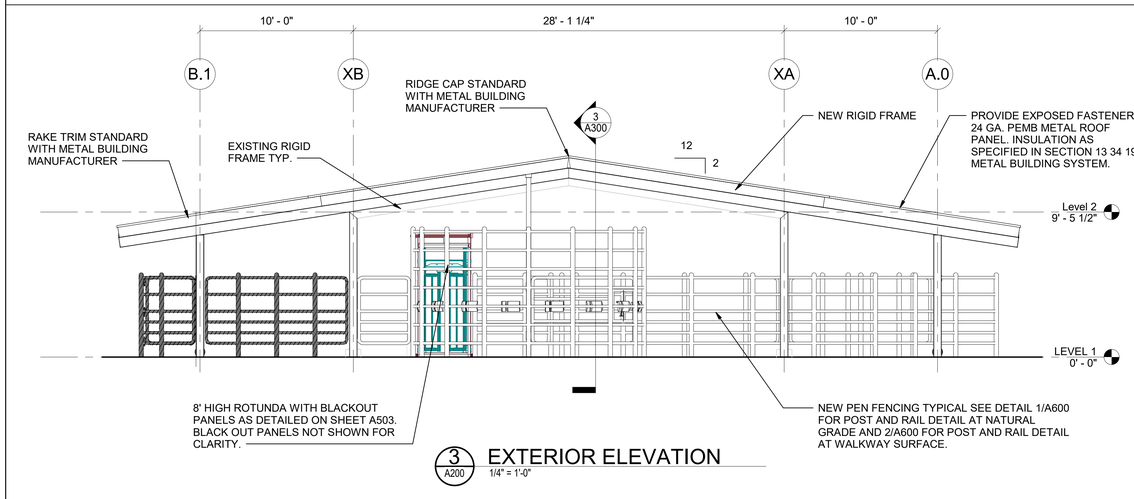
SHEET A112
12 OF 32



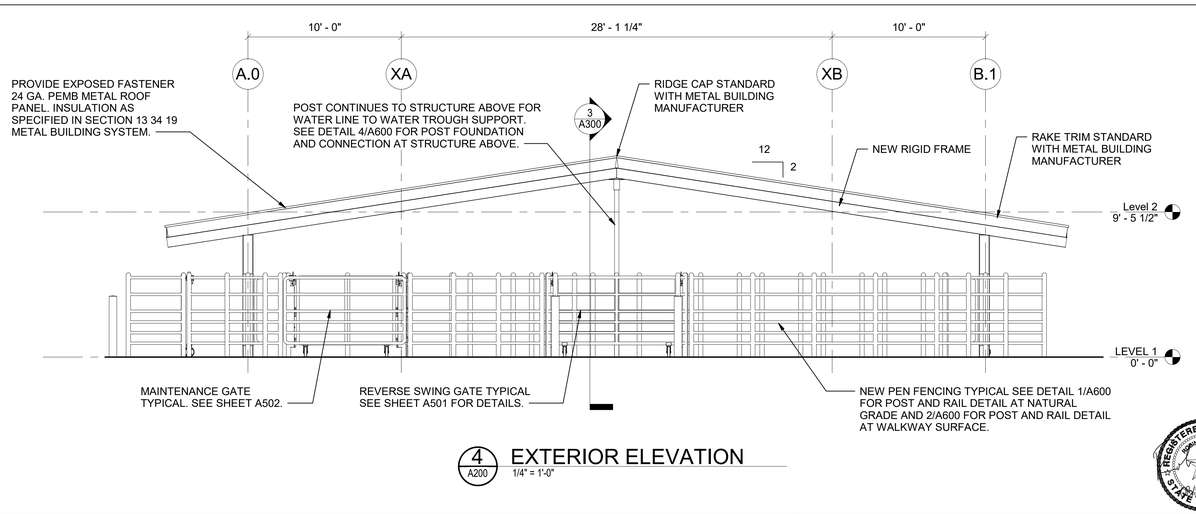
1 A200 EXTERIOR ELEVATION
1/4" = 1'-0"



2 A200 EXTERIOR ELEVATION
1/4" = 1'-0"



3 A200 EXTERIOR ELEVATION
1/4" = 1'-0"



4 A200 EXTERIOR ELEVATION
1/4" = 1'-0"

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NO.	DATE	BY	DESCRIPTION
			REVISIONS

4 0 4 8
SCALE: 1/4" = 1'-0"

A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	xxxxx
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	xxxxx

USDA Agricultural Research Service

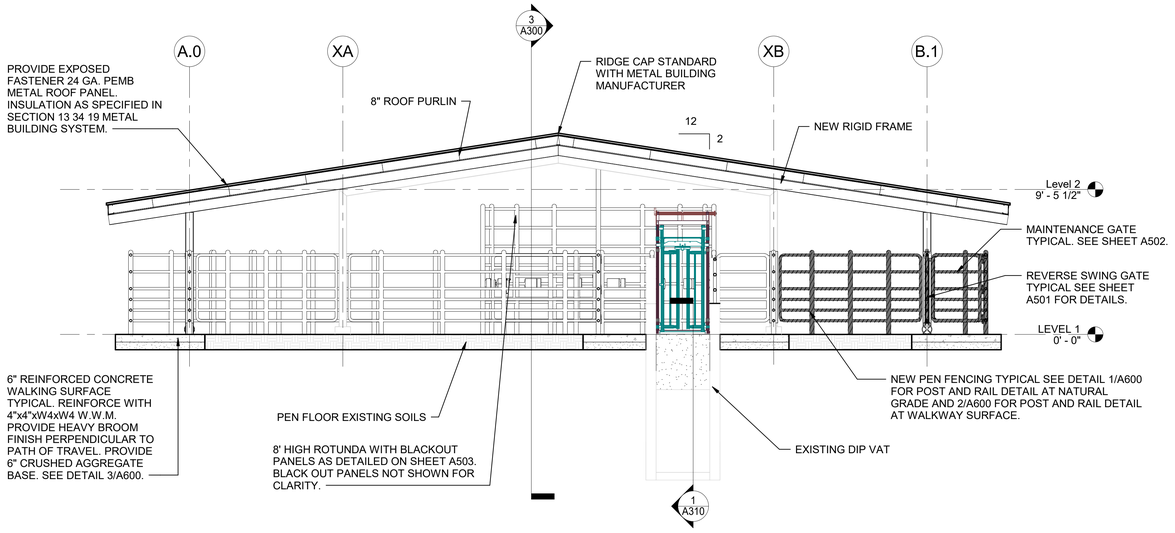
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

EXTERIOR ELEVATIONS

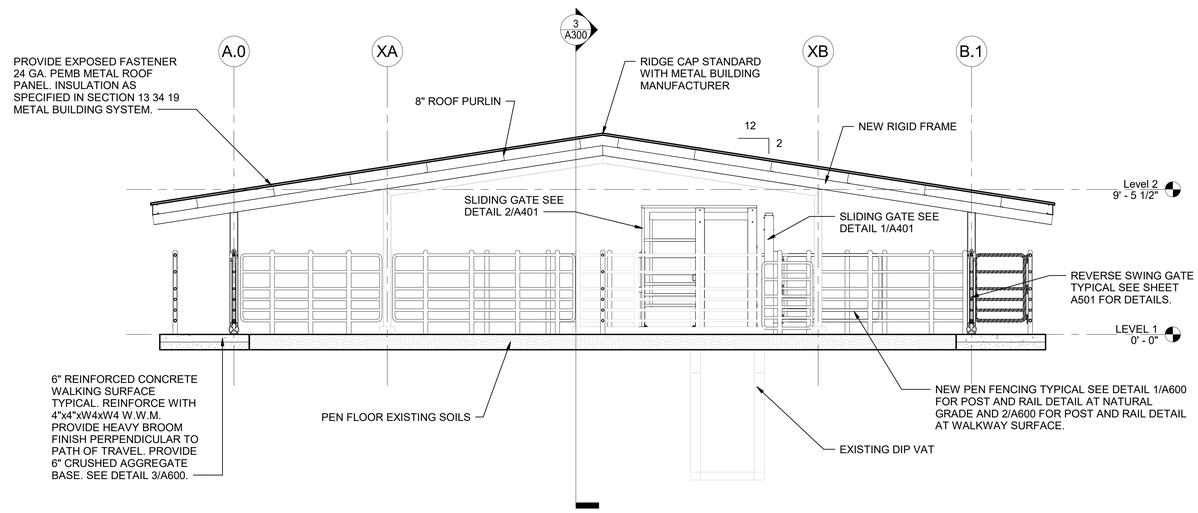
PROJECT NO.	SHEET
xxxxx	A200
DATE	13 OF 32
08/31/2026	

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD 20705

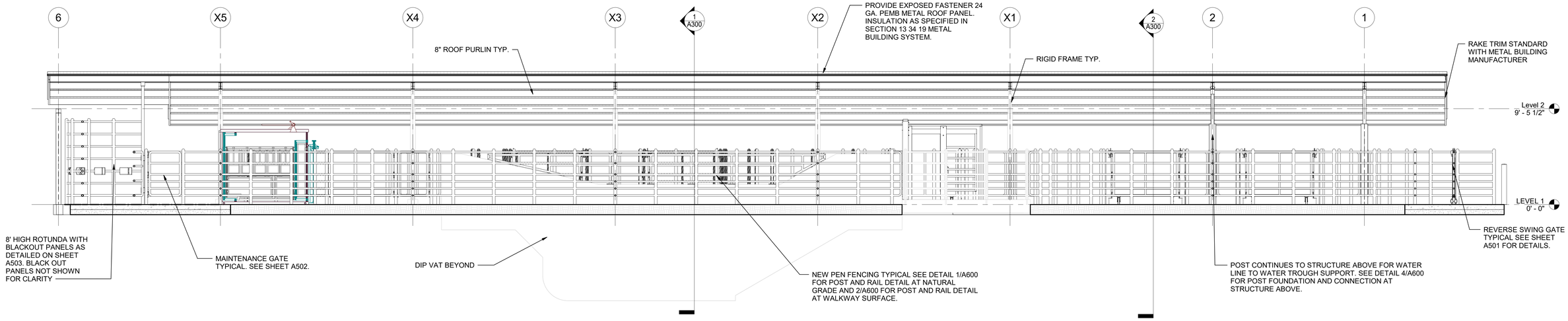
REGISTERED ARCHITECT — JOHN S. JOHNSON — STATE OF TEXAS



1 A300 BUILDING SECTION
1/4" = 1'-0"



2 A300 BUILDING SECTION
1/4" = 1'-0"



3 A300 BUILDING SECTION
1/4" = 1'-0"

4 0 4 8
SCALE: 1/4" = 1'-0"

REGISTERED ARCHITECT JOSEPH R. ANDERSON STATE OF TEXAS

Johnson Miller Adams PC
ARCHITECTS/ENGINEERS - PLANNING - SURVEYING WWW.JMAENT
P.O. BOX 512, SHEPHERD, MISSISSIPPI - (940) 459-4543

NO.	DATE	BY	DESCRIPTION
			REVISIONS

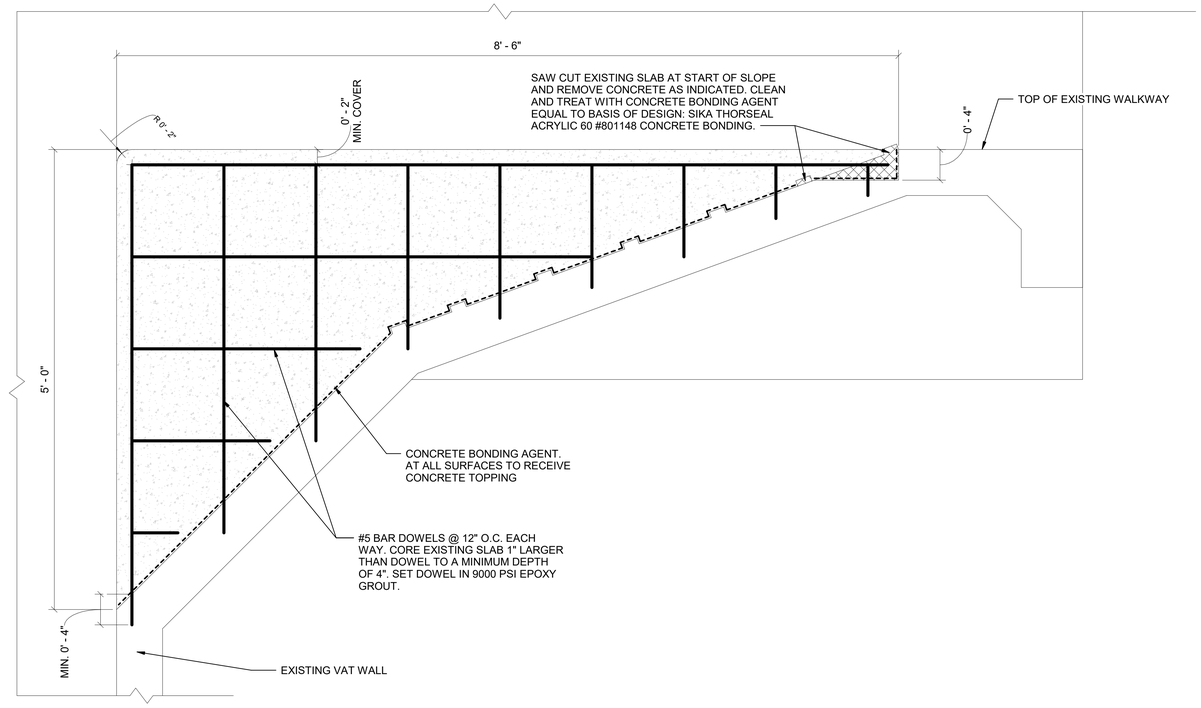
A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	xxxxx
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	xxxxx

USDA Agricultural Research Service

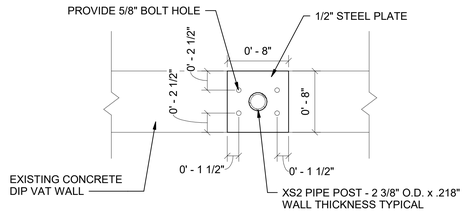
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

BUILDING SECTIONS

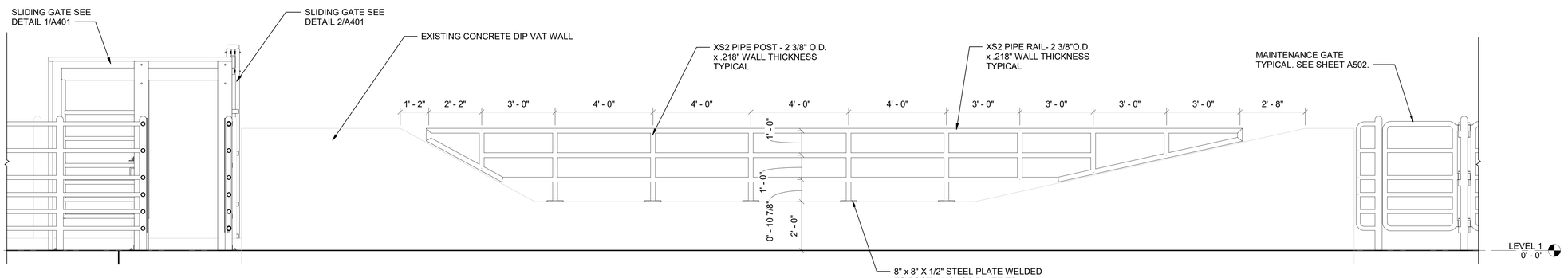
PROJECT NO. xxxxx	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD 20705	SHEET A300
DATE 08/31/2026		14 OF 32



1 A310 VAT ACCESS CAP DETAIL
1 1/2" = 1'-0"



3 A310 POST BASE PLATE DETAIL
1 1/2" = 1'-0"



2 A310 DIP VAT GUARD RAIL
1/2" = 1'-0"

WELDING NOTE – CATTLE RAILING AND FENCING

PROVIDE WELDED CONNECTIONS IN ACCORDANCE WITH AWS D1.1. WELDERS SHALL BE QUALIFIED FOR THE WELDING PROCESSES AND MATERIALS USED. UNLESS OTHERWISE INDICATED, PROVIDE CONTINUOUS 3/16-INCH FILLET WELDS AROUND THE FULL PERIMETER OF EACH RAIL-TO-POST, FRAME, BRACKET, AND PLATE CONNECTION. WELDS SHALL BE UNIFORM, SMOOTH, AND FREE OF CRACKS, POROSITY, UNDERCUT, SHARP EDGES, AND PROJECTIONS THAT COULD INJURE ANIMALS OR PERSONNEL. GRIND WELDS AND CUT EDGES SMOOTH WHERE EXPOSED TO ANIMAL CONTACT. REMOVE SLAG AND WELD SPATTER AND PREPARE WELDED AREAS FOR THE SPECIFIED PROTECTIVE COATING. PROVIDE VENT AND DRAIN HOLES IN CLOSED OR HOLLOW MEMBERS AS REQUIRED FOR HOT-DIP GALVANIZING (ADDITIVE ALTERNATE #2). REPAIR DAMAGED GALVANIZED COATINGS AFTER FIELD WELDING IN ACCORDANCE WITH ASTM A780. DO NOT FIELD-WELD OR MODIFY MEMBERS WITHOUT APPROVAL OF THE CONTRACTING OFFICERS REPRESENTATIVE.

2 0 2 4
SCALE: 1/2" = 1'-0"
1 0 1
SCALE: 1 1/2" = 1'-0"

Johnson Mirmiran Thompson Adams PA
ARCHITECTS/ENGINEERS PLANNING SURVEYING MANAGEMENT
P.O. BOX 313, GREENWOOD, MISSISSIPPI (662) 455-4643

NO.	DATE	BY	DESCRIPTION

REVISIONS

USDA Agricultural Research Service

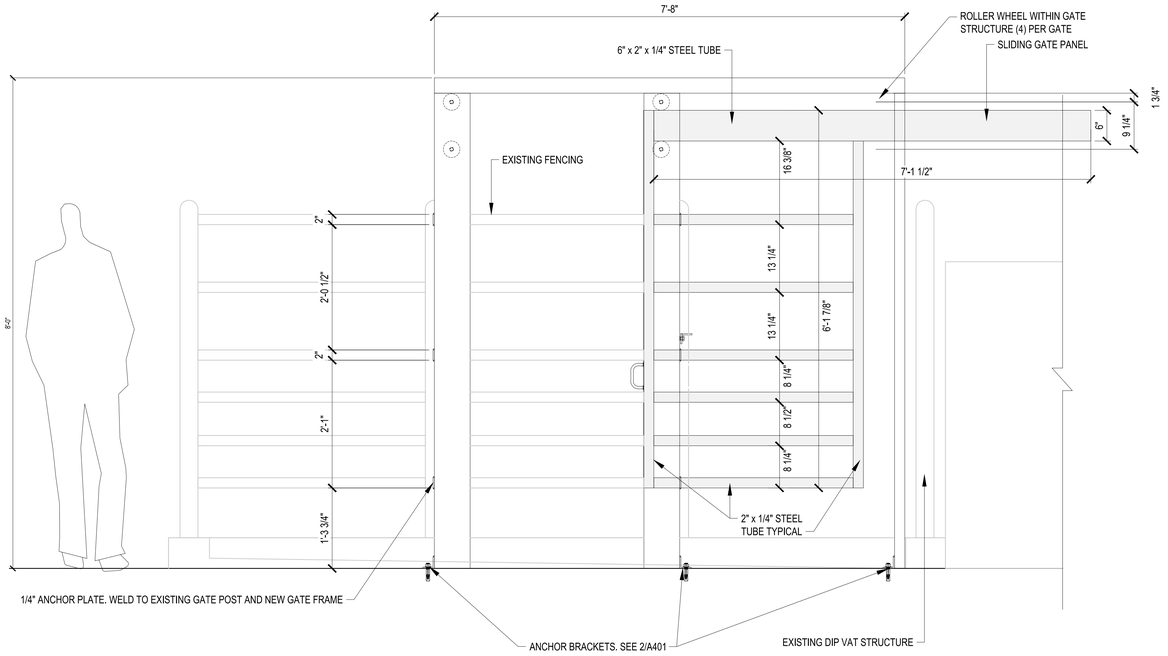
A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	xxxxx
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	xxxxx

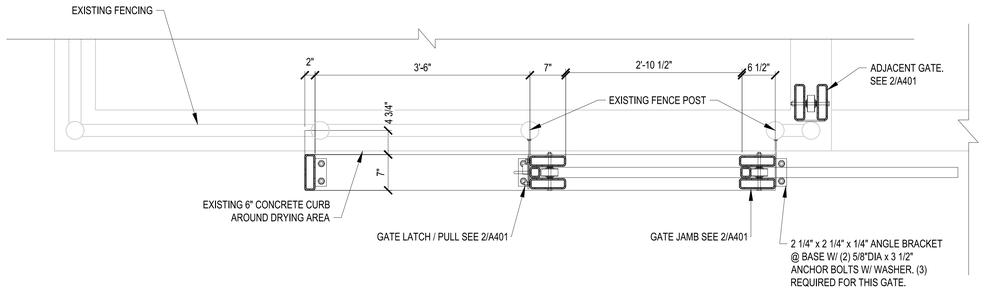
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

VAT ACCESS CAP DETAIL

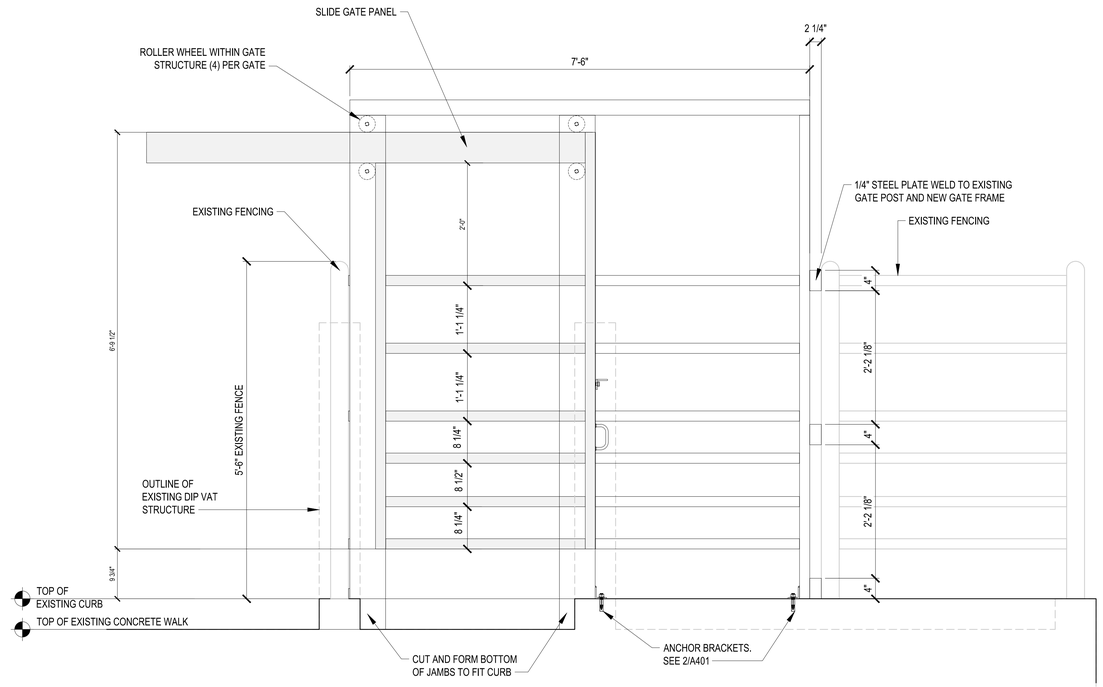
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DATE		15 OF 32
08/31/2026		

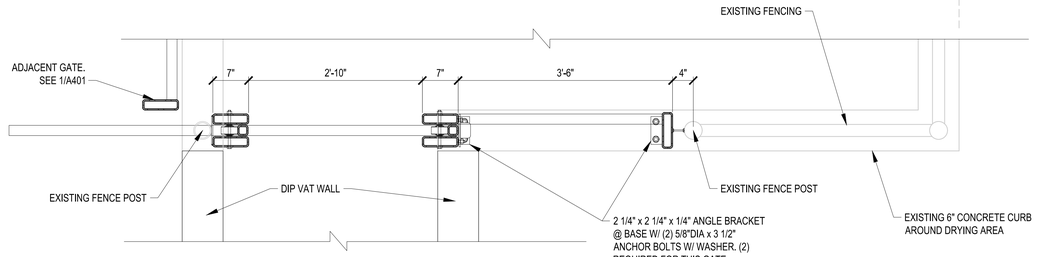
REGISTERED ARCHITECT JOHN E. MIRMIRAN 19747 STATE OF TEXAS



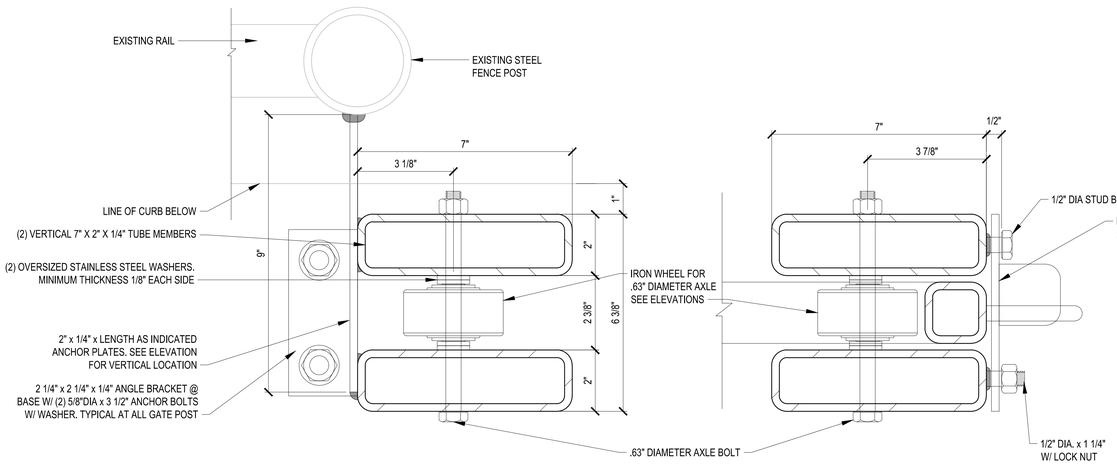


1 A401 SLIDING GATE AT WALKWAY DETAIL
SCALE: 1" = 1'-0"

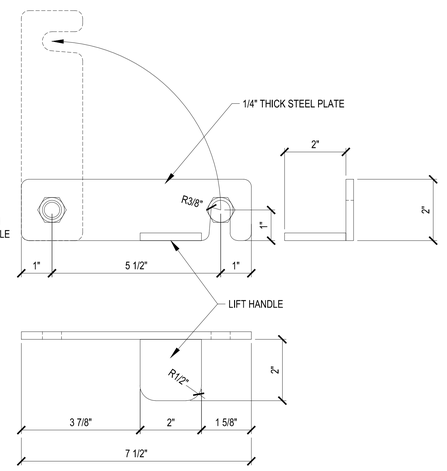




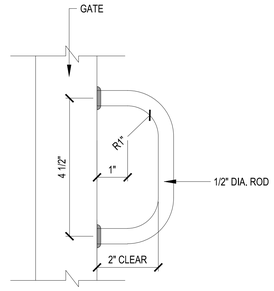
2 A401 SLIDING GATE AT CURB DETAIL
SCALE: 1" = 1'-0"



3 A401 TYPICAL SLIDE GATE SECTION
SCALE: 6" = 1'-0"



GATE LATCHING HARDWARE



GATE PULL

WELDING NOTE - CATTLE RAILING AND FENCING

PROVIDE WELDED CONNECTIONS IN ACCORDANCE WITH AWS D1.1. WELDERS SHALL BE QUALIFIED FOR THE WELDING PROCESSES AND MATERIALS USED. UNLESS OTHERWISE INDICATED, PROVIDE CONTINUOUS 3/16-INCH FILLET WELDS AROUND THE FULL PERIMETER OF EACH RAIL-TO-POST, FRAME, BRACKET, AND PLATE CONNECTION. WELDS SHALL BE UNIFORM, SMOOTH, AND FREE OF CRACKS, POROSITY, UNDERCUT, SHARP EDGES, AND PROJECTIONS THAT COULD INJURE ANIMALS OR PERSONNEL. GRIND WELDS AND CUT EDGES SMOOTH WHERE EXPOSED TO ANIMAL CONTACT. REMOVE SLAG AND WELD SPATTER AND PREPARE WELDED AREAS FOR THE SPECIFIED PROTECTIVE COATING. PROVIDE VENT AND DRAIN HOLES IN CLOSED OR HOLLOW MEMBERS AS REQUIRED FOR HOT-DIP GALVANIZING (ADDITIVE ALTERNATE #2). REPAIR DAMAGED GALVANIZED COATINGS AFTER FIELD WELDING IN ACCORDANCE WITH ASTM A780. DO NOT FIELD-WELD OR MODIFY MEMBERS WITHOUT APPROVAL OF THE CONTRACTING OFFICERS REPRESENTATIVE.

0 1 1/2" 3" 6" SCALE: 6"=1'-0"

0 1' 2' 3' SCALE: 1"=1'-0"

NO.	DATE	BY	DESCRIPTION
			REVISIONS

Johnson McAdams FIRM

A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	XXX
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	XXX

USDA Agricultural Research Service

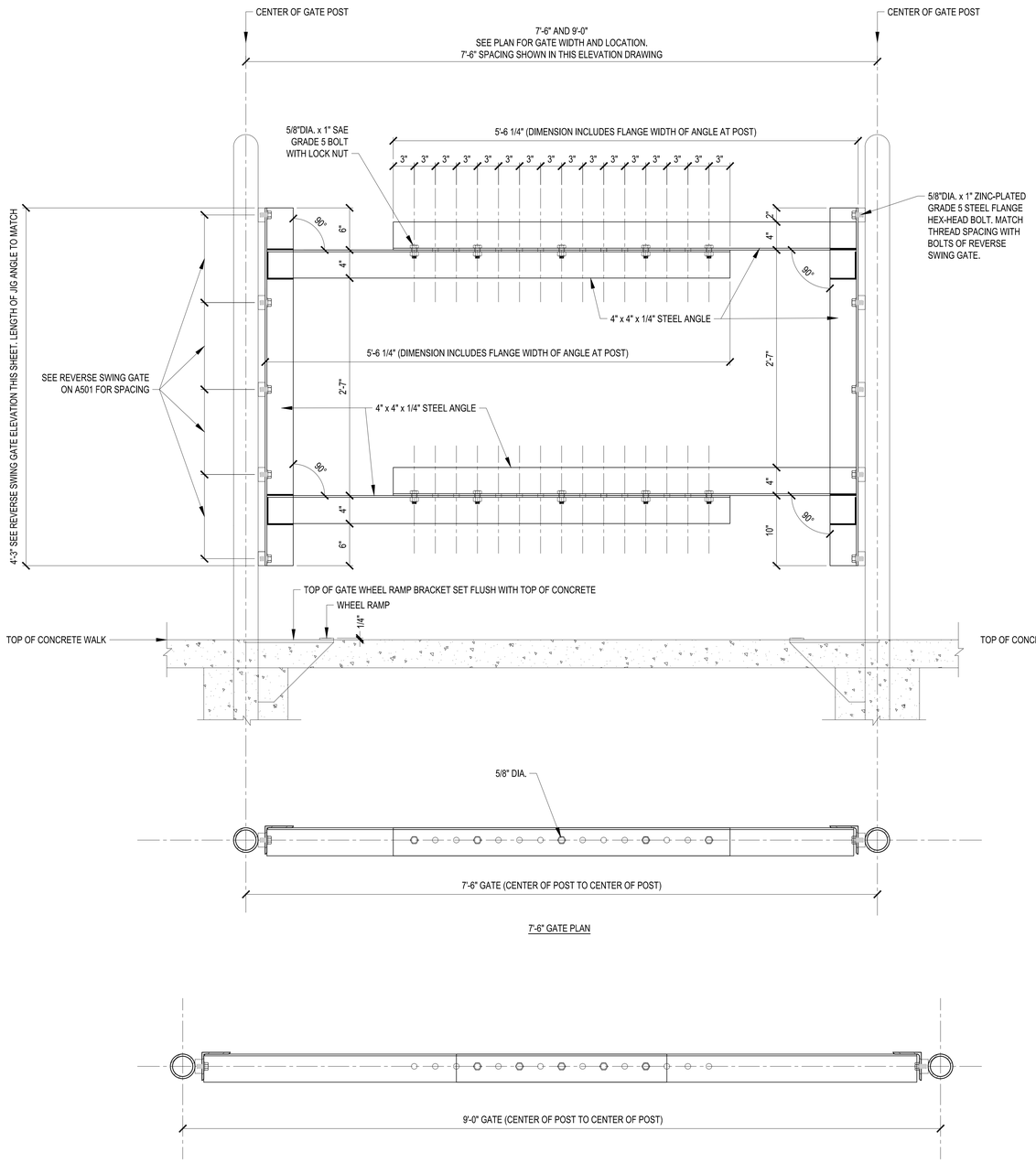
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG. 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

SLIDING GATE DETAILS

PROJECT NO.	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD, 20705	SHEET
XXX		A401
DATE		16 OF 32
08/31/2026		

REGISTERED ARCHITECT JOHN L. FRANCIS STATE OF TEXAS

J:\greenwood\2025 Jobs\X25 USDA Bld 5\e Vet Redesign Edinburg TX\06 Submittals\[PS] submittal\drawings\A401.dwg
Aug 31, 2026 - 2:53pm



7'-6" GATE PLAN

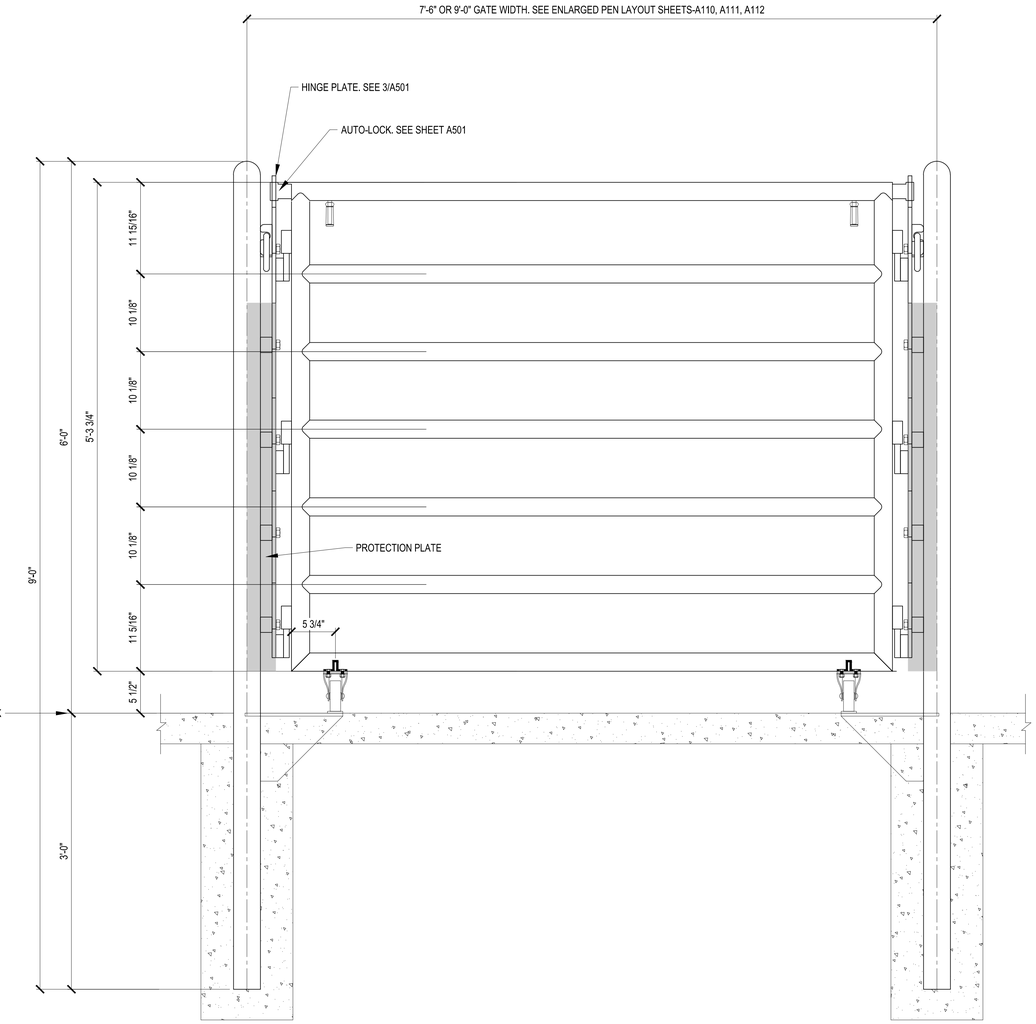
9'-0" GATE PLAN

1 A500 REVERSE SWING GATE INSTALLATION JIG DETAIL

SCALE: 1 1/2" = 1'-0"

JIG INTENT:

THE JIG IS DESIGNED TO AID IN INSTALLATION OF THE REVERSE SWING GATE. THE OBJECTIVE IS TO ENSURE THE GATE POST ARE VERTICALLY ALIGN WITH EACH OTHER. THIS IS CRITICAL TO THE GATES OPERATION. THE JIG CAN BE USED FOR BOTH GATE WIDTHS AND CAN BE USED FOR EACH INDIVIDUAL GATE INSTALLATION. PRIOR TO REMOVAL OF THE JIG, ENSURE POST CONCRETE FOUNDATION HAS CURED. PROVIDE BRACING FOR POSTS DURING INITIAL CONCRETE CURING AND AS REQUIRED TO MAINTAIN POST ALIGNMENTS. DELIVER JIG TO FACILITY PERSONNEL AS DIRECTED.



2 A500 REVERSE SWING GATE ELEVATION

SCALE: 1 1/2" = 1'-0"

WELDING NOTE - CATTLE RAILING AND FENCING

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NO.	DATE	BY	DESCRIPTION
			REVISIONS

Johnson, Mirmiran & Thompson, Inc. — Architecture/Engineering, Planning, Surveying, Management — P.O. Box 5-5, Greenwood, Mississippi — (662) 455-6443

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	XXX
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	XXX

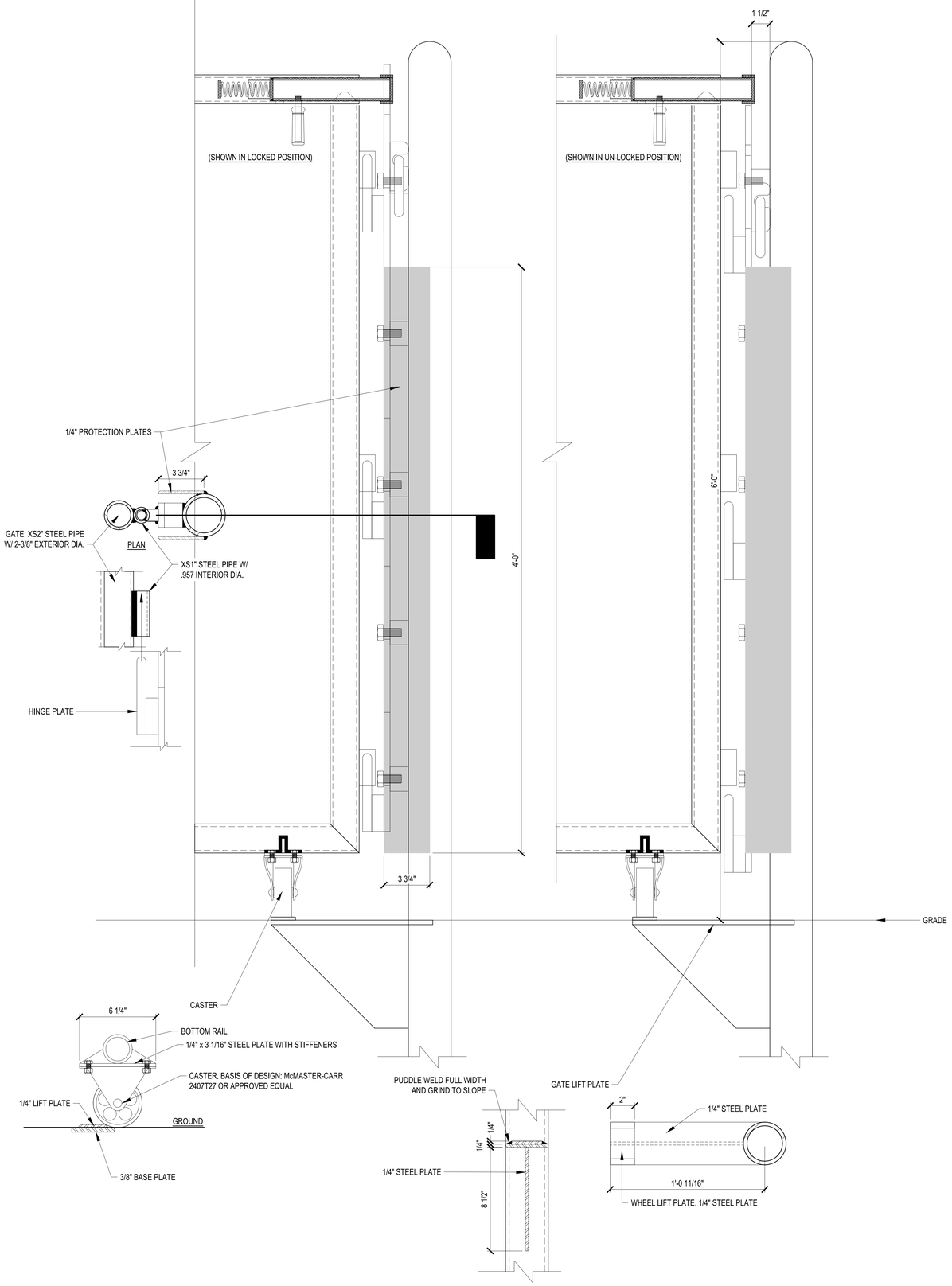
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EDINBURG - KERRVILLE BLDG. 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS
REVERSE SWING GATE
POST INSTALLATION DETAILS

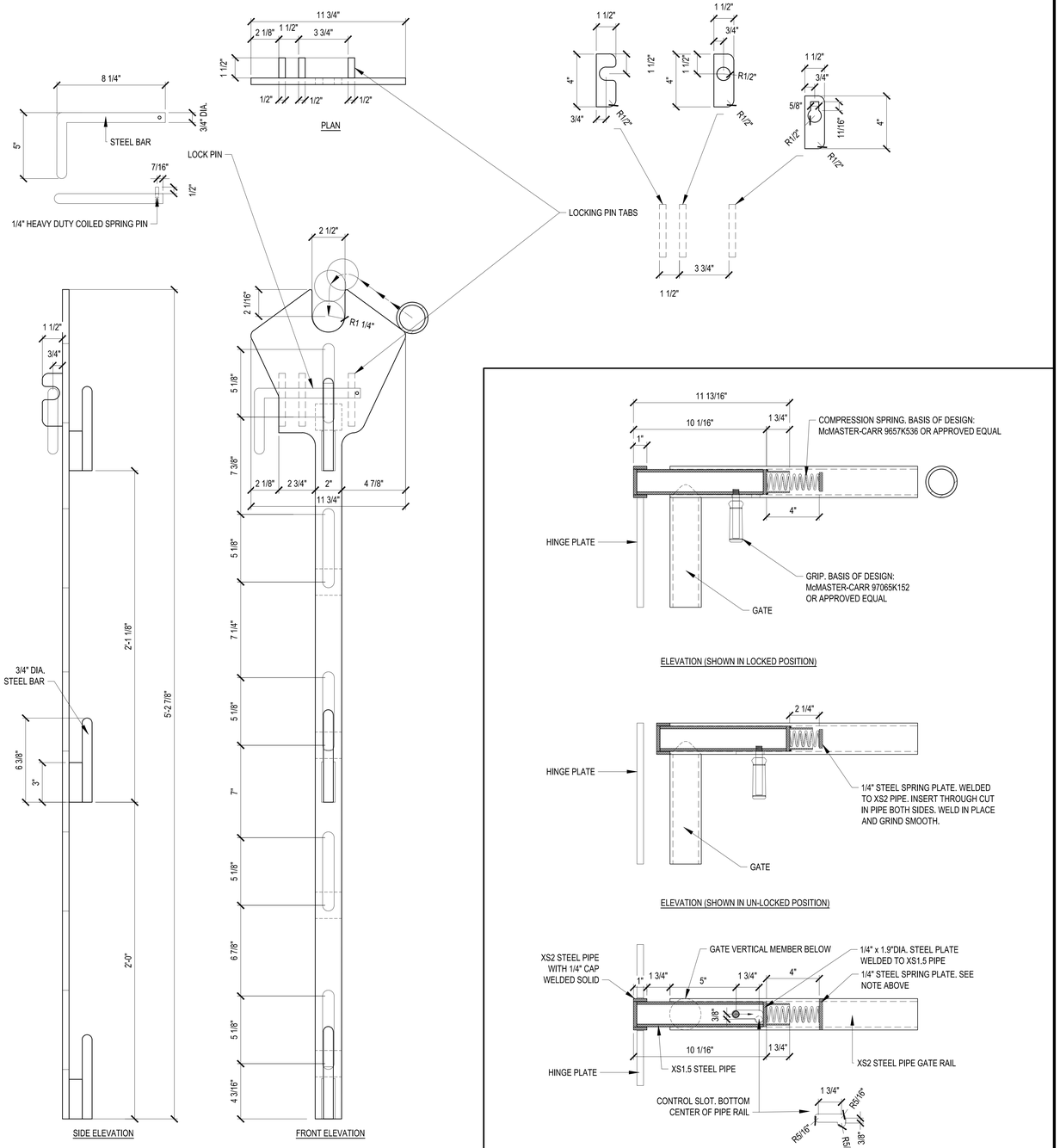
PROJECT NO. XXX	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD, 20705	SHEET A500
DATE 3		17 OF 32

REGISTERED ARCHITECT — JOHN L. FERGUSON — STATE OF TEXAS

J:\greenwood\2025 Jobs\XFDS USDA BR Vet Redesign Edinburg TX\08 Submittals\DSA submittal\drawings\A500.dwg
Aug 20, 2025 - 12:52pm



1 A501 GATE HINGE DETAIL
SCALE: 3" = 1'-0"



2 A501 HINGE PLATE DETAIL
SCALE: 3" = 1'-0"

3 A501 AUTO-LOCK DETAIL
SCALE: 3" = 1'-0"

WELDING NOTE - CATTLE RAILING AND FENCING

PROVIDE WELDED CONNECTIONS IN ACCORDANCE WITH AWS D1.1. WELDERS SHALL BE QUALIFIED FOR THE WELDING PROCESSES AND MATERIALS USED. UNLESS OTHERWISE INDICATED, PROVIDE CONTINUOUS 3/16-INCH FILLET WELDS AROUND THE FULL PERIMETER OF EACH RAIL-TO-POST, FRAME, BRACKET, AND PLATE CONNECTION. WELDS SHALL BE UNIFORM, SMOOTH, AND FREE OF CRACKS, POROSITY, UNDERCUT, SHARP EDGES, AND PROJECTIONS THAT COULD INJURE ANIMALS OR PERSONNEL. GRIND WELDS AND CUT EDGES SMOOTH WHERE EXPOSED TO ANIMAL CONTACT. REMOVE SLAG AND WELD SPATTER AND PREPARE WELDED AREAS FOR THE SPECIFIED PROTECTIVE COATING. PROVIDE VENT AND DRAIN HOLES IN CLOSED OR HOLLOW MEMBERS AS REQUIRED FOR HOT-DIP GALVANIZING (ADDITIVE ALTERNATE #2). REPAIR DAMAGED GALVANIZED COATINGS AFTER FIELD WELDING IN ACCORDANCE WITH ASTM A780. DO NOT FIELD-WEL OR MODIFY MEMBERS WITHOUT APPROVAL OF THE CONTRACTING OFFICERS REPRESENTATIVE.

0 3" 6" 1'
SCALE: 3"=1'-0"

Johnson, McAdams PSC
P.O. BOX 513, GREENWOOD, MISSISSIPPI - (662) 455-6943

NO.	DATE	BY	DESCRIPTION
			REVISIONS

A-E FIRM	USDA
PROJECT MANAGER: REH	EPM: B. DEAN
DESIGNER: JLF	PPM: XXX
CHECKED BY: DRL	SAFETY & HEALTH: D. CORDRAY
JOB #: 4125	REAL PROPERTY: XXX

USDA Agricultural Research Service

USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG. 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

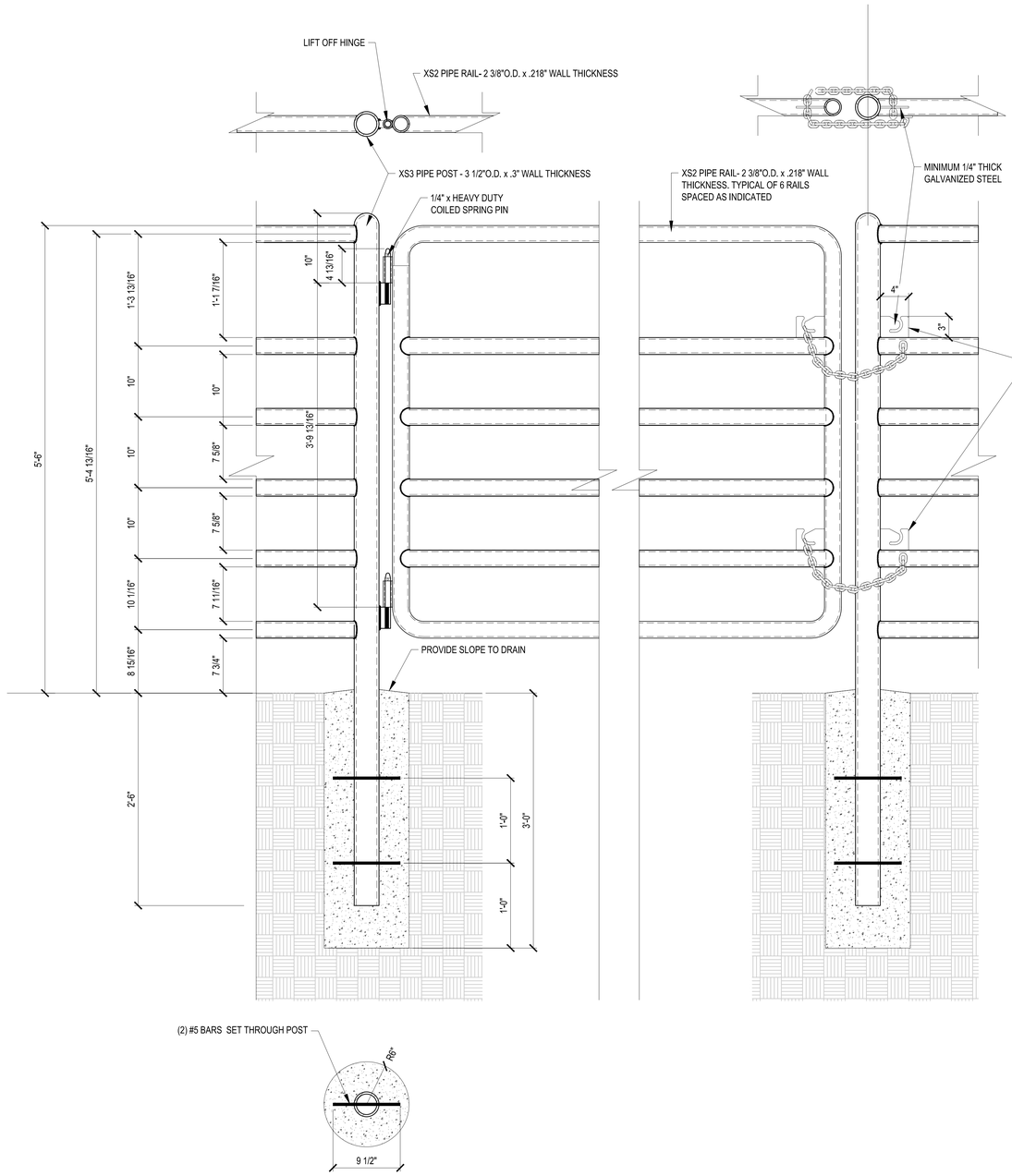
REVERSE SWING GATE DETAILS

PROJECT NO. XXX
DATE 08/31/2026

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD. 20705

SHEET A501
18 OF 32

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Aug 31, 2026 - 12:29pm



PROVIDE 5/16-INCH MINIMUM, GRADE 43, WELDED-LINK CARBON-STEEL CHAIN, HOT-DIP GALVANIZED AFTER FABRICATION (ADDITIVE ALTERNATE #2), WITH GALVANIZED CAPTIVE GATE-LATCHING HARDWARE. CHAIN AND HARDWARE SHALL BE FREE OF SHARP EDGES, PROJECTING HOOKS, AND EXCESSIVE SLACK THAT COULD INJURE OR ENTANGLE LIVESTOCK. CHAIN SHALL BE WELDED TO EXTERIOR SIDE OF THE FENCE RAIL AS INDICATED AND BE LONG ENOUGH TO ENGAGE LATCH AT GATE THEN THE FENCE RAIL LATCH. THE INTENT IS TO SECURE THE GATE IN LINE WITH THE FENCE AS TIGHT AS POSSIBLE.



1 A502 MAINTENANCE GATE DETAIL
SCALE: 1-1/2" = 1'-0"

WELDING NOTE - CATTLE RAILING AND FENCING

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0 6" 1' 2'
SCALE: 1 1/2"=1'-0"

NO.	DATE	BY	DESCRIPTION
			REVISIONS

Johnson Mirmiran Thompson P.A.
ARCHITECTS/ENGINEERS PLANNING SURVEYING MANAGEMENT
P.O. BOX 313, GREENWOOD, MISSISSIPPI (662) 455-6643

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	XXX
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	XXX

USDA Agricultural Research Service

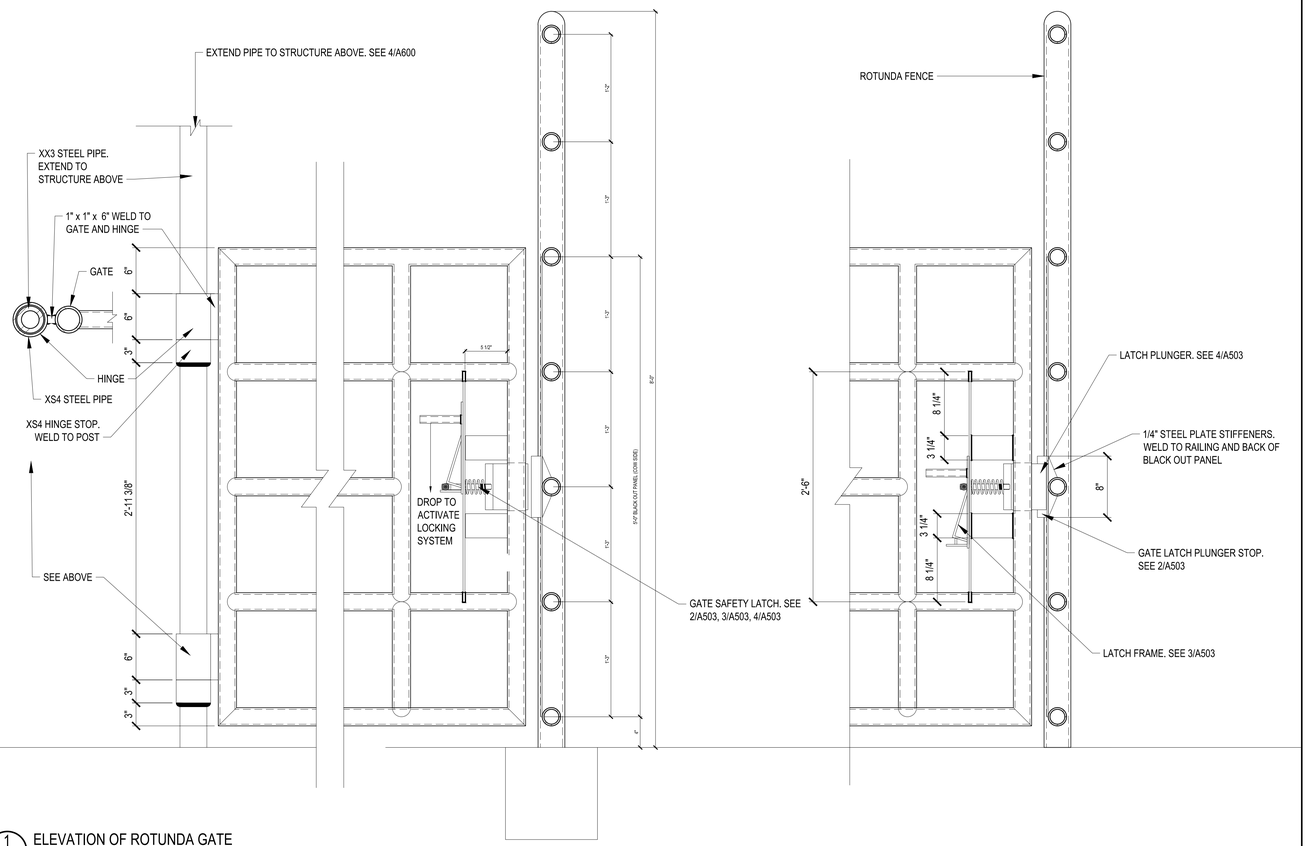
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG. 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

MAINTENANCE GATE DETAILS

PROJECT NO. XXX	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD, 20705	SHEET A502
DATE 08/31/2026		19 OF 32

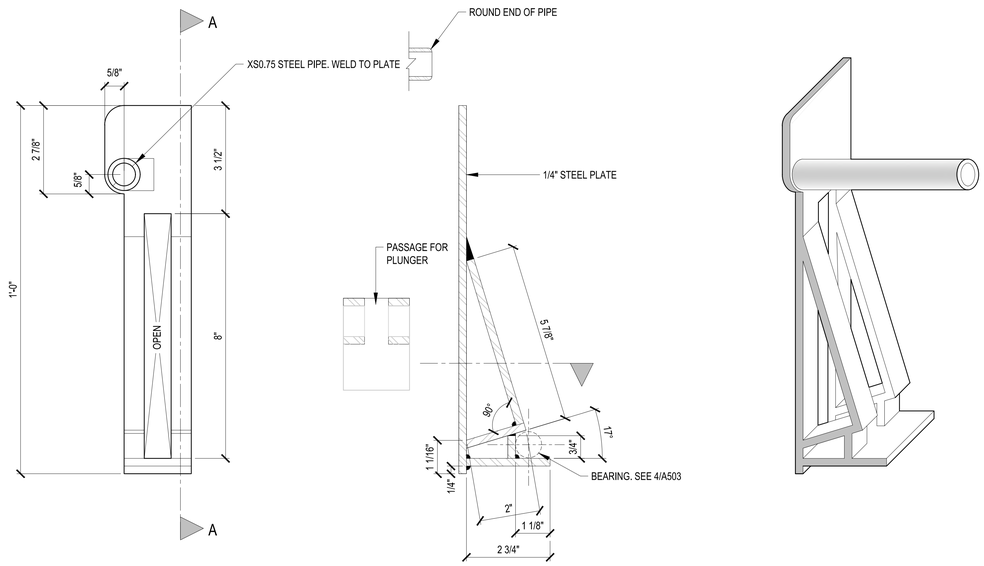
REGISTERED ARCHITECT JOHN L. FRANKLIN STATE OF TEXAS

J:\greenwood\2025 Jobs\4125 USDA 84 Vet Redesign Edinburg TX\08 Submittals\95% submittal (drawings)\A502.dwg
Aug 30, 2026 - 12:53pm



1 A503 ELEVATION OF ROTUNDA GATE

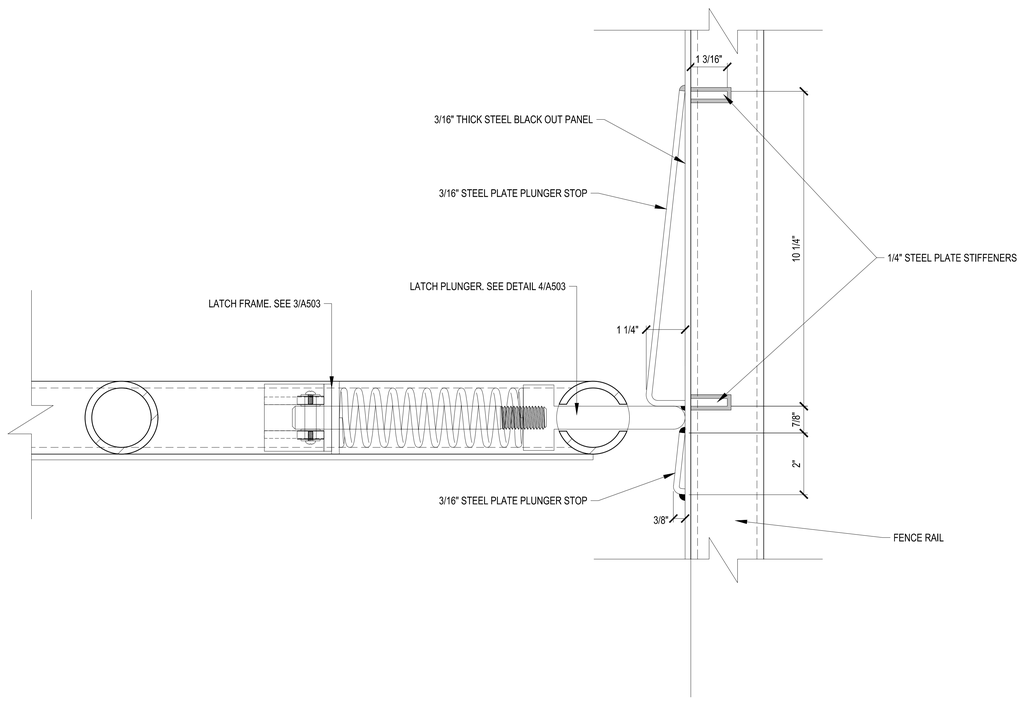
SCALE: 1 1/2" = 1'-0"



ELEVATION SECTION ISOMETRIC

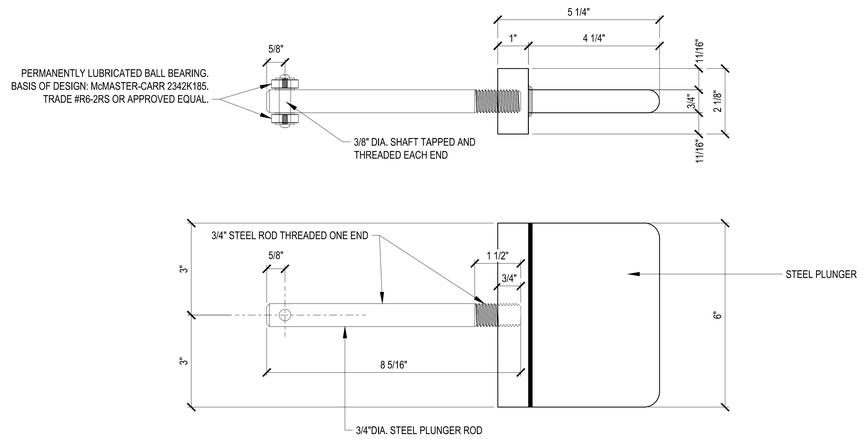
3 A503 LATCH FRAME

SCALE: 6" = 1'-0"



2 A503 PLAN DETAIL OF SAFETY LATCH

SCALE: 6"=1'-0"



4 A503 LATCH PLUNGER DETAIL

SCALE: 6" = 1'-0"

WELDING NOTE - CATTLE RAILING AND FENCING

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Johnson, Mirmiran & Thompson, Inc.
ARCHITECTS/ENGINEERS PLANNING SURVEYING MANAGEMENT
P.O. BOX 313, GREENWOOD, MISSISSIPPI (662) 455-6422

A-E FIRM	USDA
PROJECT MANAGER REH	EPM B.DEAN
DESIGNER JLF	PPM XXX
CHECKED BY DRL	SAFETY & HEALTH D. CORDRAY
JOB # 4125	REAL PROPERTY XXX

NO.	DATE	BY	DESCRIPTION
			REVISIONS

USDA Agricultural Research Service

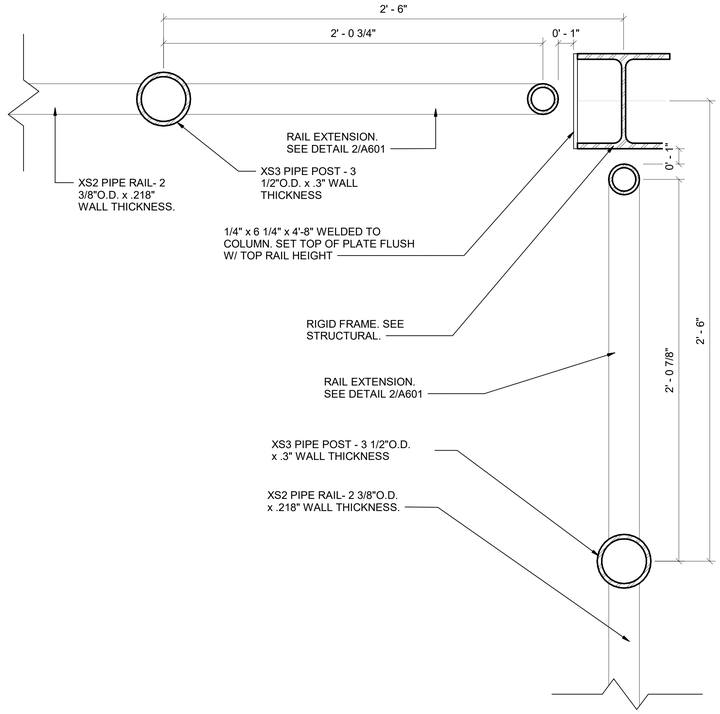
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG. 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

ROTUNDA GATE DETAIL

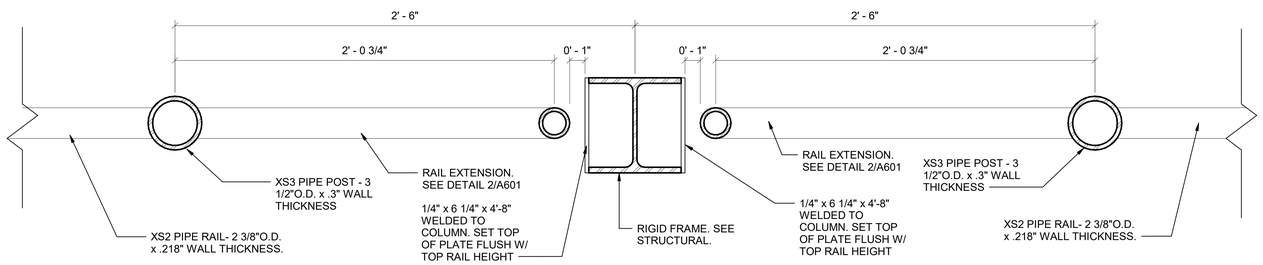
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DATE 08/31/2026		20 OF 32

REGISTERED ARCHITECT JAMES L. FERGUSON 23,001 STATE OF TEXAS

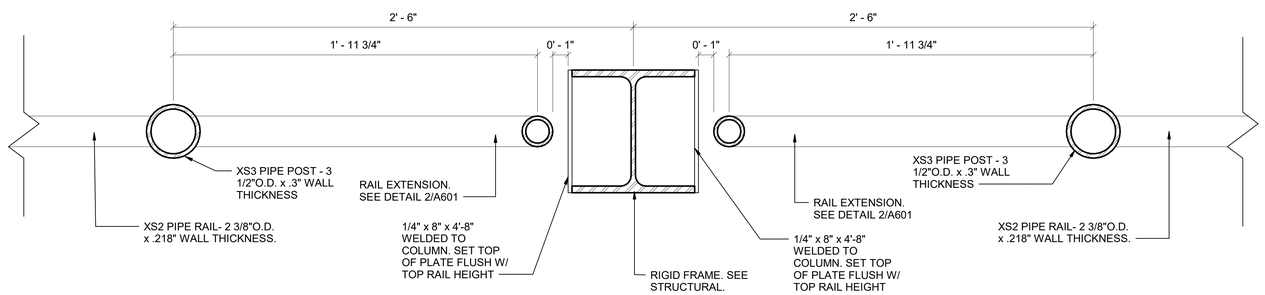
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Aug 31, 2026 - 12:53pm



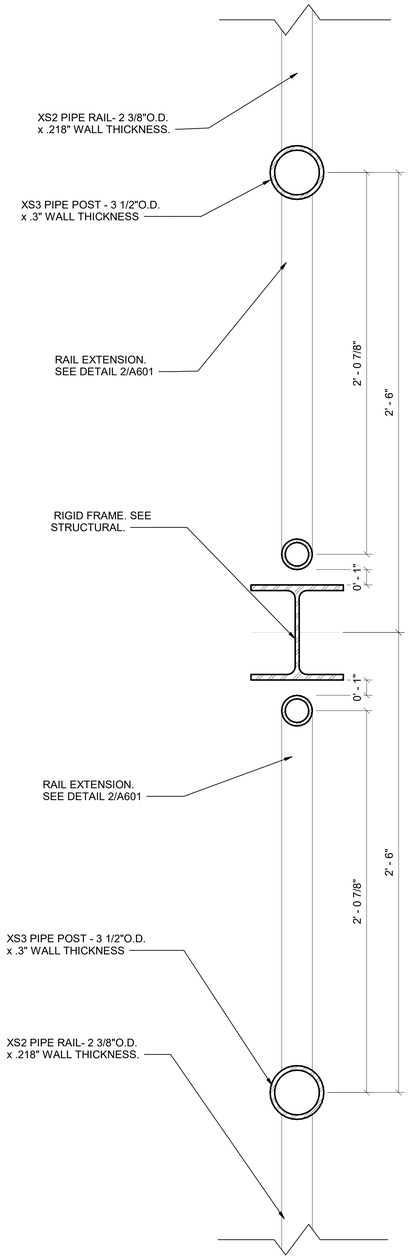
1 A510 PLAN DETAIL
3" = 1'-0"



2 A510 PLAN DETAIL
3" = 1'-0"



3 A510 PLAN DETAIL
3" = 1'-0"



4 A510 PLAN DETAIL
3" = 1'-0"

WELDING NOTE – CATTLE RAILING AND FENCING

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0 1
SCALE: 3" = 1'-0"

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DIP.VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

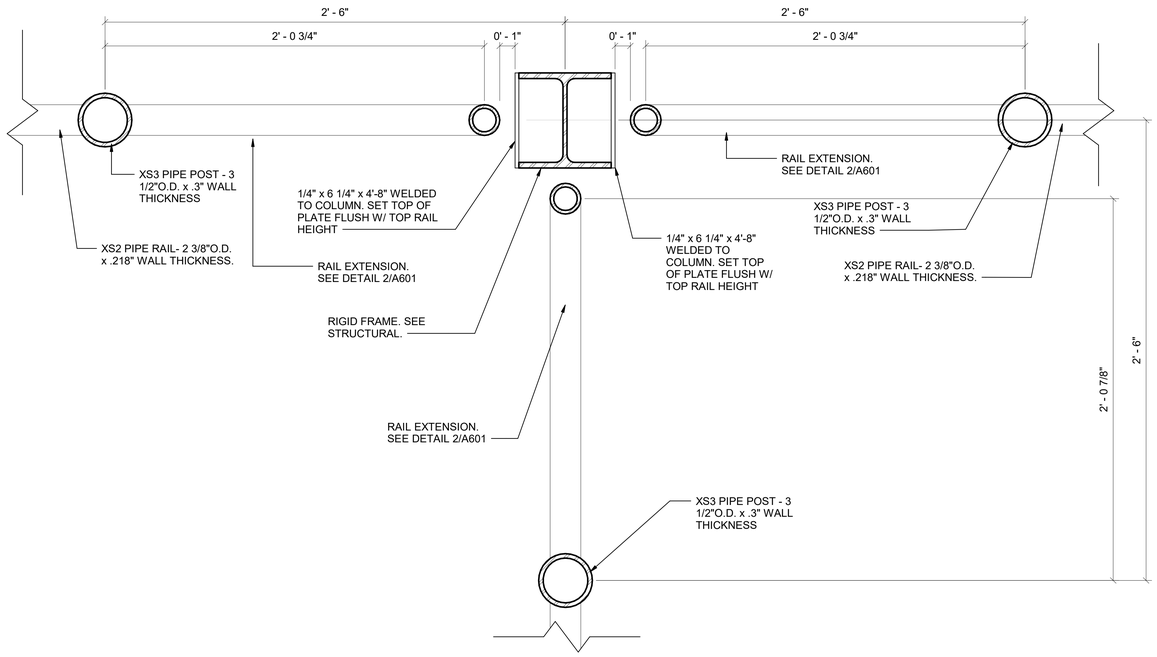
PLAN DETAILS

A-E FIRM	USDA
PROJECT MANAGER	EPM
REH	B. DEAN
DESIGNER	PPM
JLF	xxxxx
CHECKED BY	SAFETY & HEALTH
DRL	D. CORDRAY
JOB #	REAL PROPERTY
4125	xxxxx

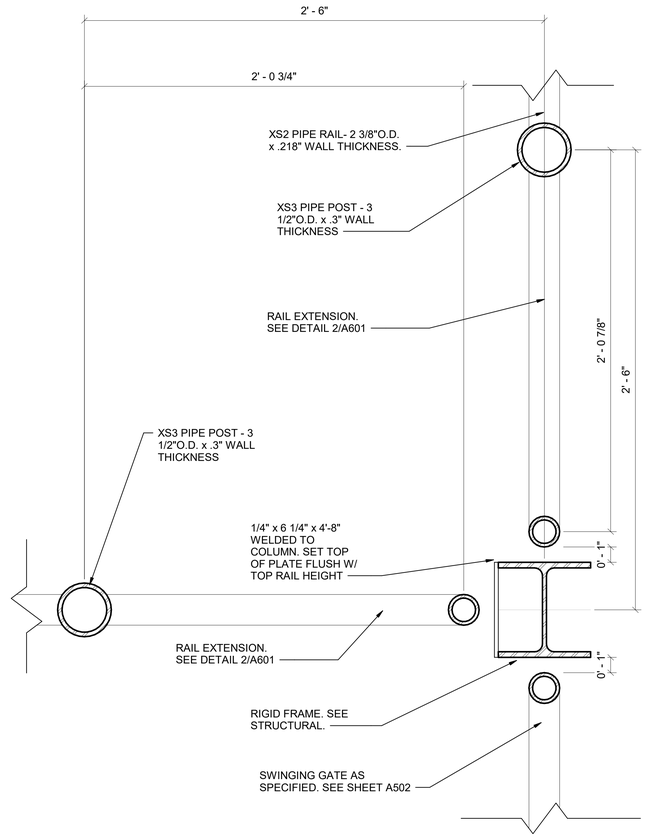
PROJECT NO.	xxxxx
DATE	08/31/2026

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD 20705

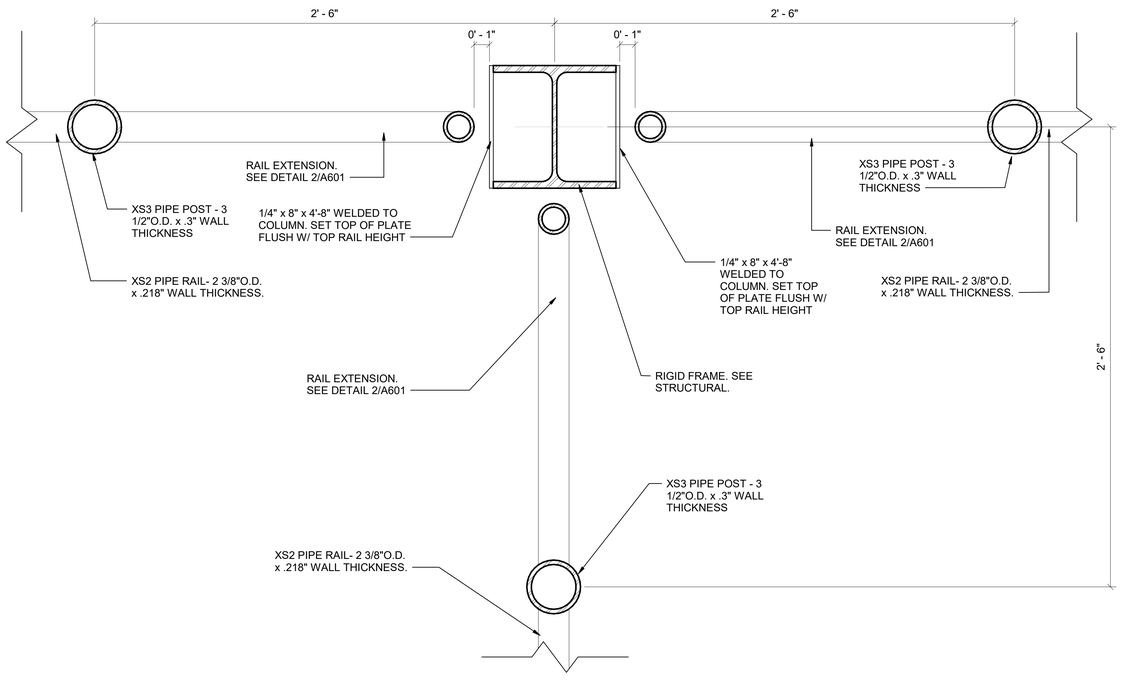
SHEET
A510
21 OF 32



1 A511 PLAN DETAIL
3" = 1'-0"



2 A511 PLAN DETAIL
3" = 1'-0"



3 A511 PLAN DETAIL
3" = 1'-0"

WELDING NOTE – CATTLE RAILING AND FENCING

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0 1

SCALE: 3" = 1'-0"

REGISTERED ARCHITECT JOHN L. FRANCIS STATE OF TEXAS

Johnson M Adams PA
ARCHITECTS/ENGINEERS – PLANNING – SURVEYING www.jmapa.com
P.O. BOX 913, GREENWOOD, MISSISSIPPI – (662) 453–4943

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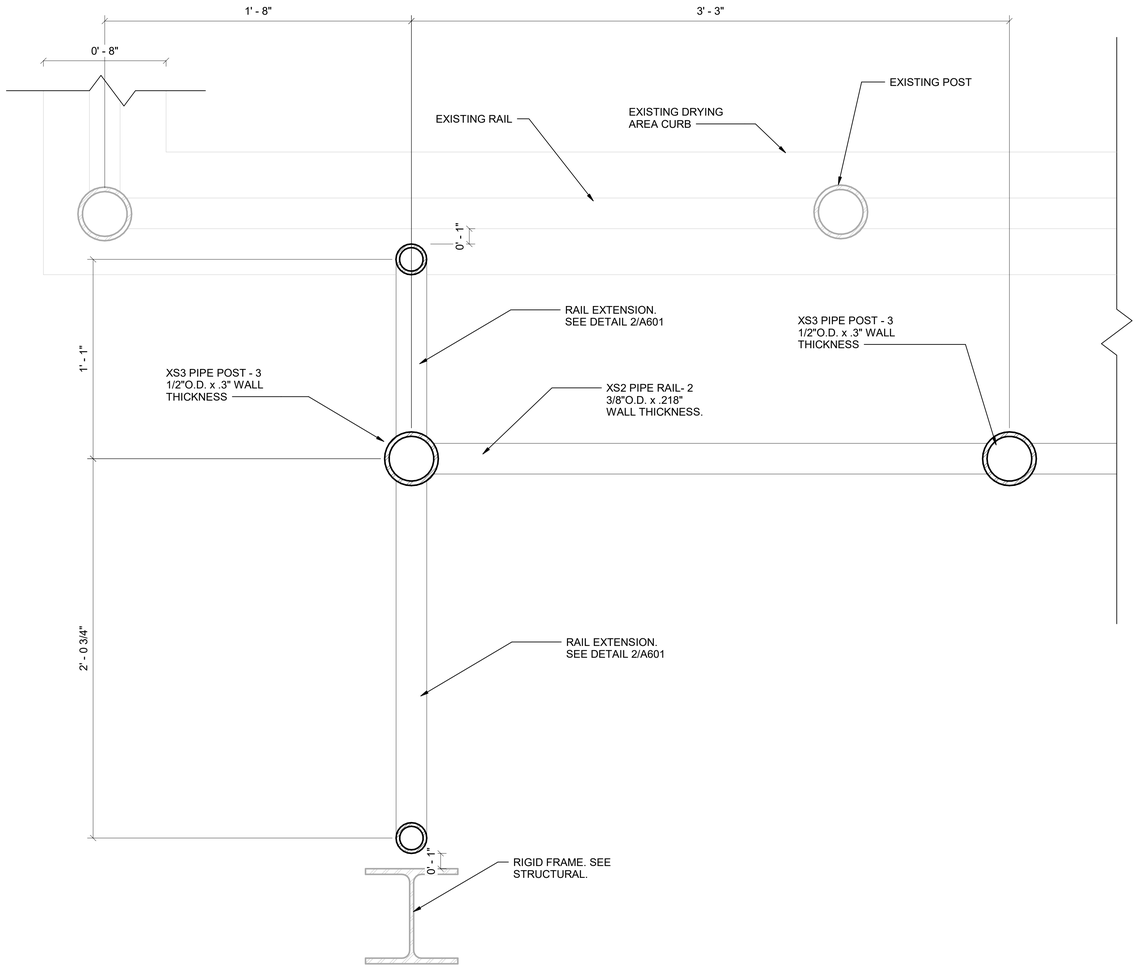
A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx

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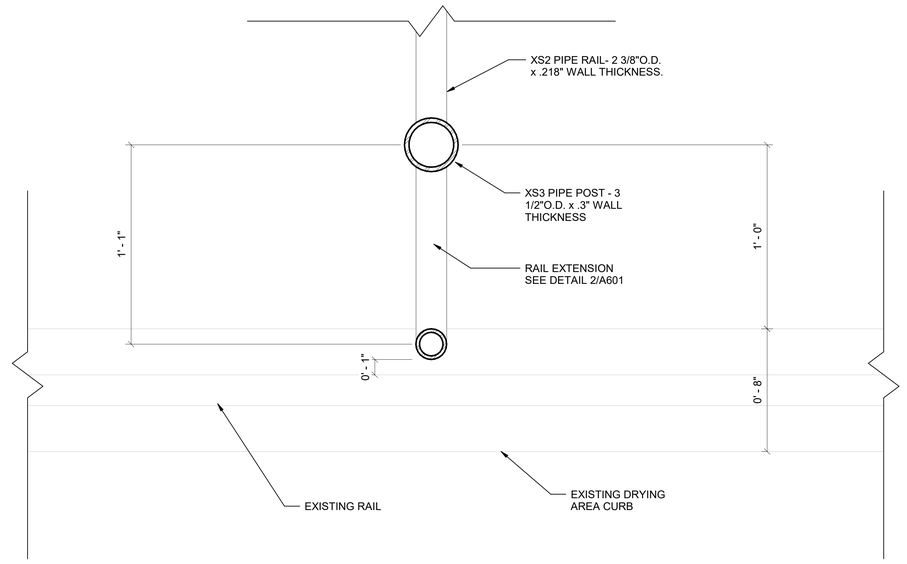
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

PLAN DETAILS

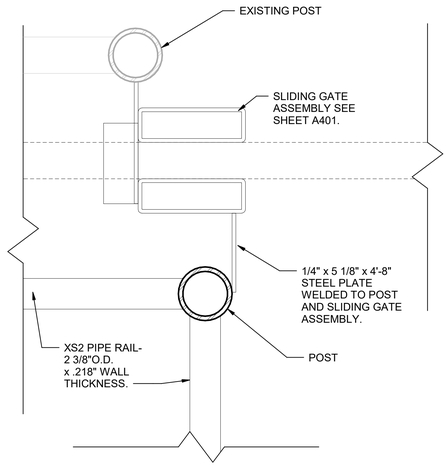
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DATE 08/31/2026		22 OF 32



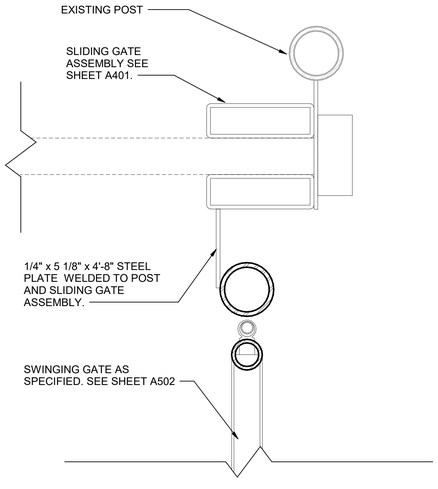
1 A512 PLAN DETAIL
3" = 1'-0"



2 A512 PLAN DETAIL
3" = 1'-0"



3 A512 PLAN DETAIL
3" = 1'-0"



4 A512 PLAN DETAIL
3" = 1'-0"

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0 1
SCALE: 3" = 1'-0"

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ARCHITECTS/ENGINEERS • PLANNING • SURVEYING • WAREHOUSE
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NO.	DATE	BY	DESCRIPTION

REVISIONS

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx

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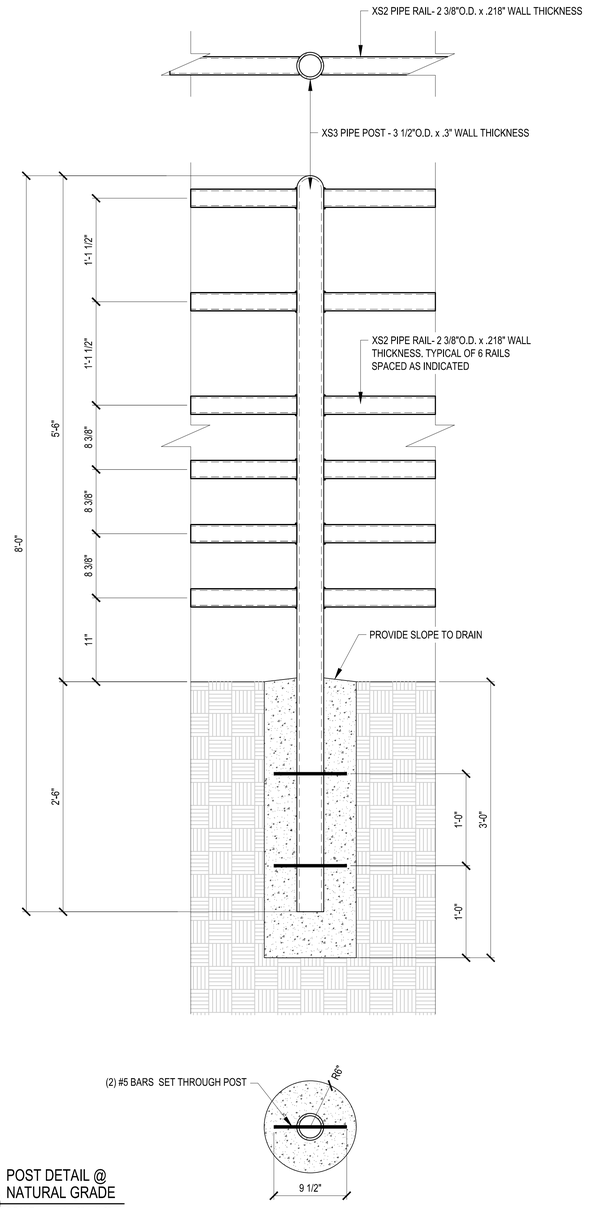
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

PLAN DETAILS

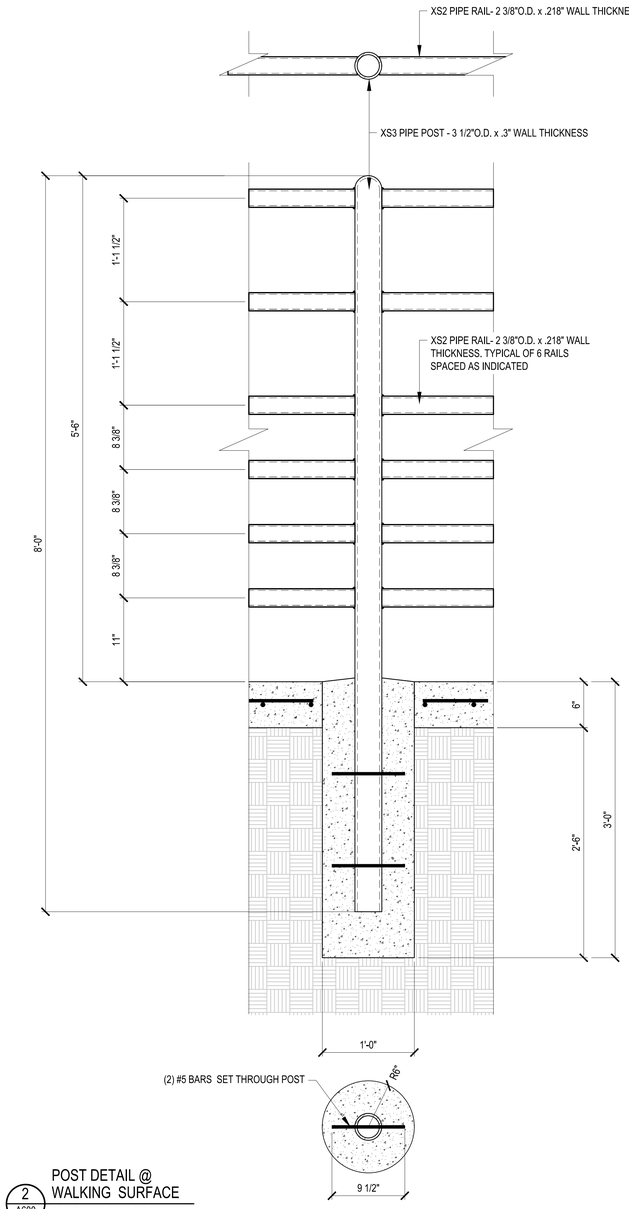
PROJECT NO.	xxxxx
DATE	08/31/2026

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD 20705

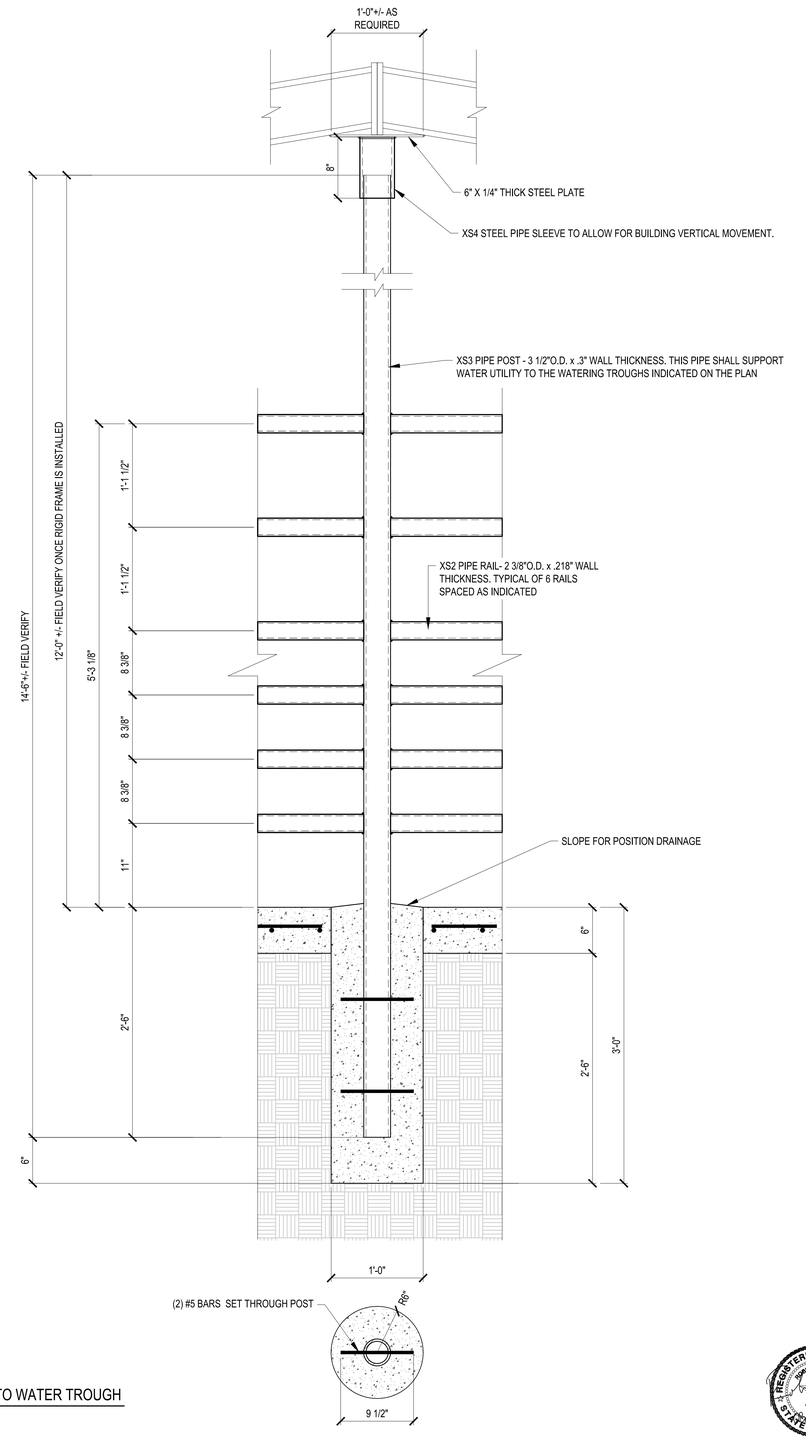
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23 OF 32



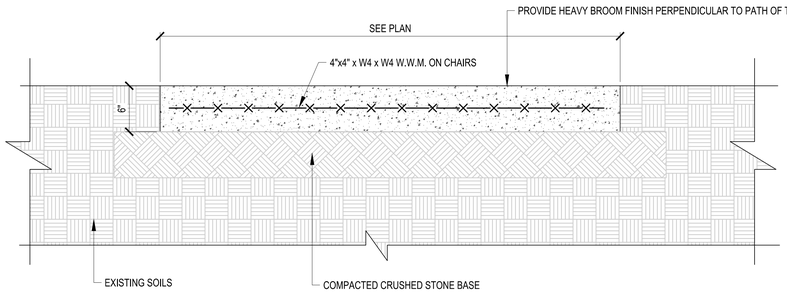
1 A600 POST DETAIL @ NATURAL GRADE
SCALE: 1 1/2" = 1'-0"



2 A600 POST DETAIL @ WALKING SURFACE
SCALE: 1 1/2" = 1'-0"



4 A600 POST DETAIL @ WATER DROPS TO WATER TROUGH
SCALE: 1 1/2" = 1'-0"



3 A600 TYPICAL WALKING SURFACE DETAIL
SCALE: 1 1/2" = 1'-0"

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0 6" 1' 2'
SCALE: 1 1/2"=1'-0"

Johnson Mirmiran Adams — ARCHITECTS/ENGINEERS/CEO PLANNING SURVEYING MANAGEMENT
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NO.	DATE	BY	DESCRIPTION

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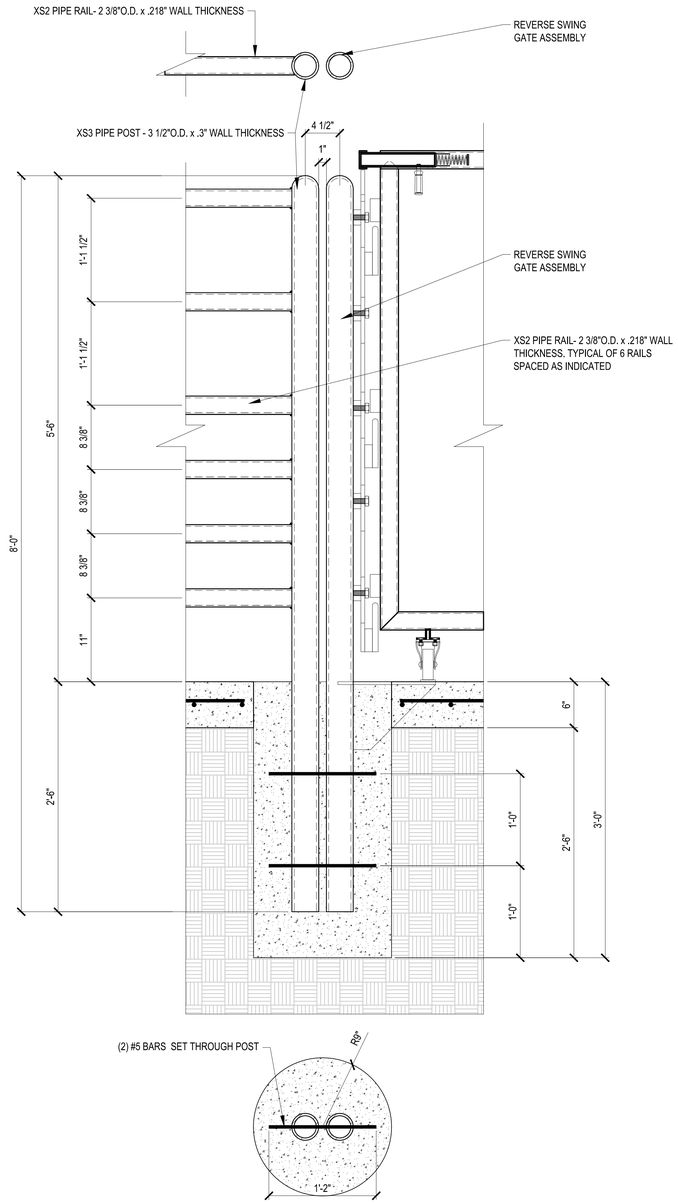
USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG. 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

DETAILS

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B.DEAN
DESIGNER	JLF
PPM	XXX
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	XXX

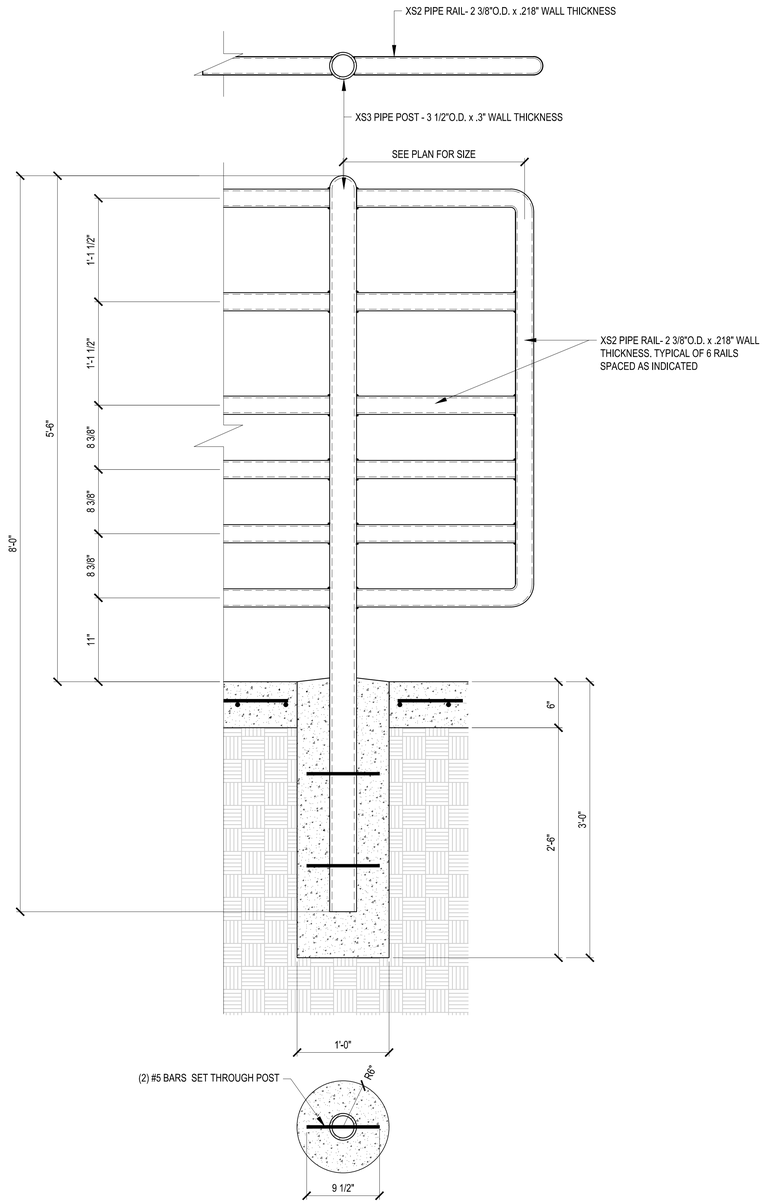
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DATE	08/31/2026	
AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD, 20705		SHEET A600 24 OF 32

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Aug 31, 2026 - 12:23pm



1 A601 POST DETAIL @ REVERSE SWING GATE

SCALE: 1 1/2" = 1'-0"



2 A601 RAIL EXTENSION DETAIL

SCALE: 1 1/2" = 1'-0"

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0 6" 1' 2'

SCALE: 1 1/2"=1'-0"

Johnson McAdams PLLC
ARCHITECTS/ENGINEERS PLANNING SURVEYING MANAGEMENT
P.O. BOX 513, GREENWOOD, MISSISSIPPI - (662) 455-8421

NO.	DATE	BY	DESCRIPTION

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USDA AGRICULTURAL RESEARCH SERVICE
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DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

DETAILS

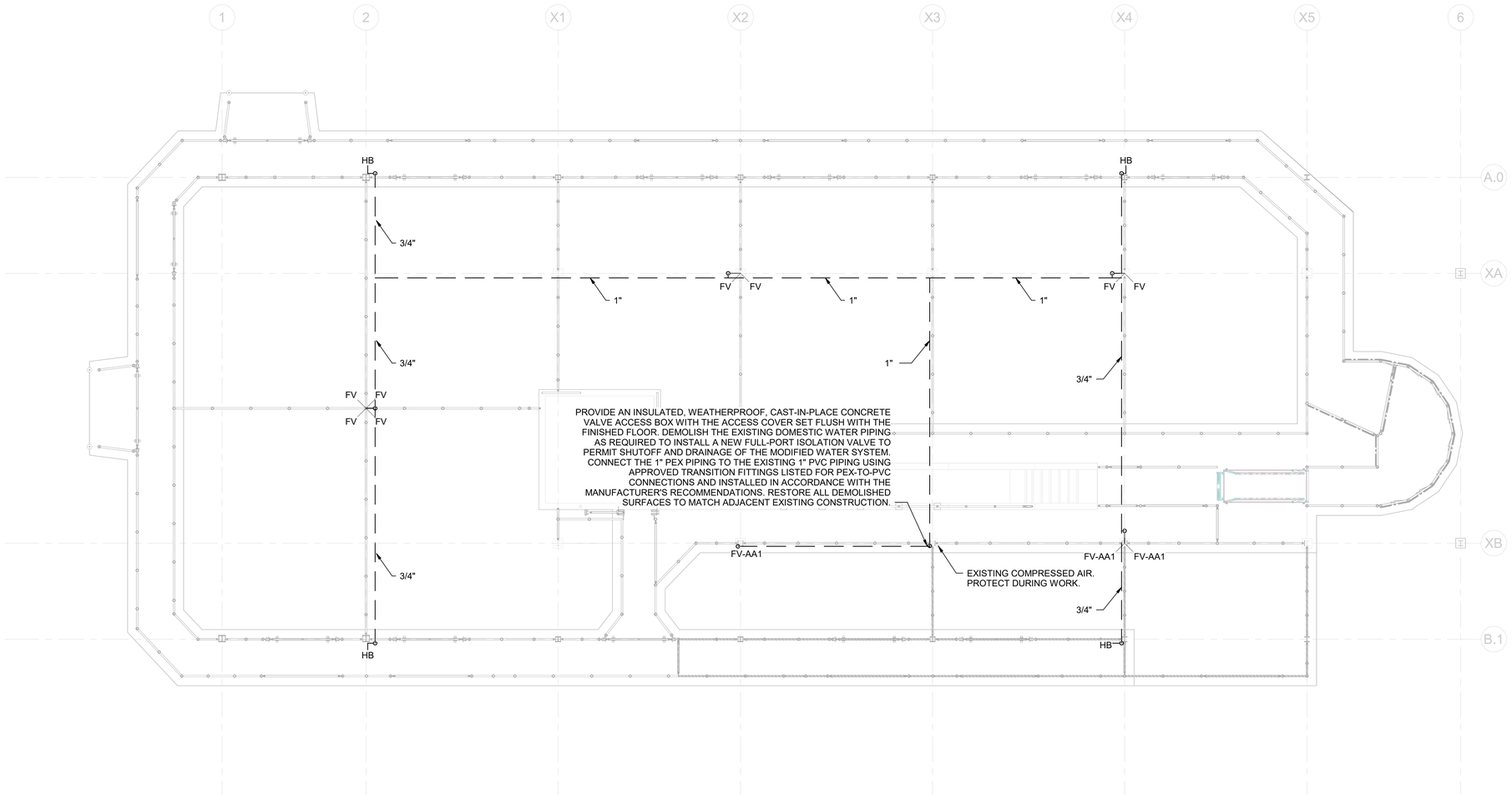
A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	JLF
PPM	XXX
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	XXX

PROJECT NO. XXX
DATE 08/31/2026

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD, 20705

SHEET A601
25 OF 32

J:\greenwood\2025 Jobs\4125 USDA 84 Dip Vat Redesign Edinburg TX\05 Submittals\PDF submittal\drawings\A601.dwg
Aug 31, 2026 - 12:53pm



LEGEND

HB	PROVIDE PRIER P-164 OR EQUAL FREEZELESS, ANTI-SIPHON, WALL HYDRANT WITH INTEGRAL VACUUM BREAKER. DROP DOWN FROM OVERHEAD TO 6' ABOVE WALK SURFACE.
FV	PROVIDE JOBE TOPAZ COMPACT OR EQUAL FLOAT VALVE INTEGRAL MANUAL LOCK-OFF SWITCH, CORROSION-RESISTANT CONSTRUCTION, AND PROTECTIVE CHAMBER. COORDINATE FLOAT VALVE WITH OWNER PROVIDED WATER TROUGH. DROP DOWN FROM STRUCTURE ABOVE TO FINISHED GRADE.
FV-AA1	PROVIDE FLOAT VALVE AS PART OF ADDITIVE ALTERNATE #1. ALL PIPING PART OF BASE BID PROVIDE CAP.
— —	PROVIDE UV-RESISTANT PEX TUBING, FITTINGS, AND SUPPORTS LISTED FOR POTABLE WATER SERVICE. ROUTE PIPING ALONG THE ROOF STRUCTURE AND DROP TO SERVE FIXTURES AT THE LOCATIONS INDICATED. ROUTE ALL VERTICAL DROPS ALONG STRUCTURAL MEMBERS WHERE PRACTICAL. INSTALL PIPING TO MAINTAIN CLEARANCE FROM ALL LIVESTOCK AND TO PREVENT DAMAGE FROM ANIMAL CONTACT OR NORMAL FACILITY OPERATIONS.

1 / P100 PLUMBING FLOOR PLAN
3/16" = 1'-0"

PLAN NORTH

4 0 4 8 12
SCALE: 3/16" = 1'-0"

Johnson Adams PA
ARCHITECTS/ENGINEERS PLANNING SURVEYING MANAGEMENT
P.O. BOX 513, GREENWOOD, MISSISSIPPI (662) 455-4943

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USDA Agricultural Research Service

USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

PLUMBING PLAN

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	DKM
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
BELTSVILLE MD 20705

SHEET **P100**
26 OF 32

ELECTRICAL GENERAL NOTES:

1. THE WORK MUST CONFORM WITH ALL REQUIREMENTS OF:
   - NFPA 70, NATIONAL ELECTRICAL CODE (NEC) (2026)
   - NFPA 101, LIFE SAFETY CODE (2024)
   - UFC 1-200-01, DOD BUILDING CODE (GENERAL BUILDING REQUIREMENTS),
   - UFC 1-200-02, HIGH PERFORMANCE AND SUSTAINABLE BUILDING REQUIREMENTS
   - UFC 3-501-01, ELECTRICAL ENGINEERING
   - UFC 3-520-01 INTERIOR ELECTRICAL SYSTEMS
   - UFC 3-530-01, INTERIOR AND EXTERIOR LIGHTING AND CONTROLS
   - UFC 3-550-01, EXTERIOR ELECTRICAL POWER DISTRIBUTION
   - UFC 3-560-01, OPERATION AND MAINTENANCE: ELECTRICAL SAFETY
   - ASHRAE 90.1, ENERGY STANDARD FOR BUILDINGS EXCEPT LOW-RISE RESIDENTIAL BUILDINGS (2025)

   *CRITERIA AND STANDARDS LISTED ALSO INCLUDE SPECIFIED STANDARDS AND RELATED SECTIONS AS PART OF THE DESIGN.

2. MINIMUM RACEWAY SIZE MUST BE 3/4". INCREASE RACEWAY SIZE AS REQUIRED TO LIMIT RACEWAY FILL RATIO TO LESS THAN 40% FULL.

3. THE ELECTRICAL CONTRACTOR MUST CAREFULLY COORDINATE WORK WITH OTHER TRADES THROUGH THE GENERAL CONTRACTOR AND MUST BE RESPONSIBLE FOR SECURING SPACE REQUIREMENTS FOR ELECTRICAL EQUIPMENT CLEARANCE FOR RECESSED LUMINAIRES AND CORRECT ROUGH-IN LOCATIONS OF ELECTRICAL CONNECTIONS.

4. ALL FEEDERS AND BRANCH CIRCUITS MUST INCLUDE A GREEN INSULATED GROUND CONDUCTOR, SIZE PER NATIONAL ELECTRICAL CODE, OR AS SHOWN, CONNECTED TO EACH DEVICE AND OUTLET BOX ON THE CIRCUIT AND TO THE PANELBOARD GROUND BUS. PROVIDE A DEDICATED NEUTRAL CONDUCTOR FOR EACH BRANCH CIRCUIT. MULTIPLE BRANCH CIRCUITS IN ONE RACEWAY REQUIRE ONLY ONE GROUND CONDUCTOR.

5. NEW WORK MUST BE MADE TO TIE INTO THE EXISTING IN A UNIFORM MANNER, SIMILAR ITEMS OF NEW WORK MUST BE CHECKED AGAINST EXISTING WORK.

6. PROVIDE IS AN INCLUSIVE TERM USED TO DESCRIBE ASPECTS OF THE WORK TO BE ACCOMPLISHED, AND IS HEREBY DEFINED TO REQUIRE TO STORE, FURNISH, INSTALL, MOUNT, CONNECT, CONTROL AND POWER EQUIPMENT INDICATED, AS WELL AS ALL APPURTENANCES REQUIRED TO MAKE ELECTRICAL SYSTEMS OPERATE AS INDICATED WITHIN THESE DRAWINGS AND SPECIFICATIONS AND TO FULFILL THE SCOPE OF WORK.

7. DEMOLISH IS AN INCLUSIVE TERM USED TO DESCRIBE ASPECTS OF THE WORK TO BE ACCOMPLISHED, AND IS HEREBY DEFINED TO REQUIRE CONTRACTOR TO DISCONNECT EQUIPMENT FROM ALL CONNECTIONS, REMOVE FROM THE OWNER'S SITE, AND DISPOSE OF IN ACCORDANCE WITH FEDERAL, STATE AND LOCAL LAWS AND ORDINANCES. COST OF DISPOSAL IS ENTIRELY THE CONTRACTOR'S RESPONSIBILITY.

8. ALL CONDUCTORS ARE COPPER UNLESS SPECIFICALLY NOTED AS ALUMINUM.

9. CONTRACTOR MUST FIELD VERIFY ALL EXISTING CONDITIONS, DIMENSIONS AND ELEVATIONS BEFORE PROCUREMENT OF ANY MATERIALS AND DEVELOPMENT OF ANY SHOP DRAWINGS OR SUBMITTALS.

10. PROVIDE LABELS ON ALL RECEPTACLES, WALL MOUNTED LIGHT SWITCHES/OCCUPANCY SENSORS AND JUNCTION BOXES INDICATING THE SOURCE PANEL & CIRCUIT(S). TEXT MUST BE BLACK, 1/4" IN HEIGHT PRINTED ON A CLEAR LABEL. HANDWRITTEN LABELS ARE NOT PERMITTED EXCEPT FOR JUNCTION BOXES LOCATED ABOVE FINISHED CEILING WHICH CAN BE HANDWRITTEN WITH AN INDELIBLE MARKER.

ELECTRICAL ABBREVIATIONS

A OR AMP	AMPERE(S)
ACC	AIR COOLED CHILLER
AC	ALTERNATING CURRENT
AC	AIR COMPRESSOR / AIR CURTAIN
AF	AMP FRAME
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AHU	AIR HANDLING UNIT
AIC	AMPERE INTERRUPTING CAPACITY
AL	ALUMINUM
AM	AMMETER
ASYM	ASYMMETRICAL
AT	AMP TRIP
ATS	AUTOMATIC TRANSFER SWITCH
AUTO	AUTOMATIC
AWG	AMERICAN WIRE GAUGE
B	BOILER
BC	BRANCH CONTROLLER
BCW	BARE COPPER WIRE
BFF	BELOW FINISHED FLOOR
BFG	BELOW FINISHED GRADE
BLDG	BUILDING
BMGB	BUILDING MASTER GROUNDING BUSBAR
C	CONDUIT
CAT	CATEGORY
CB	CIRCUIT BREAKER
CCT	CORRELATED COLOR TEMPERATURE
CCTV	CLOSED CIRCUIT TELEVISION
CFCI	CONTRACTOR FURNISHED CONTRACTOR INSTALLED
CH	CHILLER
CHWCP	CHILLED WATER CIRCULATING PUMP IN A PRIMARY-SECONDARY SYSTEM
CHWP	CHILLED WATER PUMP
CHWSP	CHILLED WATER SUPPLY PUMP IN A PRIMARY-SECONDARY SYSTEM
CKT	CIRCUIT
CMH	COMMUNICATIONS MANHOLE
CMS	COMBINATION MOTOR STARTER
CP	CONDENSATE PUMP
CPT	CONTROL POWER TRANSFORMER
CRAC	COMPUTER ROOM AC UNIT
CRI	COLOR RENDERING INDEX
CT	CURRENT TRANSFORMER
CT	COOLING TOWER
CU	COPPER OR CONDENSING UNIT
CWP	CONDENSER WATER PUMP
D	DEPTH
DC	DIRECT CURRENT
DF	DESTRATIFICATION FAN
DH or DUH	DUCT HEATERS
DHWCP	DOMESTIC HOT WATER CIRCULATING PUMP
DISC	DISCONNECT SWITCH
DM	TO BE DEMOLISHED
DPST	DOUBLE POLE SINGLE THROW
DPDT	DOUBLE POLE DOUBLE THROW
DSCU	DUCTLESS SPLIT CONDENSING UNIT (POWERS THE DSS INDOOR UNIT)
DSHP	DUCTLESS SPLIT HEAT PUMP (POWERS THE DSS INDOOR UNIT)
DSS	DUCTLESS SPLIT SYSTEM (INDOOR UNIT)
DWBP	DOMESTIC WATER BOOSTER PUMP
DX	DIRECT EXPANSION (DX) COOLING COIL
EC	EMPTY CONDUIT
ECB	ENCLOSED CIRCUIT BREAKER
EF	EXHAUST FAN
ELEC	ELECTRICAL
EMH	ELECTRICAL MANHOLE
EMT	ELECTRICAL METALLIC TUBING
E or EMER	EMERGENCY
EPA	EFFECTIVE PROJECTED AREA
EQUIP	EQUIPMENT
ERV	ENERGY RECOVERY VENTILATOR
ETR	EXISTING TO REMAIN
EXIST or EX	EXISTING
EUH	ELECTRIC UNIT HEATER
EXP	EXPLOSION PROOF
EWC	ELECTRIC WATER COOLER
EWH	ELECTRIC WATER HEATER
F	FUSE
FACP	FIRE ALARM CONTROL PANEL
FCU	FAN COIL UNIT
FLEX	FLEXIBLE
FWE	FURNISHED WITH EQUIPMENT
G OR GND	GROUND
GFGI	GOVERNMENT FURNISHED GOVERNMENT INSTALLED
GFI	GROUND FAULT INTERRUPTER
GRS	GALVANIZED RIGID STEEL CONDUIT
GUH	GAS UNIT HEATER
GWH	GAS WATER HEATER
H or HT	HEIGHT
HAC	HEATED AIR CURTAIN
HOA	HAND-OFF AUTOMATIC
HP	HORSE POWER
HP	HEAT PUMP (OUTDOOR PORTION OF SPLIT SYSTEM HEAT PUMP, USE CU FOR STANDARD DX OUTDOOR UNIT)
HT	HEAT TRACKING
HVU	HEATING/VENTILATING UNIT
HWCP	HEATING WATER CIRCULATING PUMP IN A PRIMARY-SECONDARY SYSTEM
HWP	HEATING WATER SUPPLY IN A PRIMARY-ONLY SYSTEM
HWRP	HOT WATER RECIRCULATION PUMP
HWSP	HEATING WATER SUPPLY PUMP IN A PRIMARY-SECONDARY SYSTEM
HWUH	HOT WATER UNIT HEATER
HZ	HERTZ
IDS	INTRUSION DETECTION SYSTEM
IMC	INTERMEDIATE METAL CONDUIT
IRH	INFRARED HEATER
J OR JB	JUNCTION BOX
K	KILO
KAIC	THOUSAND AMPERE INTERRUPTING CAPACITY
KCM OR KCMIL	THOUSAND OF CIRCULAR MILS
KEF	KITCHEN EXHAUST FAN
KH	KITCHEN HOOD
KV	KILOVOLT
KVA	KILOVOLT-AMPERES
KW	KILOWATT
KWHR	KILOWATT-HOUR
L	LENGTH
LA	LIGHTNING ARRESTOR
LAN	LOCAL AREA NETWORK
LAV	LAVATORY
LM	LOCKHEED MARTIN
LTG	LIGHTING
MAU	MAKE-UP AIR UNIT
MAX	MAXIMUM
MCA	MINIMUM CIRCUIT AMPACITY
MCB or MB	MAIN CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTER
MGB	MAIN GROUND BAR
MH	MANHOLE or MOUNTING HEIGHT
MIN	MINIMUM
MLO	MAIN LUGS ONLY
MOD	MOTOR OPERATED DAMPER
MT or MTD	MOUNT or MOUNTED
MRS	MOTOR RATED SWITCH
N	NEUTRAL
NEC	NATIONAL ELECTRICAL CODE
NEMA	NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION
NFPA	NATIONAL FIRE PROTECTION ASSOCIATION
NC	NORMALLY CLOSED
NESC	NATIONAL ELECTRICAL SAFETY CODE
NL	NIGHT LIGHT
NO	NORMALLY OPEN
NTS	NOT TO SCALE
P	POLE(S) or PUMP
PA	PUBLIC ADDRESS
PBB	PRIMARY BONDING BUSBAR
PDU	POWER DISTRIBUTION UNIT
PF	POWER FACTOR
PH	PHASE
PIU	VAV TERMINAL UNIT WITH PARALLEL OR SERIES FAN (PRIMARY INDUCTION UNIT)
PMT	PAD MOUNTED TRANSFORMER
PNL	PANEL OR PANELBOARD
PTAC	PACKAGED THRU-WALL AIR CONDITIONER
PVC	POLYVINYL CHLORIDE
RBB	RACK BONDING BUSBAR
RECEPT or RECP	RECEPTACLE
RF	RETURN FAN
RH	RADIANT HEATERS (ELECTRIC)
RMC	RIGID METAL CONDUIT
RMS	ROOT MEAN SQUARE
RTU	ROOF TOP UNIT
RVNR	REDUCED VOLTAGE NON-REVERSING
SA	SURGE ARRESTOR
SBB	SECONDARY BONDING BUSBAR
SCCR	SHORT CIRCUIT CURRENT RATING
SD	SMOKE DAMPER
SF	SUPPLY FAN
S/N	SOLID NEUTRAL
SP	SUMP PUMP
SPD	SURGE PROTECTIVE DEVICE
SPDT	SINGLE POLE DOUBLE THROW
SPEC	SPECIFICATIONS
SPST	SINGLE POLE SINGLE THROW
SWBD	SWITCHBOARD
SWGR	SWITCHGEAR
TEBC	TELECOMMUNICATIONS EQUIPMENT BONDING CONDUCTOR
TMGB	TELECOMMUNICATIONS MAIN GROUNDING BUSBAR
TOL	THERMAL OVERLOAD
TP	TWISTED PAIR
TYP	TYPICAL
U	URINAL
UH	UNIT HEATER
UG	UNDERGROUND
UIO	UNLESS INDICATED OTHERWISE
UL	UNDERWRITERS LABORATORY
UTP	UNSHIELDED TWISTED PAIR
V	VOLTS
VA	VOLT AMPERES
VAV	VAV TERMINAL UNIT
VEF	VEHICLE EXHAUST FAN
VM	VOLTMETER
VRF	VARIABLE REFRIGERANT FLOW SYSTEM
W	WATTS or WIRE or WIDTH
WC	WATER CLOSET
WEF	WELDING EXHAUST FAN
WH	WATER HEATER
WHDM	WATTHOUR DEMAND METER
WSHP	WATER SOURCE HEAT PUMP
WP	WEATHERPROOF
WPIU	WEATHERPROOF IN USE
XFMR	TRANSFORMER
Z	IMPEDANCE

NOTE:

1. NOT ALL ABBREVIATIONS SHOWN WILL BE USED ON THIS PROJECT.

ELECTRICAL LEGEND

LEGEND GENERAL NOTES

1. DEVICES NOTED AS "GFI" MUST BE GROUND FAULT CIRCUIT INTERRUPTING DEVICES.
2. DEVICES NOTED AS "WP" MUST BE GFI AND WEATHERPROOF.
3. PROVIDE UNSWITCHED POWER TO EMERGENCY BATTERY PACKS.

LUMINAIRES (SEE LIGHT FIXTURE SCHEDULE)

NOTE: THE NUMBER BESIDE THE SYMBOL IS THE CIRCUIT NUMBER. THE LETTER BESIDE THE SYMBOL IS THE FIXTURE TYPE DESCRIBED IN THE LIGHT FIXTURE SCHEDULE.

- SURFACE MOUNTED/SUSPENDED LINEAR FIXTURE.
- SURFACE MOUNTED/SUSPENDED EMERGENCY FIXTURE.
- CEILING MOUNTED EXIT SIGN. PROVIDE CHEVRONS AS INDICATED BY ARROWS.
- WALL MOUNTED EXIT SIGN. PROVIDE CHEVRONS AS INDICATED BY ARROWS.
- WALL MOUNTED FIXTURE.

RECEPTACLES

- DUPLEX RECEPTACLE, NEMA 5-20R, MOUNTED 18" A.F.F. TO CENTERLINE OF BOX UNLESS NOTED OTHERWISE.
- WP — DUPLEX RECEPTACLE, NEMA 5-20R, MOUNTED 18" A.F.F. TO CENTERLINE OF BOX UNLESS NOTED OTHERWISE WITH WEATHERPROOF COVER AND GFI TYPE BREAKER.

GEAR

- xx/xx/xx — AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE. ENCLOSED CIRCUIT BREAKER. TEXT INDICATES AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE.
- xx/xx/xx — AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE. COMBINATION MOTOR STARTER. TEXT INDICATES AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE.
- xx/xx/xx — AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE. NON-FUSED DISCONNECT SWITCH. TEXT INDICATES AMPACITY/NUMBER OF POLES/ENCLOSURE TYPE.
- xx — NAME OF PANELBOARD

SWITCHES

- $ — SINGLE-POLE, SINGLE-THROW SWITCH. MOUNT CENTERLINE OF BOX AT 45"A.F.F. UNLESS NOTED OTHERWISE.
- D — LED DIMMER. MOUNT CENTERLINE OF BOX AT 45"A.F.F. UNLESS NOTED OTHERWISE.
- SP D — LED DIMMER WITH ON/OFF WALL POD. MOUNT CENTERLINE OF BOX AT 45"A.F.F. UNLESS OTHERWISE NOTED.

MISCELLANEOUS

- XX — KEYED NOTE.
- C — CONTACTOR.
- TC — ASTRONOMICAL TIME CLOCK.
- J — CEILING MOUNTED JUNCTION BOX.
- J — WALL MOUNTED JUNCTION BOX.

CONDUIT AND WIRING

- CONDUCTORS IN CONDUIT. TIC MARKS INDICATE NUMBER OF CONDUCTORS. THE EQUIPMENT GROUNDING CONDUCTOR IS NOT SHOWN, BUT MUST BE PROVIDED. SIZE THE EQUIPMENT GROUNDING CONDUCTOR AND THE CONDUIT PER THE NEC. THE ABSENCE OF TIC MARKS SIGNIFIES THAT TWO CONDUCTORS PLUS AN EQUIPMENT GROUNDING CONDUCTOR SHOULD BE PROVIDED. FOR EXAMPLE, THE MARKINGS TO THE LEFT SIGNIFY THAT THREE CONDUCTORS PLUS AN EQUIPMENT GROUNDING CONDUCTOR SHOULD BE PROVIDED.
- 10 — THE TEXT INSIDE THE ARC INDICATES THE AWG SIZE OF THE CONDUCTORS THAT MUST BE RUN IN THE CONDUIT. THE ABSENCE OF TEXT SIGNIFIES THAT THE CONDUCTORS SHOULD BE #12 AWG.
- LA-1 — HOMERUN TO PANELBOARD. ARC DENOTES CONCEALED CIRCUITRY. TEXT DENOTES PANELBOARD NAME WITH CIRCUIT NUMBER. DEVICES HAVING CIRCUIT NUMBERS LOCATED BESIDE THEM MAY NOT SHOW THE CIRCUIT NUMBERS AT THE HOMERUN ARROWS.
- LA-1 — PARTIAL HOMERUN TO PANELBOARD. COMBINE ALL PARTIAL HOMERUNS THAT ARE ON THE SAME CIRCUIT IN A JUNCTION BOX PRIOR TO ENTERING THE PANELBOARD.

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Johnson Adams — PE PE — ARCHITECTS/ENGINEERS · PLANNING · SURVEYING · WASHINGTON — P.O. BOX 513, GREENWOOD, MISSISSIPPI · (662) 455-4943

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USDA AGRICULTURAL RESEARCH SERVICE
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ELECTRICAL GENERAL NOTES & LEGEND

AGRICULTURAL RESEARCH SERVICE
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RM 3-2102
BELTSVILLE MD 20705

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	TMT
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026
SHEET	E001
	27 OF 32

GENERAL NOTES

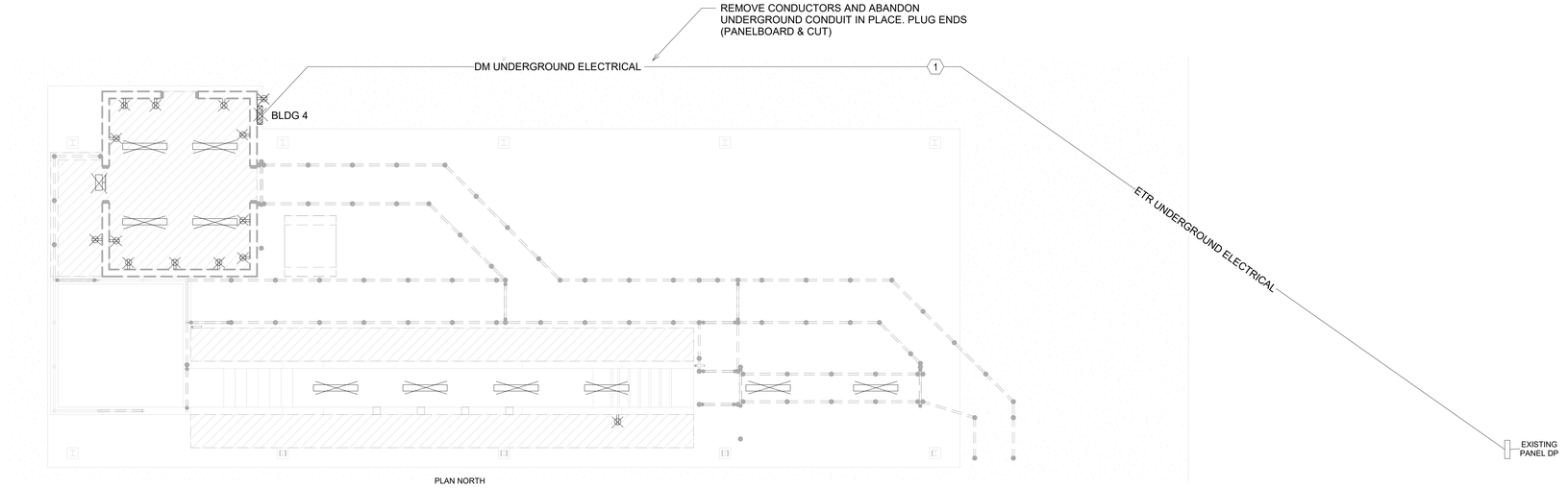
1. INFORMATION SHOWN ON THESE DRAWINGS IS PROVIDED TO SHOW THE GENERAL SCOPE OF THE WORK. ALL EQUIPMENT, LIGHTING, RACEWAYS, AND CIRCUITRY MUST BE DEMOLISHED IN THEIR ENTIRETY UNLESS OTHERWISE NOTED.
2. THE OWNER RESERVES THE FIRST RIGHT OF REFUSAL FOR ALL DEMOLISHED EQUIPMENT AND MATERIALS. REMOVE AND STORE SUCH MATERIALS THAT THE OWNER DOES NOT CHOOSE TO KEEP SHALL BE REMOVED FROM THE PROPERTY AND DISCARDED IN ACCORDANCE WITH ALL FEDERAL, STATE, AND LOCAL LAWS AND REGULATIONS.
3. SEE ARCHITECTURAL DRAWINGS FOR ADDITIONAL DEMOLITION.

DEMO LEGEND

- LIGHTING FIXTURE TO BE DEMOLISHED. DEMOLISH CONDUIT AND FEEDER BACK TO PANEL.
- LIGHTING WALL PACK TO BE DEMOLISHED. DEMOLISH CONDUIT AND FEEDERS BACK TO PANEL.
- RECEPTACLES TO BE DEMOLISHED. DEMOLISH CONDUIT AND FEEDERS BACK TO PANEL.
- PANELBOARD BLDG 4 TO BE DEMOLISHED. CONTRACTOR MUST FIELD VERIFY EXISTING CONDUIT IS UNDAMAGED AND PROPERLY SUPPORTED. PRESERVE AND PROTECT CONDUIT AND CONDUCTORS FOR INSTALLATION OF NEW PANEL. REPORT ANY DEFICIENCIES TO THE ENGINEER. REUSE OF EXISTING CONDUIT AND CONDUCTORS IS SUBJECT TO ENGINEER REVIEW AND ACCEPTANCE.

KEYED NOTES

1. CUT CONDUIT AND REROUTE TO COORDINATE WITH NEW PANEL LOCATION. REFER TO E100.



PLAN NORTH

1 ED100 ELECTRICAL DEMOLITION
3/16" = 1'-0"

4 0 4 8 12

SCALE: 3/16" = 1'-0"

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ELECTRICAL DEMOLITION

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	TMT
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026

AGRICULTURAL RESEARCH SERVICE
5601 SUNNYSIDE AVENUE
RM 3-2102
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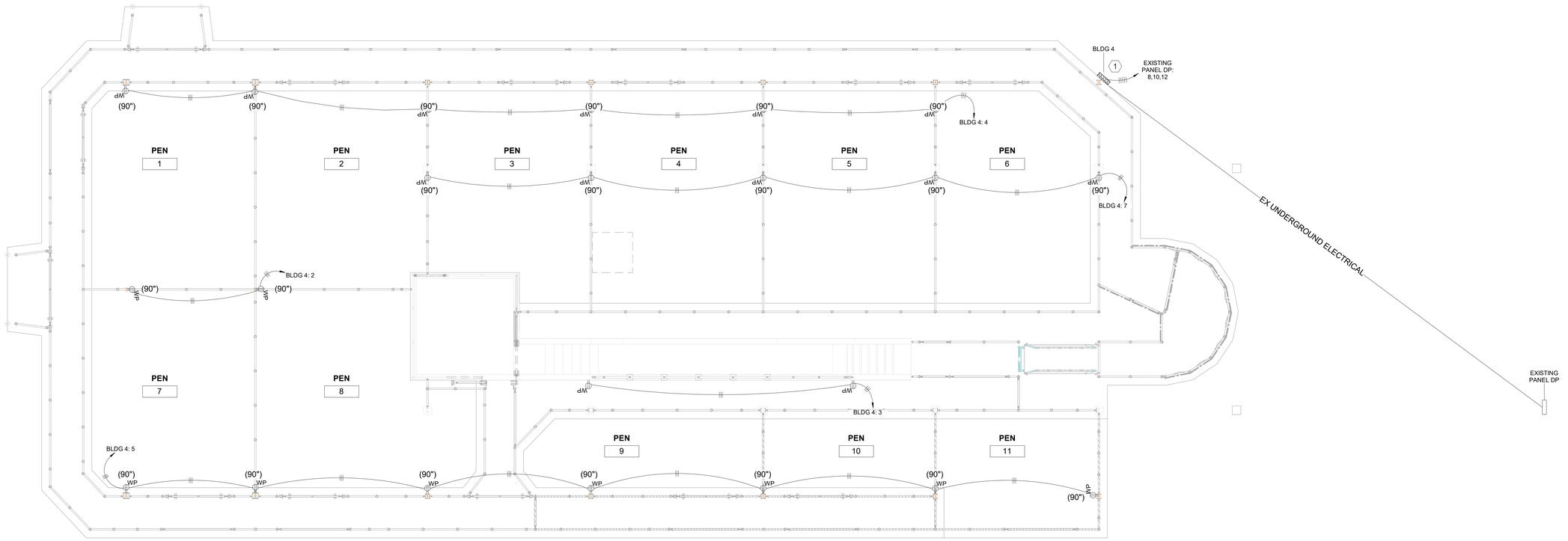
SHEET ED100
28 OF 32

GENERAL NOTES

1. REFER TO E001 FOR GENERAL NOTES AND LEGEND.
2. ALL RECEPTACLES MUST HAVE A WEATHERPROOF COVER AND HAVE A GFCI TYPE BREAKER.
3. ALL CONDUIT ABOVE GROUND MUST BE GALVANIZED RIGID CONDUIT.
4. ALL EQUIPMENT MUST BE RATED A MINIMUM OF NEMA 3R.

KEYED NOTES

1. PROVIDE 100A PANEL. CONTRACTOR MUST FIELD VERIFY THE CONDITION AND CONTINUITY OF THE EXISTING UNDERGROUND CONDUIT AND CONDUCTORS BEFORE CONSTRUCTION. NOTIFY THE ENGINEER IF THE EXISTING CONDUIT IS UNSUITABLE FOR REUSE. PROVIDE (4) #3, #8G COPPER WIRE IN 1-1/2" CONDUIT FROM EXISTING BREAKER IN DP TO NEW BLDG 4 PANEL IF EXISTING CONDUIT AND CONDUCTORS CAN NOT BE USED. TRENCH AND REPAIR CONCRETE AS REQUIRED.



1 E100 POWER PLAN
3/16" = 1'-0"

PLAN NORTH

4 0 4 8 12
SCALE: 3/16" = 1'-0"

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POWER PLAN

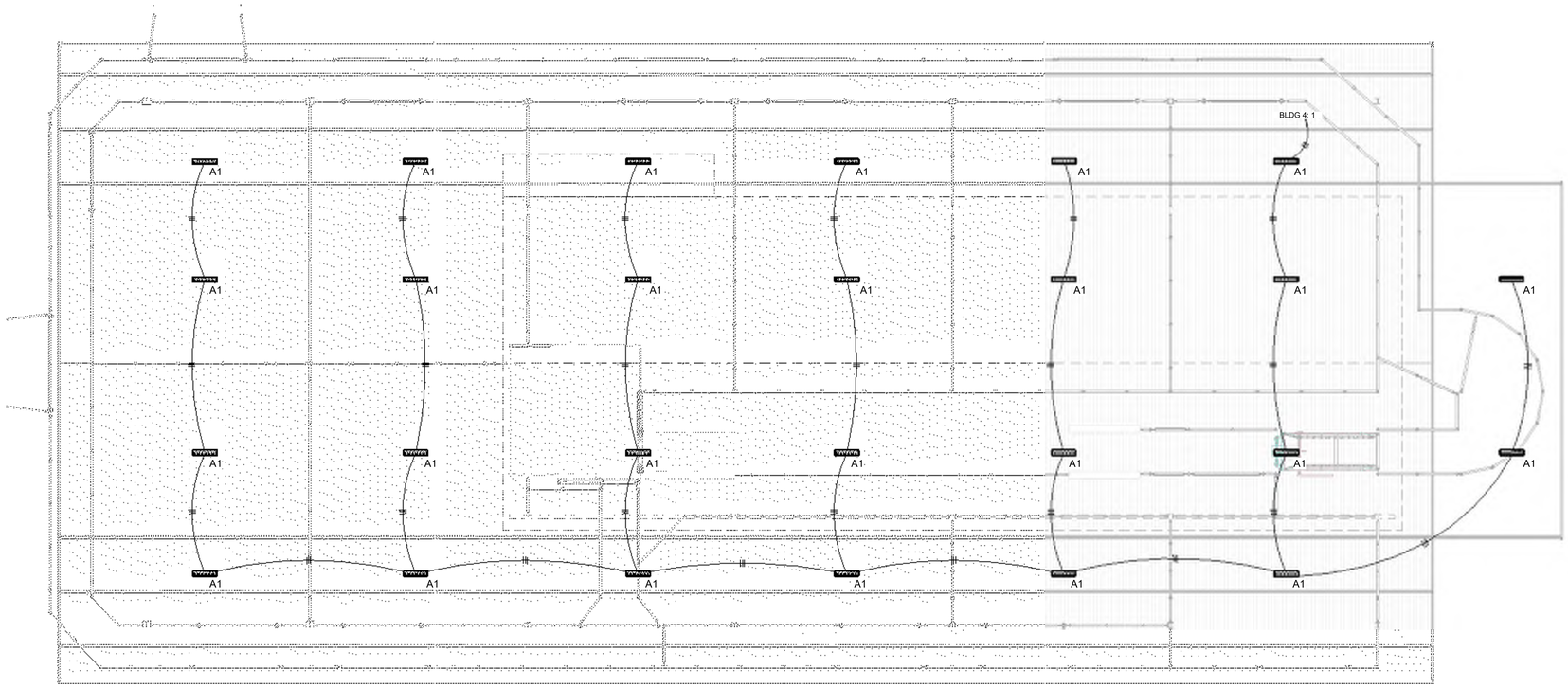
A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	TMT
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026

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SHEET E100
29 OF 32

GENERAL NOTES

1. REFER TO E001 FOR GENERAL NOTES AND LEGEND.
2. CONTROL LIGHTING VIA ASTRONOMICAL TIME CLOCK WITH 0-10V DIMMING CAPABILITIES.
3. DO NOT PENETRATE ROOF. ELECTRICAL CONTRACTOR IS TO PROVIDE UNISTRUT TO MOUNT LIGHT FIXTURES TO BUILDING STRUCTURE.



1 / EL100 LIGHTING PLAN
3/16" = 1'-0"

PLAN NORTH

4 0 4 8 12

SCALE: 3/16" = 1'-0"

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LIGHTING PLAN

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	TMT
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx
PROJECT NO.	xxxxx
DATE	08/31/2026

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SHEET **EL100**
30 OF 32

Branch Panel: BLDG 4

Location: Space 1
Supply From: EXISTING PANEL DP
Mounting: Surface
Enclosure: Type 3R

Volts: 120/208 Wye
Phases: 3
Wires: 4

A.I.C. Rating: 10 kAIC
Mains Type: MCB
Mains Rating: 100 A
MCB Rating: 100 A

Notes:

CKT	Circuit Description	Trip	Poles	A		B		C		Poles	Trip	Circuit Description	CKT
1	Lighting	20 A	1	341 VA	360 VA					1	20 A	Receptacle	2
3	Receptacle	20 A	1			360 VA	1080 VA			1	20 A	Receptacle	4
5	Receptacle	20 A	1					1260 VA	0 VA	1	20 A	Spare	6
7	Receptacle	20 A	1	900 VA	0 VA					1	20 A	Spare	8
9	Spare	20 A	1			0 VA	0 VA			1	20 A	Spare	10
11	Space	--	1					--	0 VA	1	20 A	Spare	12
13	Space	--	1	--	0 VA					3	20 A	SPD Type 2	14
15	Space	--	1			--	0 VA			--	--	--	16
17	Space	--	1					--	0 VA	--	--	--	18
		Total Load:		1601 VA		1440 VA		1260 VA					
		Total Amps:		14 A		12 A		11 A					

Legend:

Load Classification	Connected Load	Demand Factor	Estimated Demand
Receptacle	3960 VA	100.00%	3960 VA
Lighting	341 VA	100.00%	341 VA

Panel Totals	
Total Conn. Load:	4301 VA
Total Est. Demand:	4301 VA
Total Conn.:	12 A
Total Est. Demand:	12 A

Notes:



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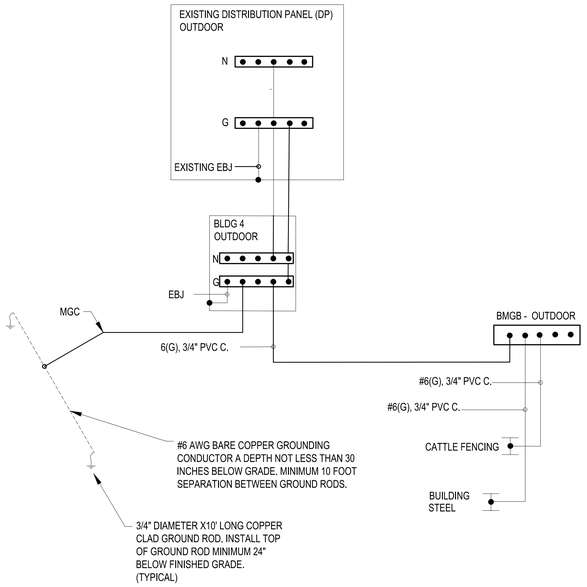
A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	TMT
PPM	xxxxx
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125
REAL PROPERTY	xxxxx

USDA Agricultural Research Service

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MOORE FIELD EDINBURG, TEXAS

PANEL SCHEDULE

PROJECT NO.	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD 20705	SHEET
xxxxx		**E600**
DATE 08/31/2026		31 OF 32



NOTES:

1. THIS DETAIL DOES NOT PRECLUDE THE REQUIREMENTS OF NEC ART. 250 OR NFPA 780. CONTRACTOR SHALL STRICTLY ADHERE TO NEC ART. 250, IN ADDITION TO THIS DETAIL.

1 E700 GROUNDING DIAGRAM
SCALE: N.T.S.

ABBREVIATIONS:

MGC	MAIN GROUNDING CONDUCTOR
SBJ	SYSTEM BONDING JUMPER
SSBJ	SUPPLY SIDE BONDING JUMPER
EGC	EQUIPMENT GROUNDING CONDUCTOR
GEC	GROUNDING ELECTRODE CONDUCTOR
NC	NEUTRAL CONDUCTOR
GE	GROUNDING ELECTRODE
GES	GROUNDING ELECTRODE SYSTEM
ISBT	INTERSYSTEM BONDING TERMINATION
EBJ	EQUIPMENT BONDING JUMPERS
BMGB	BUILDING MAIN GROUNDING BUSBAR

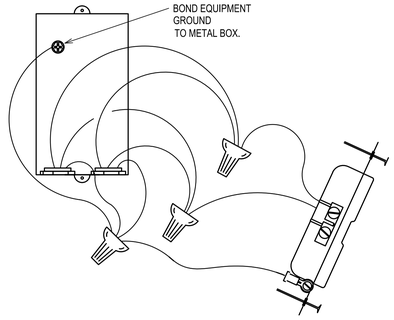
RECEPTACLE DETAIL NOTES:

A. RECEPTACLES SHOWN AS GFI MAY BE NON GFI TYPE RECEPTACLES IF FED FROM A 20/1 GFI BREAKER OR THE LOAD SIDE OF A GFI RECEPTACLE IN THE SAME ROOM, ON THE SAME CIRCUIT AND RATED 16 AMP FEED-THRU CAPACITY. COVER PLATES MUST BE CLEARLY MARKED GFI.

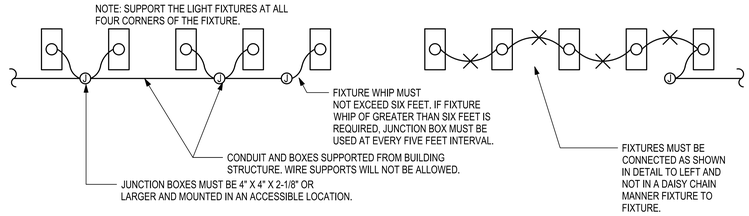
B. NON GFI RECEPTACLES MUST NOT BE CONNECTED IN A FEED-THRU MANNER. WIRE CONNECTIONS IN RECEPTACLE BOXES MUST BE MADE IN A PIGTAIL MANNER AS SHOWN IN RECEPTACLE DETAIL.

C. RECEPTACLE CONNECTIONS CANNOT BE BACK STABBED. INSTALLING CONDUCTORS LOOPED AROUND THE DEVICE SCREW IS NOT ALLOWED. CONDUCTORS MUST BE INSTALLED UNDER A BRASS PRESSURE PLATE WHICH IS PART OF THE DEVICE. THE CONDUCTORS MUST BE NICK FREE WHICH MAY BE CREATED WHEN STRIPPING THE INSULATION OFF OF THE CONDUCTOR.

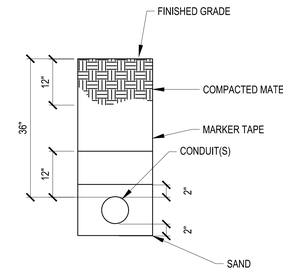
D. MANUFACTURED WIRING SYSTEMS ARE NOT ALLOWED.



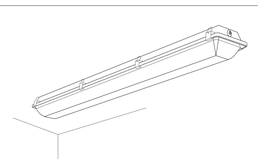
2 E700 RECEPTACLE DETAIL
SCALE: N.T.S.



3 E700 FIXTURE CONNECTION DETAIL
SCALE: N.T.S.



4 E700 TRENCH DETAIL
SCALE: N.T.S.



NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING - ONE-PIECE 5VA FIBERGLASS HOUSING WITH INTEGRAL PERIMETER CHANNEL UTILIZING CONTINUOUS POURED-IN-PLACE NEMA 4X GASKET. SIZE AS INDICATED IN LUMINAIRE SCHEDULE.
2. OPTICS - UV-STABILIZED, HIGH-IMPACT, FROSTED POLYCARBONATE LENS.
3. LIGHT SOURCE - SOLID STATE LEDS, 3000K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 90 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER - REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON/OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
5. CERTIFICATION - UL LISTED FOR WET LOCATION, ROHS COMPLIANT. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING - CEILING SURFACE-MOUNTED.

LED VAPOR TIGHT

FIXTURE TYPE: A1

LIGHTING CALCULATIONS

AVERAGE	AVG/MIN	HEIGHT
3 FC	2	0 FT

FOR AN AVERAGE OF 3 FC DIM LIGHTS TO 50%.

LUMINAIRE SCHEDULE

TYPE	DESCRIPTION	SOURCE				DIMMING	VOLTAGE	TOTAL VA	MANUFACTURER BASIS OF DESIGN		NOTES
		TYPE	LM	CCT(K)	CRI				NAME	CATALOG SERIES	
A1	LOW-PROFILE ENCLOSED & GASKETED INDUSTRIAL	LED	2000	3000	80	0-10V	120-277V	13	HOLOPHANE	EMS L24 2000LM LPAFL WD, MVOLT GZ10 30K 80CRI	

DAVID R. LEARD
REGISTERED PROFESSIONAL
ENGINEER
9569
STATE OF MISSISSIPPI

NO.	DATE	BY	DESCRIPTION

REVISIONS

Johnson, Mirmiran & Thompson, Inc.
PROJECTS/ENGINEERING PLANNING SURVEYING MANAGEMENT
P.O. BOX 4-5, GREENWOOD, MISSISSIPPI • (662) 455-6443

A-E FIRM	USDA
PROJECT MANAGER	REH
EPM	B. DEAN
DESIGNER	TMT
PPM	XXXXX
CHECKED BY	DRL
SAFETY & HEALTH	D. CORDRAY
JOB #	4125-A
REAL PROPERTY	XXXXX

USDA Agricultural Research Service

USDA AGRICULTURAL RESEARCH SERVICE
EDINBURG - KERRVILLE BLDG. 4
DIP VAT REDESIGN
MOORE FIELD EDINBURG, TEXAS

DETAILS

PROJECT NO. XXX	AGRICULTURAL RESEARCH SERVICE 5601 SUNNYSIDE AVENUE RM 3-2102 BELTSVILLE MD, 20705	SHEET E700
DATE 8/31/2026		32 OF 32

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Aug 31, 2026 - 12:59pm